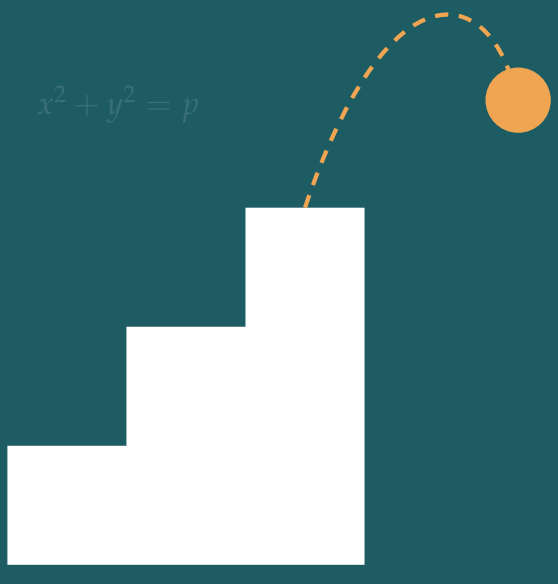
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2019 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2019 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2019 Worked Solutions

Worked solutions to all 35 questions of the 2019 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2019, Question 1

Pure • Trigonometric Optimisation

STEP I 2019, Question 1

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Everything follows from where the line cuts the axes. For the perimeter, the fixed point $P = (1, k)$ splits the hypotenuse XY into two pieces, each the hypotenuse of a small right-angled triangle with a side of known length, so no square roots need to be expanded. For the minima, the area is a sum of two terms with a constant product (an AM–GM situation), and the derivative of the perimeter collapses beautifully once we use $\cos^2\theta = (1 - \sin\theta)(1 + \sin\theta)$.

Solution

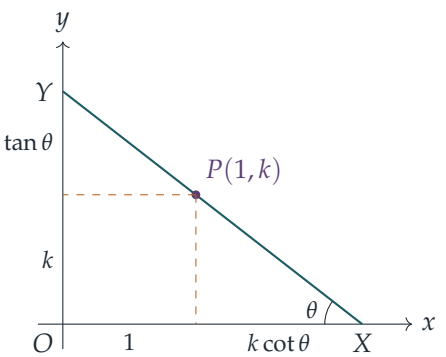
The line through $(1, k)$ with gradient $-\tan\theta$ is

$$y - k = -\tan\theta\,(x - 1).$$

Putting $y = 0$ gives $x - 1 = k\cot\theta$, and putting $x = 0$ gives $y = k + \tan\theta$, so

$$X = (1 + k\cot\theta,\ 0), \qquad Y = (0,\ k + \tan\theta).$$

Since $k > 0$ and $0 < \theta < \frac{1}{2}\pi$, both intercepts are positive, so OXY is a genuine right-angled triangle in the first quadrant, with $P = (1, k)$ on its hypotenuse.



Part (i)

The triangle has a right angle at O, so

$$\begin{aligned} A = \tfrac{1}{2}(1 + k\cot\theta)(k + \tan\theta) &= \tfrac{1}{2}\big(k + \tan\theta + k^2\cot\theta + k\big) \\ &= k + \tfrac{1}{2}\big(\tan\theta + k^2\cot\theta\big), \end{aligned}$$

using $\cot\theta\tan\theta = 1$.

For the minimum, both $\tan\theta$ and $k^2\cot\theta$ are positive and their product is the constant k^2. By the AM–GM inequality,

$$\tan\theta + k^2\cot\theta \geq 2\sqrt{\tan\theta \cdot k^2\cot\theta} = 2k,$$

with equality exactly when $\tan\theta = k^2\cot\theta$, that is, $\tan\theta = k$ (which is attained by some θ in the range, since $k > 0$). Hence $A \geq k + k = 2k$, and

$$A_{\min} = 2k.$$

(Differentiating gives the same: $\frac{\mathrm{d}A}{\mathrm{d}\theta} = \frac{1}{2}(\sec^2\theta - k^2\operatorname{cosec}^2\theta) = 0$ when $\tan^2\theta = k^2$.) At the minimum $X = (2, 0)$ and $Y = (0, 2k)$, so P is the midpoint of XY.

Part (ii)

The two legs are $OX = 1 + k\cot\theta$ and $OY = k + \tan\theta$. For the hypotenuse, split it at P as in the diagram.

- From P to Y: the horizontal run is 1 and the line makes angle θ with the horizontal, so $PY = 1/\cos\theta = \sec\theta$.
- From P to X: the vertical drop is k, so $PX = k/\sin\theta = k\operatorname{cosec}\theta$.

Adding the four lengths,

$$L = 1 + \tan\theta + \sec\theta + k\big(1 + \cot\theta + \operatorname{cosec}\theta\big),$$

as required.

> **Ask yourself** The derivative contains $\sec^2\theta + \sec\theta\tan\theta$. Is there a way to make it much simpler before setting it to zero?

Yes. Differentiating term by term,

$$\frac{\mathrm{d}L}{\mathrm{d}\theta} = \sec^2\theta + \sec\theta\tan\theta - k\big(\operatorname{cosec}^2\theta + \operatorname{cosec}\theta\cot\theta\big) = \frac{1 + \sin\theta}{\cos^2\theta} - k\,\frac{1 + \cos\theta}{\sin^2\theta}.$$

Now $\cos^2\theta = (1 - \sin\theta)(1 + \sin\theta)$ and $\sin^2\theta = (1 - \cos\theta)(1 + \cos\theta)$, and $1 + \sin\theta$, $1 + \cos\theta$ are non-zero, so we may cancel:

$$\frac{\mathrm{d}L}{\mathrm{d}\theta} = \frac{1}{1 - \sin\theta} - \frac{k}{1 - \cos\theta}.$$

This is zero exactly when $1-\cos\theta = k(1-\sin\theta)$. So the minimum occurs at $\theta = \alpha$ with

$$\frac{1-\cos\alpha}{1-\sin\alpha} = k,$$

which is what we had to show. (In fact this stationary point is the only one and is a minimum: as θ increases on $(0, \frac{1}{2}\pi)$, the first term of $\frac{\mathrm{d}L}{\mathrm{d}\theta}$ increases and the second decreases, so $\frac{\mathrm{d}L}{\mathrm{d}\theta}$ is increasing and changes sign from negative to positive once.)

Ask yourself How do we get rid of k in the minimum value?

Group the terms of L over common denominators:

$$1+\tan\alpha+\sec\alpha = \frac{1+\sin\alpha+\cos\alpha}{\cos\alpha}, \qquad 1+\cot\alpha+\operatorname{cosec}\alpha = \frac{1+\sin\alpha+\cos\alpha}{\sin\alpha}.$$

Writing $s = \sin\alpha$, $c = \cos\alpha$ and $k = \dfrac{1-c}{1-s}$, the second piece becomes

$$k\,\frac{1+s+c}{s} = \frac{(1+s+c)(1-c)}{s(1-s)} = \frac{1+s-sc-c^2}{s(1-s)} = \frac{s^2+s-sc}{s(1-s)} = \frac{1+s-c}{1-s},$$

using $1-c^2 = s^2$. Therefore

$$\begin{aligned} L_{\min} = \frac{1+s+c}{c} + \frac{1+s-c}{1-s} &= \frac{(1+s+c)(1-s)+c(1+s-c)}{c(1-s)} \\ &= \frac{1-s^2+c-sc+c+sc-c^2}{c(1-s)} \\ &= \frac{2c}{c(1-s)}, \end{aligned}$$

since $1-s^2-c^2 = 0$. So

$$L_{\min} = \frac{2}{1-\sin\alpha}.$$

Checks

Take $k = 1$. Then $\tan\theta = 1$ gives $X = (2,0)$, $Y = (0,2)$ and $A = 2 = 2k$, as predicted. The perimeter condition $(1-\cos\alpha)/(1-\sin\alpha) = 1$ gives $\alpha = \frac{1}{4}\pi$, and $2/(1-\frac{1}{\sqrt{2}}) = 4+2\sqrt{2} \approx 6.828$, which is exactly $2+2+2\sqrt{2}$ for the triangle with vertices $(0,0)$, $(2,0)$, $(0,2)$. For a range of other values of k, a computer search over θ confirmed both minimum values and the location of the perimeter minimum.

Common slips

- Expanding $\sqrt{(1+k\cot\theta)^2+(k+\tan\theta)^2}$ for the hypotenuse instead of splitting it at P.

- Setting the unsimplified derivative to zero and getting lost; cancel $1 + \sin\theta$ and $1 + \cos\theta$ first.
- Leaving k in the final answer: the question wants the minimum in terms of α alone.

Beyond STEP

The neat answer has a geometric explanation. The minimum-perimeter line through P is the one tangent at P to a circle touching both axes (an excircle of OXY), and the perimeter is then twice the tangent length from O to that circle; this is the circle of radius r through P with $r = 1/(1 - \sin\alpha)$. Optimisation problems with a constraint like "the line passes through P" are the starting point for Lagrange multipliers in a first-year calculus course. For many more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2019, Question 2

Pure • Tangents to Parametric Curves

STEP I 2019, Question 2

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The parametrisation is chosen so that the gradient at parameter t is simply t, which makes every tangent line very clean. Finding where two tangents meet is then a matter of factorising a difference of cubes, and the perpendicularity condition $pq = -1$ lets us rewrite everything in terms of the sum $p + q$. For the common points, substituting one parametrisation into the other gives a polynomial with a repeated root, and that repeated root is telling us something about the geometry.

Solution

The tangent

With $x = 3t^2$ and $y = 2t^3$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \frac{6t^2}{6t} = t \qquad (t \neq 0).$$

So the tangent at $(3p^2, 2p^3)$ has gradient p and equation

$$y - 2p^3 = p\,(x - 3p^2) \quad \Longrightarrow \quad y = px - 3p^3 + 2p^3 = px - p^3,$$

as required. (At $p = 0$ the point is the origin, a cusp; there the curve is $y = \pm 2(x/3)^{3/2}$, which is tangent to the x-axis $y = 0$, so the formula still holds.)

Where two tangents meet

For $p \neq q$, the tangents $y = px - p^3$ and $y = qx - q^3$ meet where

$$(p - q)\,x = p^3 - q^3 = (p - q)(p^2 + pq + q^2).$$

Since $p \neq q$ we may divide by $p - q$, giving $x = p^2 + pq + q^2$, and then

$$y = px - p^3 = p^3 + p^2q + pq^2 - p^3 = pq(p + q).$$

So the tangents meet at

$$(p^2 + pq + q^2,\ pq(p+q)).$$

The tangents have gradients p and q, so they are perpendicular exactly when $pq = -1$. Writing $u = p + q$, we have $p^2 + pq + q^2 = (p+q)^2 - pq = u^2 + 1$ and $pq(p+q) = -u$. Hence perpendicular tangents meet at $(u^2 + 1,\ -u)$, as required.

Ask yourself Does every point of $\widetilde{C}$ actually arise from a pair of perpendicular tangents?

Yes: given any real u, we need $p + q = u$ and $pq = -1$, so p and q are the roots of $z^2 - uz - 1 = 0$. Its discriminant $u^2 + 4$ is positive, so there are always two distinct real roots. So $\widetilde{C}$ is exactly the set of points from which two perpendicular tangents to C can be drawn.

Points on both curves

Eliminating the parameter, $\widetilde{C}$ is the parabola $x = y^2 + 1$. A point $(3t^2, 2t^3)$ of C lies on it exactly when

$$3t^2 = 4t^6 + 1 \quad \Longleftrightarrow \quad 4s^3 - 3s + 1 = 0, \quad \text{where } s = t^2.$$

Since $s = -1$ is a root, we can factorise:

$$4s^3 - 3s + 1 = (s+1)(4s^2 - 4s + 1) = (s+1)(2s-1)^2.$$

As $s = t^2 \geq 0$, the only possibility is $s = \frac{1}{2}$, so $t = \pm\frac{1}{\sqrt{2}}$. Then $x = 3t^2 = \frac{3}{2}$ and $y = 2t^3 = \pm\frac{1}{\sqrt{2}}$. The common points are

$$\left(\tfrac{3}{2},\ \tfrac{1}{\sqrt{2}}\right) \quad \text{and} \quad \left(\tfrac{3}{2},\ -\tfrac{1}{\sqrt{2}}\right).$$

Ask yourself What does the repeated factor $(2s-1)^2$ tell us?

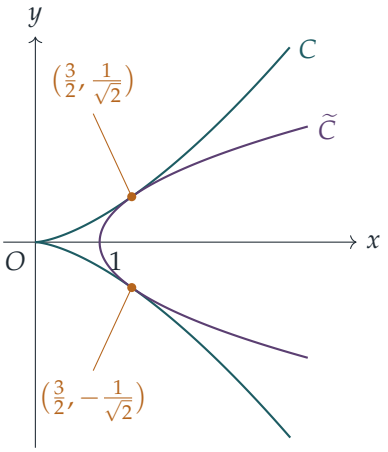
A double root usually signals tangency, and that is what happens here. At $t = \frac{1}{\sqrt{2}}$ the gradient of C is $t = \frac{1}{\sqrt{2}}$. On $\widetilde{C}$, $x = y^2 + 1$ gives $\frac{dy}{dx} = \frac{1}{2y} = \frac{1}{2/\sqrt{2}} = \frac{1}{\sqrt{2}}$, the same value. By symmetry the same is true at the lower point. So the two curves touch at both points and do not cross.

The sketch

Eliminating t from C gives $y^2 = \frac{4}{27}x^3$, with $x \geq 0$. This is a semicubical parabola, symmetric in the x-axis, with a cusp at the origin pointing left. $\widetilde{C}$ is the parabola $y^2 = x - 1$, with vertex $(1, 0)$, opening to the right. To see which curve is outside the other, compare the two values of y^2 for $x \geq 1$:

$$f(x) = \tfrac{4}{27}x^3 - (x-1), \qquad f'(x) = \tfrac{4}{9}x^2 - 1,$$

so f has its minimum on $x \geq 1$ at $x = \frac{3}{2}$, where $f(\frac{3}{2}) = \frac{1}{2} - \frac{1}{2} = 0$. Hence $f(x) \geq 0$, with equality only at $x = \frac{3}{2}$. So $\widetilde{C}$ lies inside C (closer to the x-axis) and touches it at the two common points.



Checks

- Take $p = 1$, $q = -1$ (so $pq = -1$ and $u = 0$). The tangents are $y = x - 1$ and $y = -x + 1$, which are perpendicular and meet at $(1, 0)$, the vertex of $\widetilde{C}$, as $(u^2 + 1, -u)$ predicts.
- Take $p = 2$, $q = -\frac{1}{2}$, so $u = \frac{3}{2}$. The tangents $y = 2x - 8$ and $y = -\frac{1}{2}x + \frac{1}{8}$ meet where $\frac{5}{2}x = \frac{65}{8}$, so $x = \frac{13}{4} = u^2 + 1$ and $y = -\frac{3}{2} = -u$.
- The point $(1.5, 0.7071)$ satisfies both $y^2 = \frac{4}{27}x^3$ (since $\frac{4}{27} \cdot \frac{27}{8} = \frac{1}{2}$) and $y^2 = x - 1$. A computer check also confirmed that $3t^2 - 4t^6 - 1$ never goes above zero, so there are no other common points.

Common slips

- Forgetting that $s = t^2$ cannot be negative, and trying to use the root $s = -1$.
- Sketching the parabola crossing C at the common points instead of touching it there.

Beyond STEP

The curve $\widetilde{C}$ is the *orthoptic* of C: the locus of points from which two perpendicular tangents to C can be drawn; for a parabola it is the directrix. Describing a curve by its family of tangent lines leads to envelopes and to duality in projective geometry. Siklos, *Advanced Problems in Mathematics*, has more questions of this kind.

WORKED SOLUTION

STEP I 2019, Question 3

Pure • Integration

STEP I 2019, Question 3

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Multiplying top and bottom by $1-\sin x$ turns $1+\sin x$ into $1-\sin^2 x=\cos^2 x$, and everything over a power of $\cos x$ splits into standard pieces: $\sec^2 x$, $\sec x\tan x$, $\tan^2 x\sec^2 x$ and $\sin x/\cos^4 x$, each of which we can integrate on sight. The other two integrals are the same trick in disguise: the second becomes a $1+\cos x$ integral, which the substitution $x=\frac{1}{2}\pi-u$ turns into the first kind, and the third squares the factor $1-\sin x$.

Solution

The first integral

Since $(1+\sin x)(1-\sin x)=1-\sin^2 x=\cos^2 x$,

$$\frac{1}{1+\sin x}=\frac{1-\sin x}{\cos^2 x}=\sec^2 x-\sec x\tan x.$$

We know $\frac{\mathrm{d}}{\mathrm{d}x}\tan x=\sec^2 x$ and $\frac{\mathrm{d}}{\mathrm{d}x}\sec x=\sec x\tan x$, so

$$\int\frac{1}{1+\sin x}\,\mathrm{d}x=\tan x-\sec x+C. \qquad (*)$$

Using $\tan\frac{1}{4}\pi=1$, $\sec\frac{1}{4}\pi=\sqrt{2}$, $\tan 0=0$ and $\sec 0=1$:

$$\int_0^{\frac{1}{4}\pi}\frac{1}{1+\sin x}\,\mathrm{d}x=\left(1-\sqrt{2}\right)-\left(0-1\right)=2-\sqrt{2}.$$

(Multiplying by $1-\sin x$ is fine because $\sin x<1$ throughout the range, so we never multiply by zero.)

The second integral

Multiplying top and bottom by $\cos x$ (non-zero on $[\frac{1}{4}\pi, \frac{1}{3}\pi]$),

$$\frac{1}{1+\sec x} = \frac{\cos x}{\cos x + 1} = 1 - \frac{1}{1+\cos x}.$$

? **Ask yourself** How can we use $(*)$ on $\dfrac{1}{1+\cos x}$?

Since $\cos x = \sin(\frac{1}{2}\pi - x)$, substitute $u = \frac{1}{2}\pi - x$, so $\mathrm{d}x = -\mathrm{d}u$; when $x = \frac{1}{4}\pi$, $u = \frac{1}{4}\pi$, and when $x = \frac{1}{3}\pi$, $u = \frac{1}{6}\pi$. Then

$$\begin{aligned}\int_{\frac{1}{4}\pi}^{\frac{1}{3}\pi} \frac{1}{1+\cos x}\,\mathrm{d}x = \int_{\frac{1}{6}\pi}^{\frac{1}{4}\pi} \frac{1}{1+\sin u}\,\mathrm{d}u &= \Big[\tan u - \sec u\Big]_{\frac{1}{6}\pi}^{\frac{1}{4}\pi} \\ &= (1-\sqrt{2}) - \Big(\frac{1}{\sqrt{3}} - \frac{2}{\sqrt{3}}\Big) = 1 - \sqrt{2} + \frac{1}{\sqrt{3}},\end{aligned}$$

using $\tan\frac{1}{6}\pi = 1/\sqrt{3}$ and $\sec\frac{1}{6}\pi = 2/\sqrt{3}$. The interval has length $\frac{1}{3}\pi - \frac{1}{4}\pi = \frac{1}{12}\pi$, so

$$\int_{\frac{1}{4}\pi}^{\frac{1}{3}\pi} \frac{1}{1+\sec x}\,\mathrm{d}x = \frac{\pi}{12} - \Big(1 - \sqrt{2} + \frac{1}{\sqrt{3}}\Big) = \frac{\pi}{12} + \sqrt{2} - 1 - \frac{\sqrt{3}}{3}.$$

(Equally, one can multiply $\frac{1}{1+\cos x}$ by $\frac{1-\cos x}{1-\cos x}$ to get $\operatorname{cosec}^2 x - \cot x \operatorname{cosec} x$, with antiderivative $\operatorname{cosec} x - \cot x = \tan\frac{1}{2}x$; since $\tan\frac{1}{8}\pi = \sqrt{2}-1$ this gives the same answer.)

The third integral

Now we multiply top and bottom by $(1-\sin x)^2$, so the denominator becomes $\cos^4 x$:

$$\frac{1}{(1+\sin x)^2} = \frac{(1-\sin x)^2}{\cos^4 x} = \frac{1+\sin^2 x}{\cos^4 x} - \frac{2\sin x}{\cos^4 x}.$$

For the first piece, write $1 = \sin^2 x + \cos^2 x$ in the numerator:

$$\frac{1+\sin^2 x}{\cos^4 x} = \frac{\cos^2 x + 2\sin^2 x}{\cos^4 x} = \sec^2 x + 2\tan^2 x \sec^2 x.$$

? **Ask yourself** What do $\tan^2 x \sec^2 x$ and $\sin x/\cos^4 x$ integrate to?

Both are of the form $f(x)^n f'(x)$. With $f = \tan x$, $\frac{\mathrm{d}}{\mathrm{d}x}\tan^3 x = 3\tan^2 x \sec^2 x$; with $f = \cos x$, $\frac{\mathrm{d}}{\mathrm{d}x}(\cos x)^{-3} = 3\sin x\,(\cos x)^{-4}$. Hence

$$\int \frac{1}{(1+\sin x)^2}\,\mathrm{d}x = \tan x + \tfrac{2}{3}\tan^3 x - \tfrac{2}{3}\sec^3 x + C.$$

At $x = \frac{1}{4}\pi$: $\tan = 1$ and $\sec^3 = (\sqrt{2})^3 = 2\sqrt{2}$, giving $1 + \frac{2}{3} - \frac{4}{3}\sqrt{2}$. At $x = 0$: $0 + 0 - \frac{2}{3}$. Therefore

$$\int_0^{\frac{1}{4}\pi} \frac{1}{(1+\sin x)^2}\,\mathrm{d}x = \frac{5}{3} - \frac{4\sqrt{2}}{3} + \frac{2}{3} = \frac{7-4\sqrt{2}}{3}.$$

Checks

- Size: on $[0, \frac{1}{4}\pi]$ the first integrand falls from 1 to $2-\sqrt{2} \approx 0.59$, so the integral lies between 0.46 and 0.79; $2-\sqrt{2} \approx 0.586$ fits. Likewise the third integrand is the square of the first, so its integral should be smaller: $(7-4\sqrt{2})/3 \approx 0.448$.
- Differentiate your antiderivatives: $\frac{\mathrm{d}}{\mathrm{d}x}(\tan x - \sec x) = \frac{1-\sin x}{\cos^2 x} = \frac{1}{1+\sin x}$.
- Simpson's rule gives 0.585786, 0.098663 and 0.447715 for the three integrals, matching $2-\sqrt{2}$, $\frac{\pi}{12} + \sqrt{2} - 1 - \frac{\sqrt{3}}{3}$ and $\frac{1}{3}(7-4\sqrt{2})$ to ten decimal places.

Common slips

- Sign errors in $\frac{\mathrm{d}}{\mathrm{d}x}\sec x$ or $\frac{\mathrm{d}}{\mathrm{d}x}\sec^3 x$: both carry a plus sign, as $\frac{\mathrm{d}}{\mathrm{d}x}(\cos x)^{-n} = n\sin x(\cos x)^{-n-1}$.
- Forgetting the lower limit in the first and third integrals: $\sec 0 = 1$, not 0.
- Swapping the limits wrongly after $u = \frac{1}{2}\pi - x$: the minus sign from $\mathrm{d}x = -\mathrm{d}u$ is used to put them back in increasing order.

Beyond STEP

The trick of multiplying by a "conjugate" is a special case of a systematic method: the substitution $t = \tan\frac{1}{2}x$ turns any rational function of $\sin x$ and $\cos x$ into a rational function of t, which partial fractions can then integrate. (Our answer $\tan\frac{1}{2}x$ for $\int \frac{\mathrm{d}x}{1+\cos x}$ is a hint of this.) First-year analysis also asks when an integral like these exists at all, for example near $x = -\frac{1}{2}\pi$ where $1+\sin x$ vanishes. For both ideas, see Bartle and Sherbert, *Introduction to Real Analysis*, and for many more integrals in this style, Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2019, Question 4

Pure • Factorising Quartic Equations

STEP I 2019, Question 4

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

A quartic with no x^3 term has roots that add up to zero. So if we split the four roots into two pairs, the two pair-sums are negatives of each other, and that is exactly why the two quadratic factors have x-coefficients s and $-s$. Comparing coefficients then gives three equations, and eliminating p and q leaves a cubic in s^2. Part (i) is there to help with the surds at the very end.

Solution

Part (i)

Squaring $m+n\sqrt{2}$ gives $m^2+2n^2+2mn\sqrt{2}$, so we want $m^2+2n^2=3$ and $mn=1$. Taking $m=n=1$ works:

$$(1+\sqrt{2})^2 = 1+2\sqrt{2}+2 = 3+2\sqrt{2},$$

and since $1+\sqrt{2}>0$ it is the positive square root. Hence $m=1$, $n=1$:

$$\sqrt{3+2\sqrt{2}} = 1+\sqrt{2}.$$

In the same way, $(\sqrt{2}-1)^2 = 3-2\sqrt{2}$, so $\sqrt{3-2\sqrt{2}} = \sqrt{2}-1$. We shall need both.

Part (ii)

Why the factorisation exists. Let the four real roots be a, b, c, d, so $\mathrm{f}(x) = (x-a)(x-b)(x-c)(x-d)$. The coefficient of x^3 in f is $-(a+b+c+d)$, and it is zero, so $c+d=-(a+b)$. Grouping the roots in pairs,

$$\mathrm{f}(x) = \big(x^2-(a+b)x+ab\big)\big(x^2-(c+d)x+cd\big) = \big(x^2+sx+p\big)\big(x^2-sx+q\big),$$

with $s=-(a+b)$, $p=ab$ and $q=cd$, all real because the roots are real.

The three equations. Expanding,

$$(x^2+sx+p)(x^2-sx+q) = x^4+(p+q-s^2)x^2+s(q-p)x+pq.$$

Comparing with $x^4-10x^2+12x-2$:

$$\begin{aligned} p+q-s^2 &= -10,\\ s(q-p) &= 12,\\ pq &= -2. \end{aligned}$$

The equation for s. The second equation shows $s \neq 0$, so we may divide by it: $q-p = 12/s$. The first gives $p+q = s^2-10$. Now use the identity $(p+q)^2-(q-p)^2 = 4pq = -8$:

$$(s^2-10)^2 - \frac{144}{s^2} = -8.$$

Multiplying by s^2 (which is nonzero) gives

$$s^2(s^2-10)^2 + 8s^2 - 144 = 0,$$

as required.

The values of s^2. Put $u = s^2$. Expanding,

$$u(u-10)^2 + 8u - 144 = u^3 - 20u^2 + 108u - 144.$$

Trying small factors of 144, $u = 2$ gives $8-80+216-144 = 0$, so $u-2$ is a factor, and

$$u^3-20u^2+108u-144 = (u-2)(u^2-18u+72) = (u-2)(u-6)(u-12).$$

So the three possible values are $s^2 = 2$, 6 and 12.

> **Ask yourself** Why are there three values of s^2, when we only want one factorisation?

There are three ways to split four roots into two pairs, and each pairing gives its own pair of quadratic factors with $s^2 = (a+b)^2$. Any one of them will do for solving the equation; the smallest just gives the tidiest surds.

Solving $f(x) = 0$. Take $s^2 = 2$ and $s = \sqrt{2}$ (choosing $s = -\sqrt{2}$ simply swaps the two factors). Then

$$p+q = 2-10 = -8, \qquad q-p = \frac{12}{\sqrt{2}} = 6\sqrt{2},$$

so $p = -4-3\sqrt{2}$ and $q = -4+3\sqrt{2}$. (Check: $pq = 16-18 = -2$.) Hence

$$f(x) = \left(x^2+\sqrt{2}\,x-4-3\sqrt{2}\right)\left(x^2-\sqrt{2}\,x-4+3\sqrt{2}\right).$$

For the first factor, the discriminant is

$$2+4(4+3\sqrt{2}) = 18+12\sqrt{2} = 6(3+2\sqrt{2}) = 6(1+\sqrt{2})^2,$$

using part (i), so its square root is $\sqrt{6}(1+\sqrt{2}) = \sqrt{6}+2\sqrt{3}$. The roots are

$$x = \frac{-\sqrt{2} \pm (\sqrt{6}+2\sqrt{3})}{2}.$$

For the second factor, the discriminant is

$$2 - 4(-4+3\sqrt{2}) = 18 - 12\sqrt{2} = 6(3-2\sqrt{2}) = 6(\sqrt{2}-1)^2,$$

with square root $\sqrt{6}(\sqrt{2}-1) = 2\sqrt{3}-\sqrt{6}$ (positive, since $2\sqrt{3} = \sqrt{12} > \sqrt{6}$). The roots are

$$x = \frac{\sqrt{2} \pm (2\sqrt{3}-\sqrt{6})}{2}.$$

So the four solutions of $\mathrm{f}(x) = 0$ are

$$x = \tfrac{1}{2}(-\sqrt{2}+\sqrt{6}+2\sqrt{3}), \qquad x = \tfrac{1}{2}(-\sqrt{2}-\sqrt{6}-2\sqrt{3}),$$
$$x = \tfrac{1}{2}(\sqrt{2}+2\sqrt{3}-\sqrt{6}), \qquad x = \tfrac{1}{2}(\sqrt{2}-2\sqrt{3}+\sqrt{6}).$$

Checks

- Numerically the roots are about 2.2497, -3.6639, 1.2144 and 0.1998. They add to 0, as the missing x^3 term demands, and their product is about -2, the constant term.
- Substituting $x \approx 0.1998$: $x^4 - 10x^2 + 12x - 2 \approx 0.0016 - 0.3992 + 2.3976 - 2 \approx 0$.
- The sign changes of f agree: $\mathrm{f}(0) = -2$, $\mathrm{f}(1) = 1$, $\mathrm{f}(2) = -2$, $\mathrm{f}(3) = 25$, $\mathrm{f}(-4) = 46$, so there is one root in each of $(-4,0)$, $(0,1)$, $(1,2)$ and $(2,3)$.

Common slips

- Not explaining why the x-coefficients are s and $-s$: it comes from the zero x^3 coefficient, i.e. the roots summing to zero.
- Forgetting to note $s \neq 0$ before dividing by s.
- Leaving the roots as $\sqrt{18 \pm 12\sqrt{2}}$: the question asks for full simplification, which is what part (i) is for.

Beyond STEP

This is Ferrari's method for the quartic in disguise: the cubic in s^2 is the *resolvent cubic*, and its three roots correspond to the three ways of pairing up the four roots. Why a quartic can be reduced to a cubic, while a general quintic cannot be solved by radicals at all, is the subject of Galois theory, which links the solvability of an equation to the symmetries of its roots. Allenby's *Rings, Fields and Groups* is a good first-year route towards it.

WORKED SOLUTION

STEP I 2019, Question 5

Pure • Vectors in Three Dimensions

STEP I 2019, Question 5

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Two vectors are equal exactly when their components agree, so every geometric condition in this question turns into a few simple equations in p, q, r and s. Coplanarity of four points is a scalar triple product (a 3×3 determinant) equal to zero. Once the conditions are written in coordinates, each part is short algebra.

Solution

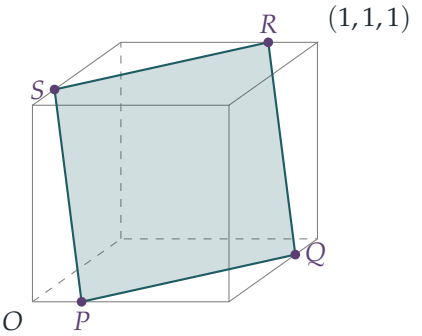
Part (i)

If $\overrightarrow{PQ} = \overrightarrow{SR}$, the opposite sides PQ and SR are parallel and of equal length, so $PQRS$ is a **parallelogram**. (Then $\overrightarrow{PS} = \overrightarrow{PQ} + \overrightarrow{QR} - \overrightarrow{SR}$ gives $\overrightarrow{PS} = \overrightarrow{QR}$ too.)

If also $|\overrightarrow{PQ}| = |\overrightarrow{PS}|$, two adjacent sides of the parallelogram are equal, and hence all four sides are equal: $PQRS$ is a **rhombus**.

Part (ii): the coplanarity condition

The points lie on edges of the unit cube, so $p, q, r, s \in [0, 1]$. Here P and Q lie in the face $z = 0$, and R and S in the face $z = 1$.



The four points are coplanar exactly when $\overrightarrow{PQ}$, $\overrightarrow{PS}$ and $\overrightarrow{PR}$ are linearly dependent, that is, when their scalar triple product is zero. With

$$\overrightarrow{PQ} = \begin{pmatrix} 1-p \\ q \\ 0 \end{pmatrix}, \qquad \overrightarrow{PS} = \begin{pmatrix} -p \\ s \\ 1 \end{pmatrix}, \qquad \overrightarrow{PR} = \begin{pmatrix} r-p \\ 1 \\ 1 \end{pmatrix},$$

expanding the determinant along the first row gives

$$\begin{vmatrix} 1-p & q & 0 \\ -p & s & 1 \\ r-p & 1 & 1 \end{vmatrix} = (1-p)(s-1) - q\big(-p-(r-p)\big) + 0$$
$$= (1-p)(s-1) + qr.$$

Setting this to zero gives $rq = (1-s)(1-p)$, as required.

If you prefer to avoid determinants, write the plane through P, Q, R as $\overrightarrow{OP} + \lambda\overrightarrow{PQ} + \mu\overrightarrow{PR}$, whose points are

$$\big(p + \lambda(1-p) + \mu(r-p),\ \lambda q + \mu,\ \mu\big).$$

For $S = (0, s, 1)$ to be on it we need $\mu = 1$ from the z-component, then $\lambda q = s - 1$ from the y-component, and $r + \lambda(1-p) = 0$ from the x-component. Multiplying the last equation by q and substituting gives $rq + (s-1)(1-p) = 0$, the same condition. (This uses P, Q, R not being collinear, which is true since R is in the top face and P, Q in the bottom.)

Part (ii)(a)

We have

$$\overrightarrow{PQ} = \begin{pmatrix} 1-p \\ q \\ 0 \end{pmatrix}, \qquad \overrightarrow{SR} = \begin{pmatrix} r \\ 1-s \\ 0 \end{pmatrix}.$$

The z-components already agree, so

$$\overrightarrow{PQ} = \overrightarrow{SR} \iff r = 1-p \text{ and } q = 1-s \iff p + r = 1 \text{ and } q + s = 1.$$

The centroid is

$$\tfrac{1}{4}(\overrightarrow{OP}+\overrightarrow{OQ}+\overrightarrow{OR}+\overrightarrow{OS}) = \tfrac{1}{4}\begin{pmatrix} p+1+r \\ q+1+s \\ 2 \end{pmatrix},$$

and the centre of the cube is $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$. The third components always agree, so the centroid is at the centre if and only if $p+1+r=2$ and $q+1+s=2$, which are the same two conditions $p+r=1$, $q+s=1$. Hence the two statements are equivalent.

(Notice that when $r=1-p$ and $s=1-q$ the coplanarity condition $rq=(1-s)(1-p)$ becomes $(1-p)q=q(1-p)$, which holds automatically: no configuration is ruled out.)

Part (ii)(b)

From (a), $\overrightarrow{PQ}=\overrightarrow{SR}$ gives $r=1-p$ and $s=1-q$. For the second condition compare squared lengths, using $\overrightarrow{PS}=(-p,\ s,\ 1)$ with $s=1-q$:

$$\begin{aligned} |\overrightarrow{PQ}|^2 = |\overrightarrow{PS}|^2 &\iff (1-p)^2+q^2 = p^2+(1-q)^2+1 \\ &\iff 1-2p = 2-2q \\ &\iff q = p+\tfrac{1}{2}. \end{aligned}$$

So

$$q = p+\tfrac{1}{2}, \qquad r = 1-p, \qquad s = \tfrac{1}{2}-p.$$

Since $q\le 1$ and $s\ge 0$, the points lie on the edges only for $0\le p\le \frac{1}{2}$.

Ask yourself Which two vectors give the angle PQR?

The angle at Q is between $\overrightarrow{QP}$ and $\overrightarrow{QR}$ (both pointing *away* from Q). Using the values just found,

$$\overrightarrow{QP} = \begin{pmatrix} p-1 \\ -p-\frac{1}{2} \\ 0 \end{pmatrix}, \qquad \overrightarrow{QR} = \begin{pmatrix} r-1 \\ 1-q \\ 1 \end{pmatrix} = \begin{pmatrix} -p \\ \frac{1}{2}-p \\ 1 \end{pmatrix}.$$

Then

$$\begin{aligned} \overrightarrow{QP}\cdot\overrightarrow{QR} &= -p(p-1)-(p+\tfrac{1}{2})(\tfrac{1}{2}-p) = p-p^2-(\tfrac{1}{4}-p^2) = p-\tfrac{1}{4}, \\ |\overrightarrow{QP}|^2 &= (1-p)^2+(p+\tfrac{1}{2})^2 = 2p^2-p+\tfrac{5}{4}, \\ |\overrightarrow{QR}|^2 &= p^2+(\tfrac{1}{2}-p)^2+1 = 2p^2-p+\tfrac{5}{4}. \end{aligned}$$

The two lengths are equal, as they must be in a rhombus, so

$$\cos PQR = \frac{p-\frac{1}{4}}{2p^2-p+\frac{5}{4}} = \frac{4p-1}{5-4p+8p^2},$$

on multiplying top and bottom by 4. This is the required result.

The square. A rhombus is a square exactly when its angles are right angles, i.e. $\cos PQR = 0$. The denominator $8p^2 - 4p + 5 = 8(p - \frac{1}{4})^2 + \frac{9}{2}$ is always positive, so this happens only for $4p - 1 = 0$:

$$p = \tfrac{1}{4}, \qquad q = \tfrac{3}{4}, \qquad r = \tfrac{3}{4}, \qquad s = \tfrac{1}{4}.$$

The side length squared is then

$$2 \cdot \tfrac{1}{16} - \tfrac{1}{4} + \tfrac{5}{4} = \tfrac{9}{8},$$

so each side has length $\sqrt{9/8} = \frac{3\sqrt{2}}{4}$. To compare with $\frac{21}{20}$, compare squares of these positive numbers:

$$\left(\tfrac{21}{20}\right)^2 = \tfrac{441}{400} < \tfrac{450}{400} = \tfrac{9}{8}.$$

Hence the side of the square is greater than $\frac{21}{20}$. (In fact it is about 1.0607.)

Checks

- At $p = \frac{1}{4}$ the four points are $(\frac{1}{4}, 0, 0)$, $(1, \frac{3}{4}, 0)$, $(\frac{3}{4}, 1, 1)$, $(0, \frac{1}{4}, 1)$; their average is $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, as (a) predicts, and $\overrightarrow{QP} \cdot \overrightarrow{QR} = \frac{3}{16} - \frac{3}{16} = 0$ directly.
- At $p = 0$ the formula gives $\cos PQR = -\frac{1}{5}$. Directly, $\overrightarrow{QP} = (-1, -\frac{1}{2}, 0)$ and $\overrightarrow{QR} = (0, \frac{1}{2}, 1)$, with dot product $-\frac{1}{4}$ and squared lengths $\frac{5}{4}$, so $\cos PQR = -\frac{1}{5}$. ✓
- The coplanarity identity, the equivalence in (a) and the formula for $\cos PQR$ were all confirmed by computer with exact arithmetic on many random values.

Common slips

- Using $\overrightarrow{QP}$ and $\overrightarrow{RQ}$ (one vector pointing in, one out) for the angle at Q: this flips the sign of the cosine.
- Mixing up the order of letters: $\overrightarrow{SR}$ is $R - S$, not $S - R$.
- Claiming $\frac{3\sqrt{2}}{4} > \frac{21}{20}$ from a calculator value without an exact comparison; squaring both sides keeps it rigorous because both are positive.

Beyond STEP

The determinant test for coplanarity is the three-dimensional case of a general fact: vectors are linearly dependent exactly when the determinant of the matrix they form vanishes. The plane sections of a cube are a rich source of examples too (a plane through the centre perpendicular to a long diagonal cuts out a regular hexagon). Strang's *Introduction to Linear Algebra* develops determinants, volumes and linear dependence from this geometric point of view.

WORKED SOLUTION

STEP I 2019, Question 6

Pure • Inequalities and Bounds

STEP I 2019, Question 6

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Each equation looks hopeless to solve directly, but it can be split as "something = something else" where the left-hand side can never be smaller than some number and the right-hand side can never be bigger than the same number. Then the equation holds only when *both* sides sit exactly at that number, and each of those conditions is easy to solve. The early parts of each section are there to build the two bounds.

Solution

Part (i)

Completing the square,

$$\begin{aligned} 9x^2 - 12x\cos\theta + 4 &= (3x - 2\cos\theta)^2 - 4\cos^2\theta + 4 \\ &= (3x - 2\cos\theta)^2 + 4\sin^2\theta. \end{aligned}$$

A square is never negative, so the minimum value is $4\sin^2\theta$, attained exactly when $x = \frac{2}{3}\cos\theta$.

Treating the quartic as a quadratic in x^2,

$$12x^2\sin\theta - 9x^4 = -\left(3x^2 - 2\sin\theta\right)^2 + 4\sin^2\theta.$$

So $12x^2\sin\theta - 9x^4 \leq 4\sin^2\theta$, with equality exactly when $x^2 = \frac{2}{3}\sin\theta$. Since $0 < \theta < \pi$ we have $\sin\theta > 0$, so this value of x^2 really is achieved, and the maximum value is $4\sin^2\theta$.

Now split the x^2 term of the given equation as $9x^2 - 12x^2\sin\theta$ and rearrange:

$$\underbrace{9x^2 - 12x\cos\theta + 4}_{\geq 4\sin^2\theta} = \underbrace{12x^2\sin\theta - 9x^4}_{\leq 4\sin^2\theta}.$$

Ask yourself How can a quantity that is at least $4\sin^2\theta$ equal one that is at most $4\sin^2\theta$?

Only if both equal $4\sin^2\theta$. So the equation holds if and only if *both*

$$x = \tfrac{2}{3}\cos\theta \qquad \text{and} \qquad x^2 = \tfrac{2}{3}\sin\theta.$$

Eliminating x: $\frac{4}{9}\cos^2\theta = \frac{2}{3}\sin\theta$, so $2\cos^2\theta = 3\sin\theta$, i.e. $2 - 2\sin^2\theta = 3\sin\theta$. Hence

$$2\sin^2\theta + 3\sin\theta - 2 = 0 \quad \Longrightarrow \quad (2\sin\theta - 1)(\sin\theta + 2) = 0.$$

As $\sin\theta \neq -2$, we get $\sin\theta = \frac{1}{2}$, so $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$. Then $x = \frac{2}{3}\cos\theta$ gives $x = \frac{1}{\sqrt{3}}$ or $x = -\frac{1}{\sqrt{3}}$ respectively, and in both cases $x^2 = \frac{1}{3} = \frac{2}{3}\sin\theta$, as required. The solutions are

$$(x,\theta) = \left(\tfrac{1}{\sqrt{3}}, \tfrac{\pi}{6}\right) \quad \text{and} \quad (x,\theta) = \left(-\tfrac{1}{\sqrt{3}}, \tfrac{5\pi}{6}\right).$$

Part (ii)

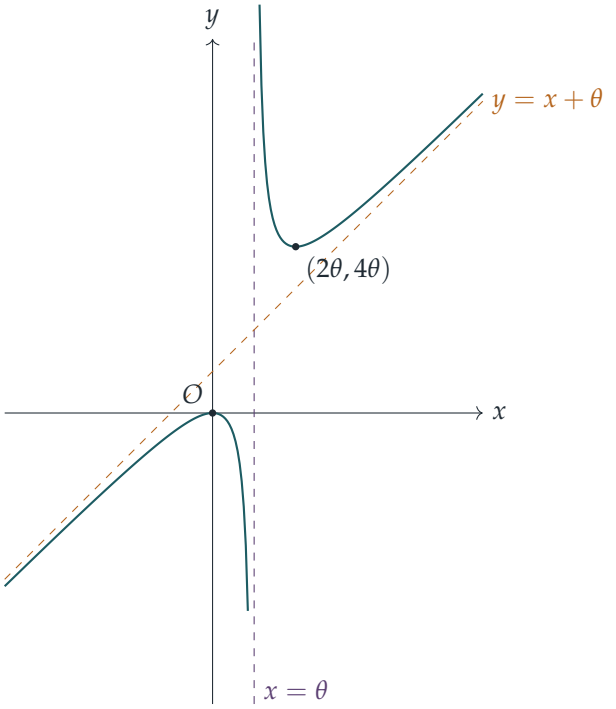
The sketch. Here $\theta > 0$. The curve $y = \dfrac{x^2}{x-\theta}$ passes through the origin (and meets the axes nowhere else), and has a vertical asymptote $x = \theta$. Dividing,

$$\frac{x^2}{x-\theta} = x + \theta + \frac{\theta^2}{x-\theta},$$

so $y = x + \theta$ is an oblique asymptote; the curve is above it for $x > \theta$ and below it for $x < \theta$. Differentiating with the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2x(x-\theta) - x^2}{(x-\theta)^2} = \frac{x(x-2\theta)}{(x-\theta)^2},$$

so the stationary points are $(0,0)$, a local maximum, and $(2\theta, 4\theta)$, a local minimum (the gradient changes sign from $+$ to $-$ at $x = 0$ and from $-$ to $+$ at $x = 2\theta$).



The deduction. If $x < \theta$, the numerator $x^2 \geq 0$ and the denominator is negative, so $y \leq 0$. If $x > \theta$, the branch has its lowest point at $(2\theta, 4\theta)$, so $y \geq 4\theta$. Algebraically, for $x > \theta$,

$$x^2 - 4\theta(x - \theta) = (x - 2\theta)^2 \geq 0 \quad \Longrightarrow \quad \frac{x^2}{x - \theta} \geq 4\theta,$$

dividing by the positive quantity $x - \theta$. So the curve never takes values strictly between 0 and 4θ.

The trigonometric bound. The numerator satisfies $\sin^2\theta \cos^2 x \leq 1 \cdot 1 = 1$, and the denominator satisfies $1 + \cos^2\theta \sin^2 x \geq 1$. A number at most 1 divided by a number at least 1 (and positive) is at most 1:

$$\frac{\sin^2\theta \cos^2 x}{1 + \cos^2\theta \sin^2 x} \leq 1.$$

Equality needs both $\sin^2\theta \cos^2 x = 1$ and $\cos^2\theta \sin^2 x = 0$. Note also that the fraction is ≥ 0, being a square over a positive number.

The final equation. Write $L = \dfrac{x^2}{4\theta(x - \theta)}$ and R for the right-hand side. Dividing the earlier result by $4\theta > 0$, either $L \leq 0$ or $L \geq 1$. But $0 \leq R \leq 1$.

Ask yourself Where can a number that is ≤ 0 or ≥ 1 meet a number in $[0, 1]$?

Only at 0 or at 1. So we need $L = R = 0$ or $L = R = 1$.

- $L = 0$ forces $x = 0$, but then $R = \sin^2\theta > 0$ since $0 < \theta < \pi$. No solutions.
- $L = 1$ means $x^2 = 4\theta(x - \theta)$, i.e. $(x - 2\theta)^2 = 0$, so $x = 2\theta$. $R = 1$ needs $\sin^2\theta\cos^2 x = 1$, which forces $\sin^2\theta = 1$ and $\cos^2 x = 1$. With $0 < \theta < \pi$ this gives $\theta = \frac{\pi}{2}$, so $x = \pi$, and indeed $\cos^2\pi = 1$ and $\cos^2\theta\sin^2 x = 0$.

The only solution is $x = \pi$, $\theta = \frac{\pi}{2}$.

Checks

- Substitute $x = \frac{1}{\sqrt{3}}$, $\theta = \frac{\pi}{6}$ into the quartic: $9 \cdot \frac{1}{9} + (9 - 6) \cdot \frac{1}{3} - 12 \cdot \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} + 4 = 1 + 1 - 6 + 4 = 0$.
- In the final equation with $x = \pi$, $\theta = \frac{\pi}{2}$: $L = \pi^2/(2\pi \cdot \frac{\pi}{2}) = 1$ and $R = 1 \cdot 1/(1 + 0) = 1$.
- A computer search over a fine grid of (x, θ) found no other near-solutions of either equation, apart from spurious ones creeping in at the excluded endpoints $\theta \to 0$ and $\theta \to \pi$ (where $\sin\theta \to 0$ and the bounds collapse).

Common slips

- Finding $\sin\theta = \frac{1}{2}$ and stopping: x must satisfy $x = \frac{2}{3}\cos\theta$, which fixes its *sign* for each θ, so $(\frac{1}{\sqrt{3}}, \frac{5\pi}{6})$ is not a solution.
- Forgetting the case $L = R = 0$ in the final part; it has to be ruled out, and it is ruled out only because $\sin\theta \neq 0$.
- Claiming the fraction is ≤ 1 "because $\sin^2$ and $\cos^2$ are at most 1" without saying why the denominator doesn't spoil it.

Beyond STEP

The trick of trapping an equation between a lower and an upper bound that meet at a single value is a staple of analysis: it is how one proves that equality holds in inequalities such as AM–GM or Cauchy–Schwarz, and how the squeeze theorem pins down limits. Learning to ask "when is this inequality tight?" is often the quickest route to a solution. For more in this spirit, see Abbott, *Understanding Analysis*, or the problem collection by Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2019, Question 7

Pure • Irrationality and Congruences

STEP I 2019, Question 7

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

If $\sqrt{2}+\sqrt{3} = a/b$ in lowest terms, squaring twice clears the surds and leaves a polynomial equation in the integers a and b. Then we look at that equation modulo a small prime. The equation forces the prime to divide both a and b or neither. "Both" contradicts lowest terms, and remainders rule out "neither". The whole art is choosing a prime for which the remainders really do clash.

Solution

Part (i)

Step 1. Every integer is $3k$, $3k+1$ or $3k-1$ for some integer k, so if $3 \nmid a$ then $a = 3k \pm 1$. In either case

$$a^2 = 9k^2 \pm 6k + 1 = 3(3k^2 \pm 2k) + 1,$$

which is one more than a multiple of 3.

Step 3. Squaring $\dfrac{a}{b} = \sqrt{2}+\sqrt{3}$ gives

$$\frac{a^2}{b^2} = 2 + 2\sqrt{6} + 3, \qquad \text{so} \qquad a^2 - 5b^2 = 2\sqrt{6}\,b^2.$$

The surd is now on its own, so squaring again removes it:

$$a^4 - 10a^2b^2 + 25b^4 = 24b^4,$$
$$a^4 + b^4 = 10a^2b^2,$$

as required. (We only need the implication in this direction: any a, b from step 2 must satisfy this equation.)

Step 4. Suppose $3 \mid a$. Rearranging step 3,

$$b^4 = 10a^2b^2 - a^4 = a^2\,(10b^2 - a^2),$$

and the right-hand side is divisible by 3 (even by 9). If $3 \nmid b$, then step 1 gives $b^2 = 3m+1$, so $b^4 = (b^2)^2 = 9m^2+6m+1$ is *not* a multiple of 3, a contradiction. Hence $3 \mid b$.

Ask yourself Step 4 handles the case $3 \mid a$. What about the case where 3 divides neither a nor b?

Step 5. The equation in step 3 is symmetric in a and b, so step 4 also works with the roles swapped: if $3 \mid b$ then $3 \mid a$. Since a and b have no common factor greater than one, they are not both multiples of 3, so by step 4 neither of them is. This case is not covered by steps 2–4, and it is where step 1 comes in. Write $a^2 = 3r+1$ and $b^2 = 3s+1$. Then a^4, b^4 and a^2b^2 are all one more than multiples of 3 (a product of two such numbers is again of this form). Working modulo 3 in step 3,

$$a^4 + b^4 \equiv 1+1 = 2, \qquad 10a^2b^2 \equiv 1\cdot 1\cdot 1 = 1 \pmod 3,$$

since $10 \equiv 1$. The two sides of step 3 leave different remainders on division by 3, which is impossible. So no such a and b exist. Also, $\sqrt{2}+\sqrt{3} > 0$, so if it were rational it could be written with a, b positive and coprime. Hence $\sqrt{2}+\sqrt{3}$ is irrational.

Part (ii)

Suppose that $\sqrt{6}+\sqrt{7} = a/b$ with a, b positive integers having no common factor greater than one. As before,

$$\frac{a^2}{b^2} = 13 + 2\sqrt{42} \quad \Longrightarrow \quad (a^2 - 13b^2)^2 = 168\, b^4,$$

which expands to $a^4 - 26a^2b^2 + 169b^4 = 168b^4$, that is,

$$a^4 + b^4 = 26a^2b^2.$$

Squares modulo 5. If $5 \nmid a$ then $a = 5k \pm 1$ or $a = 5k \pm 2$, giving

$$a^2 = 5(5k^2 \pm 2k) + 1 \qquad \text{or} \qquad a^2 = 5(5k^2 \pm 4k) + 4.$$

So $a^2 \equiv \pm 1 \pmod 5$, and therefore $a^4 \equiv 1 \pmod 5$.

Divisibility. If $5 \mid a$, then $b^4 = a^2(26b^2 - a^2)$ is a multiple of 5. But if $5 \nmid b$ then $b^4 \equiv 1 \pmod 5$, so we must have $5 \mid b$. By symmetry, $5 \mid b$ implies $5 \mid a$. Since a and b are coprime, neither is divisible by 5.

Contradiction. Now $a^4 \equiv b^4 \equiv 1$, and $a^2b^2 \equiv (\pm 1)(\pm 1) = \pm 1 \pmod 5$. Since $26 \equiv 1$,

$$a^4 + b^4 \equiv 2, \qquad 26a^2b^2 \equiv \pm 1 \pmod 5.$$

But $2 \not\equiv \pm 1 \pmod 5$, so the equation cannot hold. Hence $\sqrt{6}+\sqrt{7}$ is irrational.

Ask yourself Which step breaks if we try to use 3 here instead of 5?

Why 3 fails. The first half of the argument still works modulo 3: if $3 \mid a$ then $3 \mid b^4$, so $3 \mid b$, and so neither a nor b is a multiple of 3. But then $a^4 \equiv b^4 \equiv a^2b^2 \equiv 1 \pmod 3$, and since $26 \equiv 2 \pmod 3$,

$$a^4 + b^4 \equiv 2 \qquad \text{and} \qquad 26a^2b^2 \equiv 2 \pmod 3.$$

The two sides agree, so there is no contradiction. Modulo 3 the equation $a^4 + b^4 = 26a^2b^2$ is satisfied by *every* pair of non-multiples of 3, so divisibility by 3 cannot rule it out. (In part (i) the coefficient was $10 \equiv 1$, and $2 \neq 1$ gave the clash. Here the coefficient is $26 \equiv 2$, so the clash disappears.)

Checks

- Numerically, $x = \sqrt{2} + \sqrt{3} \approx 3.1463$ gives $x^4 - 10x^2 + 1 \approx 0$ (to about 10^{-14}). Similarly $y = \sqrt{6} + \sqrt{7} \approx 5.0952$ gives $y^4 - 26y^2 + 1 \approx 0$. So the two quartic equations are right.
- Try every pair of remainders: the only solution of $a^4 + b^4 \equiv 10a^2b^2 \pmod 3$ is $a \equiv b \equiv 0$, and the same holds for $a^4 + b^4 \equiv 26a^2b^2 \pmod 5$. But $a^4 + b^4 \equiv 26a^2b^2 \pmod 3$ is also solved by $(1,1)$, $(1,2)$, $(2,1)$ and $(2,2)$. This is exactly why 3 fails in part (ii).
- A computer search found no coprime positive $a, b \leq 400$ satisfying either equation.

Common slips

- Stopping after step 4. Step 4 only shows that a and b can't be multiples of 3. You still need step 1 to deal with the case where neither is.
- Claiming $3 \mid b^4 \Rightarrow 3 \mid b$ without a reason. Step 1 supplies it: if $3 \nmid b$ then b^4 leaves remainder 1.
- In part (ii), forgetting that a square can leave remainder 4 (that is, -1) modulo 5, so a^2b^2 can be $\equiv \pm 1$, not just 1.

Beyond STEP

The number $\sqrt{2} + \sqrt{3}$ is a root of $x^4 - 10x^2 + 1$, and the argument here is really a proof that this polynomial has no rational roots. The rational root theorem gives a quicker route: any rational root must be ± 1, and neither works. Reducing a polynomial modulo a prime to show that it has no roots or factors is a standard tool in first-year algebra (for example, in irreducibility tests). Curiously, $x^4 - 10x^2 + 1$ *factorises* modulo every prime, even though it is irreducible over the rationals, and this is a famous example. Allenby's *Rings, Fields and Groups* develops these ideas, and Davenport's *The Higher Arithmetic* is a lovely introduction to congruences.

WORKED SOLUTION

STEP I 2019, Question 8

Pure • Integration by Substitution

STEP I 2019, Question 8

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

We are told not to evaluate f, so every part is about *reshaping* an integral until it looks exactly like

$$\int_1^X \sqrt{\frac{t-1}{t+1}}\,\mathrm{d}t = \mathrm{f}(X)$$

for some upper limit X. Two moves do almost all the work: a *stretch* ($u = kt$), which turns $\frac{u-k}{u+k}$ into $\frac{t-1}{t+1}$, and a *shift* ($u = t + c$), which makes the two numbers in the fraction symmetric about zero. The last part needs one extra substitution to turn a square root of a quotient into a function of the right shape.

Solution

Throughout, we use the fact that a substitution changes the integrand, the differential *and* the limits; the upper limit is where the answer appears.

Part (i)

The integrand $\sqrt{\frac{u-2}{u+2}}$ becomes $\sqrt{\frac{t-1}{t+1}}$ if we put $u = 2t$, since then $\frac{2t-2}{2t+2} = \frac{t-1}{t+1}$. With $\mathrm{d}u = 2\,\mathrm{d}t$, and limits $u = 2 \mapsto t = 1$, $u = x \mapsto t = \frac{1}{2}x$,

$$\int_2^x \sqrt{\frac{u-2}{u+2}}\,\mathrm{d}u = \int_1^{x/2} \sqrt{\frac{t-1}{t+1}}\,2\,\mathrm{d}t = 2\,\mathrm{f}\left(\tfrac{1}{2}x\right).$$

The condition $x > 2$ is exactly what makes $\frac{1}{2}x > 1$, so $\mathrm{f}(\frac{1}{2}x)$ is defined. This is the required result.

Part (ii)

Here the numbers in the fraction are 0 and -4, which are not symmetric about zero. Their midpoint is -2, so we shift by 2: put $u = v - 2$. Then $\frac{u}{u+4} = \frac{v-2}{v+2}$, $\mathrm{d}u = \mathrm{d}v$, and

the limits $u = 0, x$ become $v = 2, x+2$. Now part (i), with $x+2$ (which exceeds 2) in place of x, finishes it:

$$\int_0^x \sqrt{\frac{u}{u+4}}\,\mathrm{d}u = \int_2^{x+2} \sqrt{\frac{v-2}{v+2}}\,\mathrm{d}v = 2\,\mathrm{f}\big(\tfrac{1}{2}x+1\big).$$

Part (iii)

The numbers in the fraction are 5 and -1, with midpoint 2. So put $u = v+2$: then

$$\frac{u-5}{u+1} = \frac{v-3}{v+3},$$

and the limits $u = 5, x$ become $v = 3, x-2$. Now stretch by 3, putting $v = 3t$, so that $\frac{v-3}{v+3} = \frac{t-1}{t+1}$ and $\mathrm{d}v = 3\,\mathrm{d}t$, with limits $t = 1, \frac{1}{3}(x-2)$:

$$\begin{aligned}\int_5^x \sqrt{\frac{u-5}{u+1}}\,\mathrm{d}u = \int_3^{x-2} \sqrt{\frac{v-3}{v+3}}\,\mathrm{d}v &= \int_1^{(x-2)/3} \sqrt{\frac{t-1}{t+1}}\,3\,\mathrm{d}t \\ &= 3\,\mathrm{f}\Big(\frac{x-2}{3}\Big).\end{aligned}$$

Since $x > 5$, the argument $\frac{x-2}{3}$ is greater than 1, as it must be. (The two substitutions combine into the single one $u = 3t+2$.)

Part (iv)

This integrand has no square root of a quotient in sight, and the powers are wrong: u^2 on top, u^2+4 under the root.

> **?** **Ask yourself** What substitution would turn u^2+4 into the $u+4$ of part (ii)?

Put $u^2 = w$, so $u = \sqrt{w}$ (we have $u > 0$ on $[1,2]$) and $\mathrm{d}u = \dfrac{\mathrm{d}w}{2\sqrt{w}}$. The limits $u = 1, 2$ become $w = 1, 4$, and

$$\frac{u^2}{\sqrt{u^2+4}}\,\mathrm{d}u = \frac{w}{\sqrt{w+4}} \cdot \frac{\mathrm{d}w}{2\sqrt{w}} = \frac{1}{2} \cdot \frac{\sqrt{w}}{\sqrt{w+4}}\,\mathrm{d}w = \frac{1}{2}\sqrt{\frac{w}{w+4}}\,\mathrm{d}w,$$

which is exactly the integrand of part (ii). Splitting the range at 0 and using part (ii) twice (with $x = 4$ and $x = 1$),

$$\begin{aligned}\int_1^2 \frac{u^2}{\sqrt{u^2+4}}\,\mathrm{d}u &= \frac{1}{2}\left(\int_0^4 \sqrt{\frac{w}{w+4}}\,\mathrm{d}w - \int_0^1 \sqrt{\frac{w}{w+4}}\,\mathrm{d}w\right) \\ &= \frac{1}{2}\Big(2\,\mathrm{f}(3) - 2\,\mathrm{f}\big(\tfrac{3}{2}\big)\Big) = \mathrm{f}(3) - \mathrm{f}\big(\tfrac{3}{2}\big).\end{aligned}$$

Checks

- Each answer has the right behaviour at the lower limit: in (i) at $x = 2$ we get $2\,\mathrm{f}(1) = 0$; in (ii) at $x = 0$ we get $2\,\mathrm{f}(1) = 0$; in (iii) at $x = 5$ we get $3\,\mathrm{f}(1) = 0$.

- Differentiating is a quick test. For (iii), $\frac{\mathrm{d}}{\mathrm{d}x}\,3\,\mathrm{f}\left(\frac{x-2}{3}\right) = \mathrm{f}'\left(\frac{x-2}{3}\right) = \sqrt{\frac{x-5}{x+1}}$, as it should be.
- Numerically, both sides of (iv) equal 0.91007, and in (iii) with $x = 11$ both sides equal 3.19704, each found by direct numerical integration.

Common slips

- Forgetting the factor from $\mathrm{d}u = k\,\mathrm{d}t$, which gives $\mathrm{f}(\frac{1}{2}x)$ instead of $2\,\mathrm{f}(\frac{1}{2}x)$.
- Changing the integrand but not the limits, especially the upper limit, which is where the answer lives.
- In (iv), writing the result as $\frac{1}{2}\cdot 2\,\mathrm{f}(3)$ only: part (ii) gives integrals from 0, so the piece from 0 to 1 must be subtracted.

Beyond STEP

The idea here is that a whole family of integrals is really one integral seen through different affine changes of variable $u = at + b$; recognising the "standard form" hiding inside an expression is a skill used constantly in university analysis and in reducing differential equations to canonical forms. (The function f can in fact be found in closed form using hyperbolic functions, but the question shows that you can get a long way without it.) For the careful theory of substitution and the integral, see Abbott, *Understanding Analysis*; for many more problems in this style, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2019, Question 9

Mechanics • Toppling and Friction

STEP I 2019, Question 9

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

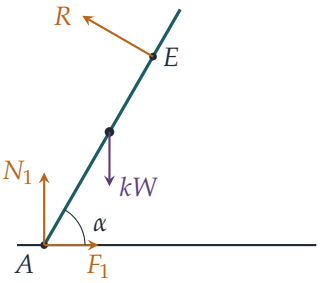
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The key idea

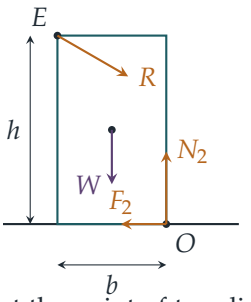
Treat the ladder and the box as two separate bodies joined by one force. Because the contact is smooth, that force is perpendicular to the ladder, so its line is fixed by α and only its size is unknown. Taking moments for the ladder about its foot gives the size; taking moments for the box about the edge it tips over gives the toppling condition. The relation $h = b\tan\alpha$ is there to make the box's moment collapse into a double-angle expression.

Solution

Model the box as a uniform rigid cuboid, the ladder as a light rod and the painter as a particle. The contact between ladder and box is smooth; both ground contacts are rough. Let E be the edge the ladder touches, A the foot of the ladder and O the bottom edge of the box furthest from the ladder (the edge about which it will tip).



ladder



box (at the point of toppling)

Part (i)

Since there is no friction at E, the force R of the box on the ladder is perpendicular to the ladder. Along the ladder, $AE = h/\sin\alpha$, because E is at height h. When the painter is at height λh they are a distance $\lambda h/\sin\alpha$ along the ladder, so their horizontal distance from A is $\lambda h\cot\alpha$.

The ladder is light, so the only moments about A come from R and the painter's weight (N_1 and F_1 act at A). Since R is perpendicular to AE, its moment arm is AE itself:

$$R\cdot\frac{h}{\sin\alpha} = kW\cdot\lambda h\,\frac{\cos\alpha}{\sin\alpha} \quad\Longrightarrow\quad R = k\lambda W\cos\alpha,$$

as required. (Strictly this holds at every stage of the climb, with λh the painter's current height.)

Part (ii)

By Newton's third law the ladder pushes on the box at E with a force R perpendicular to the ladder, directed into the box. Its components are $R\sin\alpha$ horizontally (away from the ladder, towards O) and $R\cos\alpha$ vertically downwards.

When the box is on the point of toppling about O, the ground's reaction acts entirely at O, so N_2 and F_2 have no moment about O. Take moments about O, with the clockwise sense (the way the box would tip) positive. E is a horizontal distance b from O and a height h above it, so

- the horizontal component $R\sin\alpha$, acting at height h, has moment $+Rh\sin\alpha$;
- the vertical component $R\cos\alpha$, acting a distance b to the ladder side of O, has moment $-Rb\cos\alpha$ (pushing down on that side resists tipping);
- the weight W, acting a distance $\frac{1}{2}b$ from O, has moment $-\frac{1}{2}Wb$.

The box starts to topple when the total moment reaches zero:

$$R(h\sin\alpha - b\cos\alpha) = \tfrac{1}{2}Wb.$$

Ask yourself How does $h = b\tan\alpha$ help here?

Substituting it,

$$h\sin\alpha - b\cos\alpha = b\,\frac{\sin^2\alpha - \cos^2\alpha}{\cos\alpha} = -\frac{b\cos 2\alpha}{\cos\alpha}.$$

Using part (i), $R = k\lambda W\cos\alpha$, the condition becomes

$$-k\lambda W\cos\alpha\cdot\frac{b\cos 2\alpha}{\cos\alpha} = \tfrac{1}{2}Wb \quad\Longrightarrow\quad 2k\lambda\cos 2\alpha + 1 = 0,$$

as required. Notice that this needs $\cos 2\alpha < 0$, i.e. $\alpha > 45°$, which agrees with $h > b$.

Part (iii)

Resolving for the box (whose ground reaction has components F_2 towards the ladder and N_2 upwards):

$$F_2 = R\sin\alpha, \qquad N_2 = W + R\cos\alpha.$$

The box doesn't slip, so $F_2 \le \mu N_2$, i.e. $\mu \ge \dfrac{R\sin\alpha}{W + R\cos\alpha}$.

? **Ask yourself** At which moment of the climb is this condition hardest to satisfy?

The ratio $\dfrac{R\sin\alpha}{W + R\cos\alpha} = \dfrac{\sin\alpha}{W/R + \cos\alpha}$ increases with R, and R grows as the painter climbs. So the greatest demand on friction is at the moment of toppling, and we need the inequality there. From (ii), $k\lambda = -\dfrac{1}{2\cos 2\alpha}$, so

$$R = -\frac{W\cos\alpha}{2\cos 2\alpha}.$$

Then

$$R\sin\alpha = -\frac{W\sin\alpha\cos\alpha}{2\cos 2\alpha} = -\frac{W\sin 2\alpha}{4\cos 2\alpha},$$

$$W + R\cos\alpha = W\left(1 - \frac{\cos^2\alpha}{2\cos 2\alpha}\right) = W\,\frac{4\cos 2\alpha - (1+\cos 2\alpha)}{4\cos 2\alpha} = W\,\frac{3\cos 2\alpha - 1}{4\cos 2\alpha},$$

using $2\cos^2\alpha = 1 + \cos 2\alpha$. Dividing,

$$\mu \ge \frac{-\sin 2\alpha}{3\cos 2\alpha - 1} = \frac{\sin 2\alpha}{1 - 3\cos 2\alpha},$$

as required. Both numerator and denominator are positive, since $45^\circ < \alpha < 90^\circ$ gives $\sin 2\alpha > 0$ and $\cos 2\alpha < 0$.

Checks

- Take $\alpha = 60^\circ$, so $\cos 2\alpha = -\frac{1}{2}$. Part (ii) gives $k\lambda = 1$, so $R = \frac{1}{2}W$. Then $F_2 = \frac{\sqrt{3}}{4}W$ and $N_2 = \frac{5}{4}W$, giving $\mu \ge \frac{\sqrt{3}}{5} \approx 0.346$, which matches $\sin 120^\circ / (1 - 3\cos 120^\circ) = \frac{\sqrt{3}}{2} / \frac{5}{2}$.
- As $\alpha \to 45^\circ$ (a cube-like box only just taller than wide), $\cos 2\alpha \to 0$ and (ii) needs $k\lambda \to \infty$: the push at E is almost along the line EO, so it has almost no turning effect. That matches the geometry: with $h = b\tan\alpha$, the line of R through E is close to passing through O when α is near 45°.
- The moment equation, the toppling height and the friction bound were also confirmed by computer for many random choices of α, b, k and W.

Common slips

- Taking the force at E as horizontal (as for a smooth wall). Here the contact is with an edge, so the force is perpendicular to the ladder.
- Forgetting that the vertical component of R adds to the box's normal reaction, so that $N_2 = W + R\cos\alpha$, not W.
- Taking moments about the wrong bottom edge of the box: it tips about the edge away from the ladder.

Beyond STEP

This is two-body statics: isolate each body, use Newton's third law at the contact, and pick moment centres that remove unknowns. The questions of whether a body slides or topples first, and which constraint fails first as loading increases, run through engineering mechanics and the study of rigid bodies in contact. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction in this spirit.

WORKED SOLUTION

STEP I 2019, Question 10

Mechanics • Projectile Motion

STEP I 2019, Question 10

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

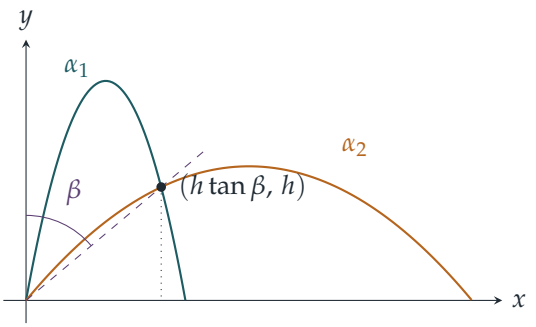
www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Write down the trajectory equation y in terms of x, put in the fixed point, and notice that $1/\sin^2\alpha = 1 + \cot^2\alpha$ turns it into a quadratic in $c = \cot\alpha$. After that, everything in (i) is the algebra of quadratics: sum and product of roots give $\alpha_1 + \alpha_2$ through the cot addition formula, and "two distinct roots" is a discriminant condition. Part (ii) is a one-line energy (or $v^2 = u^2 + 2as$) argument.

Solution

Model the particle as a point mass moving freely under gravity, with no air resistance. Take the angle α from the upward vertical, so the initial velocity has horizontal component $u\sin\alpha$ and vertical component $u\cos\alpha$.



The figure shows the two possible trajectories through the point (drawn for $\beta = 50°$, $k = 8$). The point lies on the line through O at angle β to the vertical.

Part (i)

At time t,

$$x = ut\sin\alpha, \qquad y = ut\cos\alpha - \tfrac{1}{2}gt^2.$$

Since $0 < \beta < 90^\circ$ the point has $x > 0$, so $\sin\alpha \neq 0$ and we may eliminate $t = x/(u\sin\alpha)$:

$$y = x\cot\alpha - \frac{gx^2}{2u^2\sin^2\alpha} = cx - \frac{gx^2}{2u^2}(1+c^2),$$

using $\csc^2\alpha = 1 + \cot^2\alpha$. Now put $x = h\tan\beta$, $y = h$ and divide by $h > 0$:

$$1 = c\tan\beta - \frac{gh}{2u^2}\tan^2\beta\,(1+c^2) = c\tan\beta - \frac{\tan^2\beta}{k}(1+c^2).$$

Multiplying through by $k\cot^2\beta$ gives $k\cot^2\beta = ck\cot\beta - 1 - c^2$, that is

$$c^2 - ck\cot\beta + 1 + k\cot^2\beta = 0, \qquad (*)$$

as required.

(a) The two values $c_1 = \cot\alpha_1$ and $c_2 = \cot\alpha_2$ are the two roots of the quadratic $(*)$ in c. (They are distinct because cot is one-to-one on $(0,\pi)$.) So, by the sum and product of roots,

$$\cot\alpha_1 + \cot\alpha_2 = k\cot\beta, \qquad \cot\alpha_1\cot\alpha_2 = 1 + k\cot^2\beta.$$

The first of these is the first result. For the second, use the addition formula for cot:

$$\cot(\alpha_1+\alpha_2) = \frac{\cot\alpha_1\cot\alpha_2 - 1}{\cot\alpha_1 + \cot\alpha_2} = \frac{k\cot^2\beta}{k\cot\beta} = \cot\beta.$$

? **Ask yourself** Does $\cot(\alpha_1+\alpha_2) = \cot\beta$ really force $\alpha_1 + \alpha_2 = \beta$?

Not on its own: cot has period 180°, so we also need to pin down the range of $\alpha_1 + \alpha_2$. Here both the sum $k\cot\beta$ and the product $1 + k\cot^2\beta$ are positive (as $k > 0$ and $\cot\beta > 0$), so both roots c_1, c_2 are positive. Hence both angles are acute, $0 < \alpha_1 + \alpha_2 < 180^\circ$, and on that interval cot is one-to-one. Since β is also in that interval, $\alpha_1 + \alpha_2 = \beta$.

(b) Two distinct real values of c require the discriminant of $(*)$ to be positive:

$$k^2\cot^2\beta - 4(1 + k\cot^2\beta) > 0.$$

Multiplying by $\tan^2\beta > 0$,

$$k^2 - 4k - 4\tan^2\beta > 0.$$

The quadratic in k on the left has roots $k = 2 \pm \sqrt{4 + 4\tan^2\beta} = 2 \pm 2\sec\beta$ (using $1 + \tan^2\beta = \sec^2\beta$ and $\sec\beta > 0$), so the inequality holds exactly when $k < 2 - 2\sec\beta$ or $k > 2 + 2\sec\beta$. But $\sec\beta > 1$, so $2 - 2\sec\beta < 0 < k$, and the first option is impossible. Therefore

$$k > 2(1 + \sec\beta).$$

Part (ii)

The particle must rise to height $h > 0$, so its initial vertical velocity $u\cos\alpha$ is positive. Vertically the motion has constant acceleration $-g$, and the vertical velocity is zero at the top, so by $v^2 = u^2 + 2as$ the greatest height is

$$H = \frac{u^2\cos^2\alpha}{2g}.$$

The particle passes through a point at height h, so $H \geq h$:

$$\frac{u^2\cos^2\alpha}{2g} \geq h \quad\Longrightarrow\quad \frac{2u^2}{gh}\cos^2\alpha \geq 4 \quad\Longrightarrow\quad k \geq 4\sec^2\alpha,$$

where we multiplied by $4/(gh) > 0$ and then divided by $\cos^2\alpha > 0$. Equality holds exactly when the point $(h\tan\beta, h)$ is the top of the trajectory.

Checks

- Take $\beta = 50°$, $k = 8$. Then $2(1+\sec 50°) \approx 5.11 < 8$, so two angles exist; solving $(*)$ gives $\alpha_1 \approx 10.29°$ and $\alpha_2 \approx 39.71°$, which add to $50°$. Stepping the motion forward numerically, both trajectories reach height $1.0000\,h$ at $x = h\tan\beta$.
- For these angles $k\cos^2\alpha \approx 7.74$ and 4.73, both at least 4, as (ii) requires.
- A computer check over thousands of random β, h, k confirmed that $(*)$ has two distinct roots exactly when $k > 2(1+\sec\beta)$, and that $\alpha_1 + \alpha_2 = \beta$ every time.
- Units: $u^2/(gh)$ is dimensionless, as k must be.

Common slips

- Measuring α from the horizontal out of habit: here the horizontal component is $u\sin\alpha$, not $u\cos\alpha$.
- Concluding $\alpha_1 + \alpha_2 = \beta$ from $\cot(\alpha_1+\alpha_2) = \cot\beta$ without showing both angles are acute.
- Keeping the root $k < 2 - 2\sec\beta$ in (b), which is negative.

Beyond STEP

The result $\alpha_1 + \alpha_2 = \beta$ says the two firing directions are symmetric about the bisector of the vertical and the line to the target, a classical fact about the "parabola of safety": the envelope of all trajectories with speed u, which is the boundary case $k = 2(1+\sec\beta)$, where the two roots merge. Envelopes of families of curves are a nice first-year topic linking calculus and geometry. For more mechanics in this spirit, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 2019, Question 11

Probability and Statistics • Geometric Distribution

STEP I 2019, Question 11

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.9, *A Fair Decision from a Biased Coin*: https://beyondstep.keithmcnulty.org/ch12/von-neumann.html

The key idea

Each round of the two-person procedure is an independent trial that either ends things or doesn't, so the round of the decision follows a geometric distribution. The bound "whatever the value of p" then comes down to how large pq can be when $p+q=1$, and the AM–GM inequality answers that. In part (ii) the second round is no longer a fresh start, so we split by what happened in the first round, and the same bound on pq finishes it.

Solution

Part (i)

In any one round the two coins are independent. A decision is made when both show heads (probability p^2) or both show tails (probability q^2), so

$$\mathrm{P}(\text{decision in a given round}) = p^2+q^2, \qquad \mathrm{P}(\text{no decision}) = pq+qp = 2pq.$$

As a check, $p^2+q^2+2pq=(p+q)^2=1$.

The rounds are independent, since fresh tosses are made each time. A decision on the nth round means $n-1$ rounds with no decision followed by one with a decision, so

$$\mathrm{P}(\text{decision on round } n) = (2qp)^{n-1}(q^2+p^2),$$

which is the required result.

For the second result, it is easiest to use the complement. No decision on or before round n means n undecided rounds in a row, with probability $(2pq)^n$. Hence

$$\mathrm{P}(\text{decision on or before round } n) = 1-(2pq)^n.$$

(Summing the geometric series $\sum_{r=1}^{n}(p^2+q^2)(2pq)^{r-1}$ gives the same thing, using $p^2+q^2=1-2pq$.)

Now we need an upper bound on pq that holds for every p. Since $p,q\geq 0$, the AM–GM inequality gives

$$\sqrt{pq}\leq \frac{p+q}{2}=\frac{1}{2}, \qquad \text{so} \qquad pq\leq \frac{1}{4},$$

with equality exactly when $p=q=\frac{1}{2}$. (Equivalently, $pq=p(1-p)=\frac{1}{4}-(p-\frac{1}{2})^2$.) Therefore $0\leq 2pq\leq \frac{1}{2}$, so $(2pq)^n\leq \frac{1}{2^n}$ and

$$\mathrm{P}(\text{decision on or before round } n)=1-(2pq)^n\geq 1-\frac{1}{2^n}$$

for every value of p, as required. Fair coins are the worst case.

Part (ii)

Let D_1 be the event of a decision in round 1 and D_2 the event of a decision in round 2. We need $\mathrm{P}(D_1)+\mathrm{P}(D_2)$, since these events are mutually exclusive.

Round 1. All three heads or all three tails: $\mathrm{P}(D_1)=p^3+q^3$.

Round 2. If round 1 did not decide, the coins showed either two heads and one tail or two tails and one head.

- **Two heads, one tail** (probability $3p^2q$, as any of the three people can be the one with the tail). The two who showed heads toss again, and the third turns the tail over to show a head. All tails is now impossible, and all heads happens exactly when both new tosses give heads, with probability p^2. This route contributes $3p^2q\cdot p^2=3p^4q$.
- **Two tails, one head** (probability $3pq^2$). By the same argument, the third coin now shows a tail, and we need both new tosses to give tails, with probability q^2. This route contributes $3pq^2\cdot q^2=3pq^4$.

The round-2 tosses are independent of round 1, which is why we can multiply. So

$$\mathrm{P}(D_2)=3p^4q+3pq^4=3pq\,(p^3+q^3).$$

? **Ask yourself** Is there a neat way to tidy up $p^3+q^3+3pq(p^3+q^3)$?

Factor out p^3+q^3, then use the sum of cubes with $p+q=1$:

$$p^3+q^3=(p+q)(p^2-pq+q^2)=(p+q)^2-3pq=1-3pq.$$

Hence

$$\begin{aligned}\mathrm{P}(\text{decision on or before round 2}) &= (p^3+q^3)(1+3pq)\\ &= (1-3pq)(1+3pq)\\ &= 1-9p^2q^2.\end{aligned}$$

From part (i), $0 \leq pq \leq \frac{1}{4}$, so $p^2q^2 \leq \frac{1}{16}$ and

$$\mathrm{P}(\text{decision on or before round 2}) \geq 1 - \frac{9}{16} = \frac{7}{16}$$

whatever the value of p, with equality when $p = \frac{1}{2}$. This completes the proof.

Checks

- With $p = 1$ (double-headed coins) the decision is immediate. In part (i), $1 - (2pq)^n = 1$, and in part (ii), $1 - 9p^2q^2 = 1$, as they should be.
- With $p = \frac{1}{2}$ in part (ii): round 1 decides with probability $\frac{1}{4}$. Otherwise ($\frac{3}{4}$) round 2 decides with probability $\frac{1}{4}$, giving $\frac{1}{4} + \frac{3}{16} = \frac{7}{16}$, so the bound is attained.
- With $p = 0.3$, the formula gives $1 - 9(0.21)^2 = 0.6031$. A computer simulation of 200 000 runs of the three-person procedure gave 0.602.

Common slips

- Forgetting the factor 3 for which person is the odd one out in round 1 of part (ii).
- Treating round 2 of part (ii) like a fresh three-coin round. The turned-over coin is fixed, so only one of "all heads" and "all tails" is possible.
- Proving the bound only at $p = \frac{1}{2}$. You need $pq \leq \frac{1}{4}$ for *every* p, and an inequality such as AM–GM gives you that.

Beyond STEP

Part (i) is the classical trick, usually credited to von Neumann, for getting a fair decision from a biased coin. In the two-person procedure, "hotel" and "tea shop" are equally likely only when $p = \frac{1}{2}$, but if one person tosses twice and HT and TH are used as the outcomes, the result is fair for every p. The number of rounds until a decision is a geometric random variable, and questions about how quickly such procedures stop lead to the study of stopping times and Markov chains. Grimmett and Welsh, *Probability: An Introduction*, covers both.

WORKED SOLUTION

STEP II 2019, Question 1

Pure • Tangents to Cubics

STEP II 2019, Question 1

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Write down the condition for the tangent at $x = a$ to pass through $(p, 0)$ and substitute $f(x) = (x - p)g(x)$: almost everything cancels, leaving a multiple of $g'(a)$. For the cubic, the tangent from the root p therefore touches the curve where the *other* quadratic factor is stationary, which is midway between the other two roots. Part (ii) then comes down to comparing two gradients, and to factorising the equation for where a tangent meets a cubic, which always has a double root at the point of contact.

Solution

The opening result

The tangent at $x = a$ is $y = f(a) + f'(a)(x - a)$. By the product rule, $f'(x) = g(x) + (x - p)g'(x)$. The tangent passes through $(p, 0)$ exactly when

$$\begin{aligned} 0 &= f(a) + f'(a)(p - a) \\ &= (a - p)g(a) + \big[g(a) + (a - p)g'(a)\big](p - a) \\ &= (a - p)g(a) - (a - p)g(a) - (a - p)^2 g'(a) \\ &= -(a - p)^2 g'(a). \end{aligned}$$

Since $a \neq p$, the factor $(a - p)^2$ is non-zero, so this holds if and only if $g'(a) = 0$. That proves the result in both directions at once.

Part (i)

Here $f(x) = (x - p)g(x)$ with $g(x) = A(x - q)(x - r) = A\big(x^2 - (q + r)x + qr\big)$, so $g'(x) = A\big(2x - (q + r)\big)$. By the opening result, and since $A \neq 0$,

$$g'(a) = 0 \iff 2a = q + r.$$

(Note that $a = \frac{1}{2}(q + r) > q > p$, so $a \neq p$ as required.)

For the gradient, $f'(a) = g(a) + (a-p)g'(a) = g(a)$ because $g'(a) = 0$. With $a - q = \frac{1}{2}(r-q)$ and $a - r = -\frac{1}{2}(r-q)$,

$$f'(a) = A \cdot \tfrac{1}{2}(r-q) \cdot \left(-\tfrac{1}{2}(r-q)\right) = -\frac{A(r-q)^2}{4}.$$

Part (ii)

Now factor out the root r instead: $f(x) = (x-r)h(x)$ with $h(x) = A(x-p)(x-q)$. The opening result applies with r in place of p (nothing in it used which root we chose), so the tangent at $x = c$ passes through $(r, 0)$ if and only if $h'(c) = 0$, that is $c = \frac{1}{2}(p+q)$. Exactly as in part (i), its gradient is

$$f'(c) = h(c) = A \cdot \tfrac{1}{2}(q-p) \cdot \left(-\tfrac{1}{2}(q-p)\right) = -\frac{A(q-p)^2}{4}.$$

So, since $A \neq 0$,

$$\text{tangents parallel} \iff (r-q)^2 = (q-p)^2 \iff r - q = q - p \iff q = \tfrac{1}{2}(p+r),$$

where the middle step uses $r - q > 0$ and $q - p > 0$, so we may take positive square roots.

Ask yourself How do we find where a tangent meets a cubic again, without solving a cubic?

We know one root already, twice over: the point of contact. At $x = q$ the gradient is $f'(q) = A(q-p)(q-r)$ (the other two terms of the product rule vanish because they contain $x - q$), and $f(q) = 0$, so the tangent there is $y = A(q-p)(q-r)(x-q)$. It meets the curve where

$$A(x-p)(x-q)(x-r) = A(q-p)(q-r)(x-q).$$

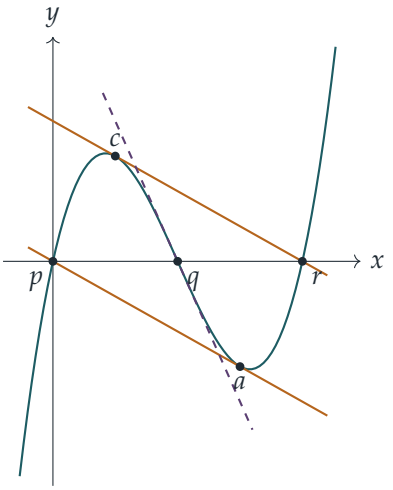
Dividing by $A \neq 0$ and taking out the factor $x - q$,

$$\begin{aligned} 0 &= (x-q)\left[(x-p)(x-r) - (q-p)(q-r)\right] \\ &= (x-q)\left[x^2 - q^2 - (p+r)(x-q)\right] \\ &= (x-q)^2\,(x+q-p-r). \end{aligned}$$

So the tangent meets the curve at $x = q$ (a double root, as expected for a point of contact) and at $x = p + r - q$, and nowhere else. It fails to meet the curve again if and only if this third root coincides with q:

$$p + r - q = q \iff q = \tfrac{1}{2}(p+r).$$

This is the same condition as before, so the tangents of parts (i) and (ii) are parallel if and only if the tangent at $x = q$ does not meet the curve again.



The sketch shows the case $q = \frac{1}{2}(p+r)$: the two tangents (amber) are parallel, and the tangent at q (dashed) crosses the curve there without meeting it anywhere else, since q is then the point of inflection.

Checks

- Take $p = 0$, $q = 1$, $r = 2$, $A = 3$, so $f(x) = 3x^3 - 9x^2 + 6x$. Then $f'(x) = 9x^2 - 18x + 6$, giving $f'(\frac{3}{2}) = -\frac{3}{4}$ and $f'(\frac{1}{2}) = -\frac{3}{4}$, both equal to $-A(r-q)^2/4 = -A(q-p)^2/4 = -\frac{3}{4}$, so the tangents are parallel, as predicted since q is the midpoint of p and r.
- The tangent at $x = \frac{3}{2}$ is $y = -\frac{9}{8} - \frac{3}{4}(x - \frac{3}{2})$, which is 0 at $x = 0$: it does pass through $(p, 0)$.
- All the identities above were also confirmed by computer in exact arithmetic for about two thousand random choices of $p < q < r$ and A.

Common slips

- Forgetting that $a \neq p$ is what allows us to cancel $(a-p)^2$; without it the "only if" direction fails, since the tangent at p itself always passes through $(p, 0)$.
- Writing $(r-q)^2 = (q-p)^2 \Rightarrow r - q = q - p$ without saying why the negative root is excluded.
- Trying to solve the cubic for the intersections from scratch, instead of using the known double root at the point of contact.

Beyond STEP

The tangent at $x = q$ is special because q is then the inflection point, the centre of symmetry of the cubic: every cubic is symmetric under a half-turn about its inflection point, which is why the two tangents become parallel. The tangent and chord constructions here are the first steps towards the geometry of cubic curves, where the third intersection of a line with a cubic is used to define an addition law on its points (elliptic curves). Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

WORKED SOLUTION

STEP II 2019, Question 2

Pure • Integrals of Inverse Functions

STEP II 2019, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

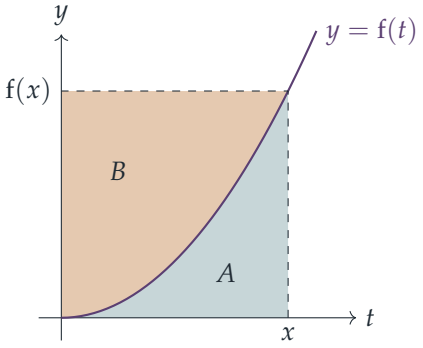
The key idea

The graph of $y = \mathrm{f}(t)$ cuts the rectangle with corners $(0,0)$ and $(x, \mathrm{f}(x))$ into two pieces. One piece is the area under the curve, and the other is the area between the curve and the y-axis, which is the integral of the inverse function. So the two integrals add up to the area of the rectangle. Parts (i) and (ii) define functions only implicitly, so we can't integrate them directly, but their *inverses* are simple cubics. The identity lets us swap a hard integral for an easy one.

Solution

The opening result

Since $\mathrm{f}'(t) > 0$ for $t > 0$ and $\mathrm{f}(0) = 0$, f is strictly increasing on $[0, x]$, so $\mathrm{f}(t) > 0$ for $t > 0$ and f has an inverse f^{-1} from $[0, \mathrm{f}(x)]$ to $[0, x]$. The same curve is described by $y = \mathrm{f}(t)$ and by $t = \mathrm{f}^{-1}(y)$.



The region A, between the curve and the t-axis for $0 \leq t \leq x$, has area

$$A = \int_0^x \mathrm{f}(t)\,\mathrm{d}t.$$

The region B, between the curve and the y-axis for $0 \leq y \leq \mathrm{f}(x)$, is made of horizontal strips. The strip at height y runs from $t = 0$ to $t = \mathrm{f}^{-1}(y)$, so

$$B = \int_0^{\mathrm{f}(x)} \mathrm{f}^{-1}(y)\,\mathrm{d}y.$$

Because the curve is increasing and goes from $(0,0)$ to $(x, \mathrm{f}(x))$, A and B fit together exactly, with no overlap, to fill the rectangle $0 \leq t \leq x$, $0 \leq y \leq \mathrm{f}(x)$. That rectangle has area $x\,\mathrm{f}(x)$, so

$$\int_0^x \mathrm{f}(t)\,\mathrm{d}t + \int_0^{\mathrm{f}(x)} \mathrm{f}^{-1}(y)\,\mathrm{d}y = x\,\mathrm{f}(x),$$

as required.

Part (i)

Write $\phi(y) = y^3 + y$. Then $\phi'(y) = 3y^2 + 1 > 0$, so ϕ is strictly increasing, and $\phi(y) \to \pm\infty$ as $y \to \pm\infty$. So for each real t there is exactly one real y with $\phi(y) = t$, and this is $\mathrm{g}(t)$. In other words, $\mathrm{g} = \phi^{-1}$.

$\mathrm{g}(0) = 0$. Putting $t = 0$ gives $\mathrm{g}(0)\left(\mathrm{g}(0)^2 + 1\right) = 0$. Since $\mathrm{g}(0)^2 + 1 \geq 1$ for real $\mathrm{g}(0)$, it follows that $\mathrm{g}(0) = 0$.

$\mathrm{g}'(t) > 0$. Differentiating $\mathrm{g}(t)^3 + \mathrm{g}(t) = t$ implicitly with respect to t,

$$\left(3\,\mathrm{g}(t)^2 + 1\right)\mathrm{g}'(t) = 1 \quad \Longrightarrow \quad \mathrm{g}'(t) = \frac{1}{3\,\mathrm{g}(t)^2 + 1} > 0,$$

since the denominator is at least 1.

? **Ask yourself** We can't write $\mathrm{g}(t)$ down, so how can we integrate it?

We don't need to. g satisfies the conditions of the opening result, and its inverse is $\phi(y) = y^3 + y$, which is easy to integrate. Applying the result with $x = 2$ needs $\mathrm{g}(2)$, the real root of $y^3 + y = 2$. This root is $y = 1$, and it is the only one because ϕ is strictly increasing. So

$$\int_0^2 \mathrm{g}(t)\,\mathrm{d}t = 2\,\mathrm{g}(2) - \int_0^1 (y^3 + y)\,\mathrm{d}y = 2 - \left(\tfrac{1}{4} + \tfrac{1}{2}\right) = \frac{5}{4}.$$

Part (ii)

Comparing the two definitions, $\mathrm{h}(t)$ is the real solution of $y^3 + y = t + 2$, so $\mathrm{h}(t) = \mathrm{g}(t+2)$. Substituting $u = t + 2$,

$$\int_0^8 \mathrm{h}(t)\,\mathrm{d}t = \int_2^{10} \mathrm{g}(u)\,\mathrm{d}u = \int_0^{10} \mathrm{g}(u)\,\mathrm{d}u - \int_0^2 \mathrm{g}(u)\,\mathrm{d}u.$$

For the first integral we use the opening result with $x = 10$. Since $2^3 + 2 = 10$, we have $\mathrm{g}(10) = 2$, and so

$$\int_0^{10} \mathrm{g}(u)\,\mathrm{d}u = 10 \times 2 - \int_0^2 (y^3 + y)\,\mathrm{d}y = 20 - (4 + 2) = 14.$$

Using part (i) for the second integral,

$$\int_0^8 \mathrm{h}(t)\,\mathrm{d}t = 14 - \frac{5}{4} = \frac{51}{4}.$$

Another way. We can also apply the picture to h directly. This time $\mathrm{h}(0) = 1$, not 0. The curve runs from $(0,1)$ to $(8,2)$, and $\mathrm{h}^{-1}(y) = y^3 + y - 2$. The area under the curve is the 8×2 rectangle minus the 0×1 rectangle (which is empty here) minus the area to the left of the curve for $1 \le y \le 2$:

$$16 - \int_1^2 (y^3 + y - 2)\,\mathrm{d}y = 16 - \left(\tfrac{15}{4} + \tfrac{3}{2} - 2\right) = 16 - \tfrac{13}{4} = \tfrac{51}{4}.$$

Checks

- $0 \le \mathrm{g}(t) \le 1$ on $[0,2]$, so the integral in (i) must lie between 0 and 2. Also, g is concave there (it is the inverse of an increasing convex function), so the integral is at least the triangle area $\frac{1}{2} \cdot 2 \cdot 1 = 1$. The answer $\frac{5}{4}$ fits.
- In (ii), $1 \le \mathrm{h}(t) \le 2$ on $[0,8]$, so the integral lies between 8 and 16, and 12.75 does too.
- Solving the cubics numerically at many points and integrating by Simpson's rule gives 1.25000 and 12.75000, which agree with the exact answers.

Common slips

- Using the wrong upper limit for the inverse integral: it is $\mathrm{g}(2) = 1$, not 2.
- Applying the opening result to h as if $\mathrm{h}(0) = 0$. Since $\mathrm{h}(0) = 1$, the picture has an extra strip, and the identity needs adjusting (or use $\mathrm{h}(t) = \mathrm{g}(t+2)$).
- Forgetting to justify that g is well defined, that is, that $y^3 + y = t$ has exactly one real root.

Beyond STEP

The opening identity is the equality case of *Young's inequality*: for increasing f with $\mathrm{f}(0) = 0$ and any $a, b \ge 0$, $ab \le \int_0^a \mathrm{f} + \int_0^b \mathrm{f}^{-1}$. Taking $\mathrm{f}(t) = t^{p-1}$ gives $ab \le a^p/p + b^q/q$, the starting point for Hölder's inequality in analysis. The fact that g is differentiable with $\mathrm{g}' = 1/\phi'(\mathrm{g})$ is the *inverse function theorem*. Abbott's *Understanding Analysis* proves this carefully.

WORKED SOLUTION

STEP II 2019, Question 3

Pure • Bounds on Polynomial Roots

STEP II 2019, Question 3

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The triangle inequality lets us bound a polynomial by replacing every term with its size, and the sizes form a geometric series when $|x| < 1$. That gives a lower bound on small roots. For large roots, the trick is to look at $1/\omega$: reversing the coefficients of f gives a polynomial of exactly the same shape, so the small-root result does the work a second time. In part (ii) the bounds cut the search for integer roots down to four candidates.

Solution

The triangle inequality

Both sides are non-negative, so it is enough to compare their squares. Since $x_1x_2 \leq |x_1x_2| = |x_1||x_2|$,

$$|x_1 + x_2|^2 = x_1^2 + 2x_1x_2 + x_2^2 \leq |x_1|^2 + 2|x_1||x_2| + |x_2|^2 = \big(|x_1| + |x_2|\big)^2,$$

and taking non-negative square roots gives $|x_1 + x_2| \leq |x_1| + |x_2|$.

For n numbers we use induction on n. The case $n = 1$ is an equality and $n = 2$ is done. If the result holds for $n - 1$ numbers, then applying the two-number case to $x_1 + \cdots + x_{n-1}$ and x_n,

$$\begin{aligned} |x_1 + \cdots + x_{n-1} + x_n| &\leq |x_1 + \cdots + x_{n-1}| + |x_n| \\ &\leq |x_1| + \cdots + |x_{n-1}| + |x_n|, \end{aligned}$$

which is the result for n numbers.

Part (i)(a)

We have $\mathrm{f}(x) - 1 = a_1 x + a_2 x^2 + \cdots + a_{n-1}x^{n-1} + x^n$. By the triangle inequality, then $|a_i| \le A$, and then $1 \le A$ for the last term,

$$\begin{aligned} |\mathrm{f}(x) - 1| &\le |a_1||x| + \cdots + |a_{n-1}||x|^{n-1} + |x|^n \\ &\le A\big(|x| + |x|^2 + \cdots + |x|^n\big). \end{aligned}$$

Because $0 \le |x| < 1$, the finite geometric sum is at most the infinite one, all of whose terms are non-negative:

$$|x| + |x|^2 + \cdots + |x|^n \le \sum_{k=1}^{\infty} |x|^k = \frac{|x|}{1 - |x|}.$$

Hence $|\mathrm{f}(x) - 1| \le \dfrac{A|x|}{1 - |x|}$, as required.

Part (i)(b)

Since $\mathrm{f}(\omega) = 0$, we have $|\mathrm{f}(\omega) - 1| = 1$, so part (a) gives

$$1 \le \frac{A|\omega|}{1 - |\omega|}.$$

As $1 - |\omega| > 0$ we may multiply through without changing the direction: $1 - |\omega| \le A|\omega|$, that is $1 \le (1 + A)|\omega|$, so

$$|\omega| \ge \frac{1}{1 + A}.$$

The upper bound is immediate: $|\omega| < 1 \le 1 + A$. So $(*)$ holds.

Part (i)(c)

Now $|\omega| \ge 1$. The lower bound is easy: $|\omega| \ge 1 > \frac{1}{1+A}$, since $A \ge 1 > 0$. The upper bound is the real content.

If $|\omega| = 1$ then $|\omega| \le 1 + A$ at once, so suppose $|\omega| > 1$. Put $\mathrm{g}(y) = y^n \mathrm{f}(1/y)$, which is

$$\mathrm{g}(y) = 1 + a_{n-1}y + a_{n-2}y^2 + \cdots + a_1 y^{n-1} + y^n.$$

? **Ask yourself** Why is g allowed to use the results of parts (a) and (b)?

It has exactly the form of f: constant term 1, leading coefficient 1, and its other coefficients are $a_{n-1}, \ldots, a_1$ in reverse order, each still of size at most A. So part (b) applies to g with the same A.

Since $\omega \ne 0$, $\mathrm{g}(1/\omega) = \omega^{-n}\mathrm{f}(\omega) = 0$, so $1/\omega$ is a real root of g with $|1/\omega| < 1$. Part (b) for g gives

$$\frac{1}{|\omega|} \ge \frac{1}{1 + A}, \qquad \text{so} \qquad |\omega| \le 1 + A.$$

Therefore $(*)$ holds for every real root of f, whatever its size. (Note that 0 is never a root, since $\mathrm{f}(0) = 1$.)

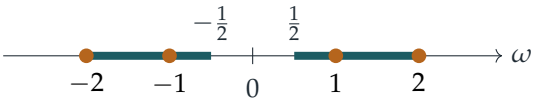
Part (ii)

Dividing by 135 and writing the terms in ascending powers, the equation is $\mathrm{f}(x) = 0$ with

$$\mathrm{f}(x) = 1 - \tfrac{126}{135}x - \tfrac{91}{135}x^2 - \tfrac{100}{135}x^3 - x^4 + x^5.$$

This has the form of part (i) with $n = 5$: constant term and leading coefficient both 1, and the other coefficients have sizes $\frac{126}{135}, \frac{91}{135}, \frac{100}{135}, 1$, all at most $A = 1$. So every real root satisfies

$$\tfrac{1}{2} \le |\omega| \le 2.$$



The only integers in the allowed region (shaded) are ± 1 and ± 2. Evaluating the original quintic $p(x) = 135x^5 - 135x^4 - 100x^3 - 91x^2 - 126x + 135$:

$$\begin{aligned} p(1) &= 135 - 135 - 100 - 91 - 126 + 135 = -182, \\ p(-1) &= -135 - 135 + 100 - 91 + 126 + 135 = 0, \\ p(2) &= 4320 - 2160 - 800 - 364 - 252 + 135 = 879, \\ p(-2) &= -4320 - 2160 + 800 - 364 + 252 + 135 = -5657. \end{aligned}$$

So the only integer root is $x = -1$. Indeed

$$p(x) = (x+1)\left(135x^4 - 270x^3 + 170x^2 - 261x + 135\right),$$

and the quartic factor is 971 at $x = -1$, so -1 is a simple root.

Checks

- Multiply out $(x+1)(135x^4 - 270x^3 + 170x^2 - 261x + 135)$ to recover the quintic.
- The quintic has two other real roots, about 0.595 and 1.725; both lie in $[\frac{1}{2}, 2]$, as part (i) predicts.
- Try $\mathrm{f}(x) = 1 - 2x + x^2 = (1-x)^2$ with $A = 2$: the root 1 lies in $[\frac{1}{3}, 3]$. The bounds were also confirmed by computer on hundreds of random polynomials.

Common slips

- Forgetting the x^n term in part (a): its coefficient is 1, not some a_n, and it is $A \ge 1$ that lets it join the geometric series.
- In part (c), trying to reuse part (a) directly at ω when $|\omega| \ge 1$: the geometric series no longer converges. Go to $1/\omega$ instead.
- Testing all the divisors of 135 as candidate roots in part (ii): legitimate, but the bounds from part (i) leave only four to try.

Beyond STEP

Bounding the roots of a polynomial in terms of its coefficients is a classical theme; this is a version of Cauchy's bound, and the same argument works word for word for complex roots, with modulus in place of absolute value. The reversal $y^n \mathrm{f}(1/y)$ swaps small roots with large ones and is a standard tool in algebra and complex analysis. For bounds like these and much more on the geometry of complex roots, try Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP II 2019, Question 4

Pure • Telescoping Trigonometric Products

STEP II 2019, Question 4

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

A product of cosines whose angles double (or halve) each time telescopes if you multiply by the sine of the smallest angle: the double-angle formula $2\sin\theta\cos\theta = \sin 2\theta$ then swallows the factors one at a time. Once the product is in closed form, taking logarithms turns it into a sum, and differentiating turns $\ln\cos$ into $\tan$: that is where the tangent sum comes from. The infinite versions then follow by letting $n \to \infty$ with the two standard limits.

Solution

Part (i)

Call the product P. Since $\sin\frac{\pi}{9} \neq 0$, we may multiply by it and apply the identity three times:

$$\begin{aligned} 8\sin\tfrac{\pi}{9}\,P &= 4\big(2\sin\tfrac{\pi}{9}\cos\tfrac{\pi}{9}\big)\cos\tfrac{2\pi}{9}\cos\tfrac{4\pi}{9} = 2\big(2\sin\tfrac{2\pi}{9}\cos\tfrac{2\pi}{9}\big)\cos\tfrac{4\pi}{9} \\ &= 2\sin\tfrac{4\pi}{9}\cos\tfrac{4\pi}{9} = \sin\tfrac{8\pi}{9} = \sin\big(\pi - \tfrac{\pi}{9}\big) = \sin\tfrac{\pi}{9}. \end{aligned}$$

Dividing by $8\sin\frac{\pi}{9}$,

$$\cos\tfrac{\pi}{9}\cos\tfrac{2\pi}{9}\cos\tfrac{4\pi}{9} = \frac{1}{8}.$$

Part (ii)

Here the angles $x, \frac{x}{2}, \ldots, \frac{x}{2^n}$ halve, so the smallest is $x/2^n$. As $0 < x/2^n < \frac{1}{2}\pi$, its sine is non-zero. Multiplying by $\sin(x/2^n)$, the factor $\cos(x/2^n)$ pairs with it to give $\frac{1}{2}\sin(x/2^{n-1})$, which pairs with $\cos(x/2^{n-1})$, and so on up to $\cos x$. Each of the $n+1$ steps contributes a factor $\frac{1}{2}$:

$$\sin\Big(\frac{x}{2^n}\Big)\prod_{k=0}^{n}\cos\Big(\frac{x}{2^k}\Big) = \frac{1}{2^{n+1}}\sin 2x, \qquad \text{so} \qquad \prod_{k=0}^{n}\cos\Big(\frac{x}{2^k}\Big) = \frac{\sin 2x}{2^{n+1}\sin(x/2^n)}.$$

Ask yourself How can differentiation turn a product of cosines into a sum of tangents?

Take logarithms. Every factor is positive, because $0 < x/2^k < \frac{1}{2}\pi$ for each k, and $\sin 2x > 0$ too. So

$$\sum_{k=0}^{n} \ln\cos\left(\frac{x}{2^k}\right) = \ln\sin 2x - (n+1)\ln 2 - \ln\sin\left(\frac{x}{2^n}\right).$$

Differentiate with respect to x, using $\frac{\mathrm{d}}{\mathrm{d}x}\ln\cos(x/2^k) = -\frac{1}{2^k}\tan(x/2^k)$ by the chain rule:

$$-\sum_{k=0}^{n} \frac{1}{2^k}\tan\left(\frac{x}{2^k}\right) = 2\cot 2x - \frac{1}{2^n}\cot\left(\frac{x}{2^n}\right).$$

Multiplying by -1 gives exactly the required result:

$$\sum_{k=0}^{n} \frac{1}{2^k}\tan\left(\frac{x}{2^k}\right) = \frac{1}{2^n}\cot\left(\frac{x}{2^n}\right) - 2\cot 2x.$$

Part (iii)

Dividing the product formula by $\cos x \neq 0$ and using $\sin 2x = 2\sin x\cos x$,

$$\prod_{k=1}^{n} \cos\left(\frac{x}{2^k}\right) = \frac{2\sin x\cos x}{2^{n+1}\sin(x/2^n)\cos x} = \frac{\sin x}{2^n\sin(x/2^n)}.$$

Ask yourself What happens to $2^n\sin(x/2^n)$ as $n \to \infty$?

Put $\theta = x/2^n$, so that $\theta \to 0$ as $n \to \infty$. Then

$$2^n\sin\left(\frac{x}{2^n}\right) = x\cdot\frac{\sin\theta}{\theta} \longrightarrow x\cdot 1 = x,$$

and therefore

$$\prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right) = \lim_{n\to\infty}\frac{\sin x}{2^n\sin(x/2^n)} = \frac{\sin x}{x}.$$

For the series, substitute $j = k+2$: then $\frac{1}{2^{j-2}} = \frac{1}{2^k}$ and $\frac{\pi}{2^j} = \frac{\pi/4}{2^k}$, so the sum is the series of part (ii) with $x = \frac{\pi}{4}$, which lies in $(0, \frac{1}{2}\pi)$. With this x, $2\cot 2x = 2\cot\frac{\pi}{2} = 0$, so the partial sum up to $k = n$ is

$$\frac{1}{2^n}\cot\left(\frac{\pi}{4\cdot 2^n}\right) = \frac{1}{x}\cdot\frac{\theta}{\tan\theta} \qquad \left(\theta = \frac{x}{2^n},\ x = \frac{\pi}{4}\right).$$

As $n \to \infty$, $\theta \to 0$ and $\theta/\tan\theta \to 1$, so the partial sums tend to $1/x$:

$$\sum_{j=2}^{\infty} \frac{1}{2^{j-2}}\tan\left(\frac{\pi}{2^j}\right) = \frac{4}{\pi}.$$

Checks

- Part (i): $\cos 20^\circ \cos 40^\circ \cos 80^\circ \approx 0.9397 \times 0.7660 \times 0.1736 \approx 0.1250$.
- Part (ii) with $n = 0$: the product is $\cos x$ and the formula gives $\sin 2x/(2\sin x) = \cos x$. The sum identity with $n = 0$ reads $\tan x = \cot x - 2\cot 2x$, which is the double-angle formula for cot rearranged.
- Part (iii): the first terms of the series are $\tan \frac{\pi}{4} = 1$, $\frac{1}{2}\tan \frac{\pi}{8} \approx 0.2071$, $\frac{1}{4}\tan \frac{\pi}{16} \approx 0.0497$, ...; the total, confirmed by computer, is $1.27324\ldots = 4/\pi$.
- Letting $x \to 0$ in $\prod \cos(x/2^k) = \sin x/x$ gives $1 = 1$, as it should.

Common slips

- Counting the factors of $\frac{1}{2}$ wrongly: the product from $k = 0$ to n has $n + 1$ factors, so the denominator is 2^{n+1}, not 2^n.
- Taking logarithms without noting that every factor is positive; this is why the range $0 < x < \frac{1}{2}\pi$ matters.
- Forgetting that $2\cot 2x$ vanishes at $x = \frac{\pi}{4}$, or using $x = \frac{\pi}{2}$ (the $j = 1$ term would be $\tan \frac{\pi}{2}$, which is undefined).

Beyond STEP

Setting $x = \frac{\pi}{2}$ in $\prod \cos(x/2^k) = \sin x/x$ and writing each cosine with nested square roots gives Viète's formula $\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdots$, the first infinite product in the history of mathematics. Differentiating the logarithm of an infinite product term by term, as we did with the finite one, needs justification; a first course in analysis sets out when it is allowed. Euler's product $\sin x = x\prod_{n\geq 1}\left(1 - \frac{x^2}{n^2\pi^2}\right)$ grows out of the same ideas. For the rigorous background, try Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2019, Question 5

Pure • Fixed Points of Iteration

STEP II 2019, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

A constant sequence starts at a *fixed point*, a solution of $f(x) = x$. A sequence of period 2 starts at a solution of $f(f(x)) = x$ that is not a fixed point. Every fixed point also solves $f(f(x)) = x$, so $f(x) - x$ must be a factor of the quartic $f(f(x)) - x$. Once we have the other factor, which is a quadratic, everything comes down to discriminants.

Solution

A factorisation for both parts

In both parts $f(x) = x^2 - px + c$ for a constant c ($c = p$ in (i), $c = q$ in (ii)). For any x and y,

$$f(y) - f(x) = (y^2 - x^2) - p(y - x) = (y - x)(y + x - p).$$

Putting $y = f(x)$ and adding $f(x) - x$ to both sides gives

$$\begin{aligned} f(f(x)) - x &= \big[f(f(x)) - f(x)\big] + \big[f(x) - x\big] \\ &= \big(f(x) - x\big)\big(f(x) + x - p + 1\big). \end{aligned}$$

Two consequences we shall use:

- The sequence has period 2 exactly when a is a root of $Q(x) = f(x) + x - p + 1$ that is *not* a root of $f(x) - x$. Indeed, if $f(f(a)) = a$ then $u_2 = u_0$, and since each term is determined by the one before, $u_{n+2} = u_n$ for all n by induction. The sequence is then non-constant precisely when $u_1 \neq u_0$, i.e. $f(a) \neq a$.
- If x is a fixed point then $Q(x) = 2x - p + 1$. So a fixed point is also a root of Q only if $x = \frac{1}{2}(p - 1)$.

Part (i)

Here $f(x) = x^2 - px + p$, so

$$f(a) - a = a^2 - (p + 1)a + p = (a - 1)(a - p).$$

The sequence is constant exactly when $u_1 = u_0$ (then every term equals a), so the values are

$$a = 1 \quad \text{and} \quad a = p.$$

For period 2, the other factor is

$$Q(x) = x^2 - px + p + x - p + 1 = x^2 + (1-p)x + 1,$$

with discriminant

$$(1-p)^2 - 4 = (p - 1 - 2)(p - 1 + 2) = (p-3)(p+1).$$

- If $-1 < p < 3$, the discriminant is negative, Q has no real roots, and no real a gives period 2.
- If $p = 3$ or $p = -1$, Q has the single root $x = \frac{1}{2}(p-1)$, which is 1 or -1 respectively. For $p = 3$ the fixed points are 1 and 3; for $p = -1$ they are 1 and -1. In both cases the root of Q is a fixed point, so the sequence starting there is constant, not of period 2.
- If $p > 3$ or $p < -1$, Q has two distinct real roots. Neither is a fixed point: by the second bullet above it would have to be $\frac{1}{2}(p-1)$, but

  $$Q\left(\tfrac{1}{2}(p-1)\right) = \tfrac{1}{4}(p-1)^2 - \tfrac{1}{2}(p-1)^2 + 1 = 1 - \tfrac{1}{4}(p-1)^2 \neq 0$$

  because $(p-1)^2 > 4$. So taking a to be either root gives a sequence of period 2.

Hence a value of a giving period 2 exists if and only if $p > 3$ or $p < -1$.

Part (ii)

Now $\mathrm{f}(x) = x^2 - px + q$, so

$$\mathrm{f}(x) - x = x^2 - (p+1)x + q.$$

A constant sequence needs a with $\mathrm{f}(a) = a$, i.e. a real root of this quadratic.

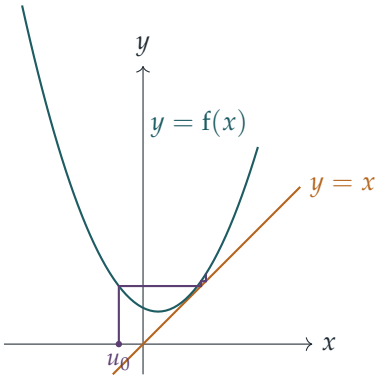
If $\mathrm{f}(x) > x$ *for all* x, then $\mathrm{f}(a) \neq a$ for every a, so no sequence is constant.

Conversely, suppose no a gives a constant sequence, so $g(x) = x^2 - (p+1)x + q$ has no real root. Then g is continuous and never zero, so it never changes sign, and since $g(x) \to +\infty$ as $x \to \infty$ it is positive everywhere. (Equivalently: the discriminant is negative and the leading coefficient positive.) That is, $\mathrm{f}(x) > x$ for all x, which completes the "if and only if".

The deduction. Suppose there is no constant sequence. Then $\mathrm{f}(x) > x$ for all x, so for any a,

$$u_{n+1} = \mathrm{f}(u_n) > u_n \quad \text{for every } n.$$

The sequence is strictly increasing, so $u_2 > u_1 > u_0$. In particular $u_2 \neq u_0$, and the sequence cannot have period 2.



The picture shows the situation: the parabola lies wholly above $y = x$, so the "staircase" of iterates can only climb.

The final part. The converse is *false*. Take $p = q = 0$, so $f(x) = x^2$. The sequence is constant for $a = 0$ and $a = 1$. For period 2 we need $f(f(a)) = a^4 = a$, i.e. $a(a-1)(a^2+a+1) = 0$. The factor $a^2 + a + 1$ has negative discriminant, so the only real solutions are $a = 0$ and $a = 1$, both fixed points. So there is no value of a giving period 2, even though some values give a constant sequence.

Ask yourself Is $f(x) = x^2$ just a lucky accident?

No. In general the fixed-point quadratic has discriminant $D = (p+1)^2 - 4q$, and $Q(x) = x^2 + (1-p)x + q - p + 1$ has discriminant

$$(1-p)^2 - 4(q-p+1) = (p+1)^2 - 4q - 4 = D - 4.$$

As in (i), a root of Q is a fixed point only when the discriminant of Q is zero. So constant sequences exist when $D \geq 0$, and period-2 sequences exist when $D > 4$. Any p, q with $0 \leq D \leq 4$ gives a counterexample.

Checks

- With $p = 4$ in (i), $Q(x) = x^2 - 3x + 1$ has roots $\frac{1}{2}(3 \pm \sqrt{5}) \approx 2.618,\ 0.382$. Starting at 2.618, $f(2.618) = 4 + (2.618 - 4)(2.618) \approx 0.382$, and $f(0.382) \approx 2.618$: period 2, as predicted.
- At $p = 3$ the root of Q is $x = 1$, and $f(1) = 3 + (1 - 3) = 1$: a fixed point, confirming that the boundary values are excluded.
- Expanding $(x^2 - (p+1)x + p)(x^2 + (1-p)x + 1)$ gives $f(f(x)) - x$; a computer check at thousands of random (p, x), together with the condition "period 2 iff $D > 4$" in (ii), found no exceptions.

Common slips

- Including $p = 3$ and $p = -1$: there $\mathrm{f}(\mathrm{f}(a)) = a$ has a repeated root, but it is a fixed point, so the sequence is constant.
- Forgetting that the factor $\mathrm{f}(x) - x$ must be removed from $\mathrm{f}(\mathrm{f}(x)) - x$ before looking for period-2 points.
- In the final part, arguing from the direction of the inequality alone; a single explicit counterexample is what settles a "Is it true that…?".

Beyond STEP

This is the first step into *discrete dynamical systems*. Every quadratic map is equivalent, by a linear change of variable, to the logistic map $x \mapsto rx(1 - x)$. As the parameter grows, the fixed point gives way to a 2-cycle, then a 4-cycle, and so on, in a period-doubling cascade that ends in chaos. The thresholds you found, where the discriminant of Q vanishes, are the first of these bifurcations. Strogatz's *Nonlinear Dynamics and Chaos* is a very readable introduction.

WORKED SOLUTION

STEP II 2019, Question 6

Pure • First-Order Differential Equations

STEP II 2019, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

In both equations the right-hand side depends on x and y only through one combination: $y+x$ in part (i), and $y-x$ in part (ii), since the quadratic is really $(y-x)^2+4(y-x)+3$. That one observation finds the straight-line solutions, pins the stationary points to lines, and (in part (i)) turns the equation into one we can separate. The note about solution curves never crossing then does the rest: a curve that starts between two straight-line solutions is trapped there for ever.

Solution

Part (i)

The straight-line solution. If $y=mx+c$ then $\frac{\mathrm{d}y}{\mathrm{d}x}=m$, so we need

$$m=mx+c+x+1=(m+1)x+(c+1) \quad \text{for all } x.$$

Comparing coefficients of x gives $m=-1$, and then the constants give $c+1=m=-1$, so $c=-2$. The solution is $y=-x-2$.

Where the stationary points are. At a stationary point of $y=y_3(x)$ we have $0=\frac{\mathrm{d}y}{\mathrm{d}x}=y+x+1$, so the point lies on the line $y=-x-1$.

Now suppose $k<-2$. The curve $y=y_3(x)$ and the line $y=-x-2$ are both solutions, and at $x=0$ the curve is below the line ($k<-2$). They never meet, so the curve stays below the line for every x: $y_3(x)<-x-2<-x-1$. Hence $y_3(x)+x+1<-1<0$ everywhere. The gradient is never zero (in fact always less than -1), so there are no stationary points.

Every stationary point is a minimum. Differentiating the equation,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=\frac{\mathrm{d}y}{\mathrm{d}x}+1.$$

At a stationary point the first derivative is 0, so the second derivative is $1>0$, and the

point is a local minimum.

Solving. With $Y = y + x$ we have $\frac{\mathrm{d}Y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}x} + 1 = Y + 2$. On the line $Y = -2$ this is the solution we already have; otherwise we separate:

$$\int \frac{\mathrm{d}Y}{Y+2} = \int \mathrm{d}x \quad \Longrightarrow \quad \ln|Y+2| = x + \text{const} \quad \Longrightarrow \quad Y + 2 = Ae^x,$$

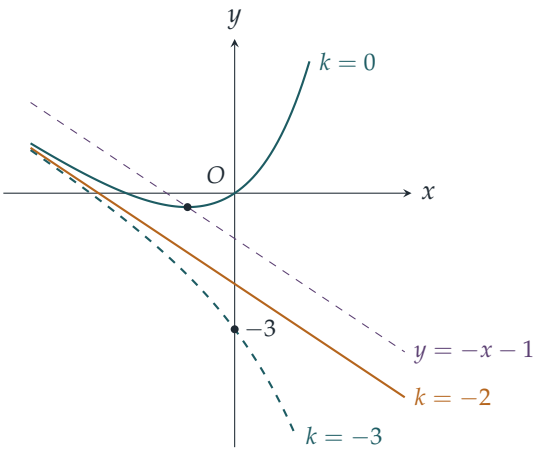
where A is a constant (and $A = 0$ covers the straight line). So $y = Ae^x - x - 2$, and $y(0) = k$ gives $A = k + 2$:

$$y_3(x) = (k+2)e^x - x - 2.$$

This agrees with what we found: $y_3'(x) = (k+2)e^x - 1$, which is never zero when $k \leq -2$, and vanishes once, at $e^x = 1/(k+2)$, when $k > -2$.

The three curves.

- $k = 0$: $y = 2e^x - x - 2$, through the origin. The minimum is where $e^x = \frac{1}{2}$, at $\big(-\ln 2,\ \ln 2 - 1\big) \approx (-0.69, -0.31)$, on the line $y = -x - 1$. As $x \to -\infty$, $2e^x \to 0$ and the curve approaches $y = -x - 2$ from above; as $x \to \infty$ the exponential takes over.
- $k = -2$: the line $y = -x - 2$ itself.
- $k = -3$: $y = -e^x - x - 2$, through $(0, -3)$, always decreasing. It approaches $y = -x - 2$ from below as $x \to -\infty$ and plunges like $-e^x$ as $x \to \infty$.



Part (ii)

Rewriting the right-hand side. Grouping the terms,

$$x^2 + y^2 - 2xy - 4x + 4y + 3 = (y-x)^2 + 4(y-x) + 3 = (y - x + 1)(y - x + 3).$$

It is worth writing $u = y - x$, so that $\frac{\mathrm{d}y}{\mathrm{d}x} = (u+1)(u+3)$.

The straight-line solutions. If $y = mx + c$ then $u = (m-1)x + c$, and we need

$$m = \big((m-1)x + c + 1\big)\big((m-1)x + c + 3\big) \quad \text{for all } x.$$

If $m \neq 1$ the right-hand side is a genuine quadratic in x and can't equal a constant, so $m = 1$. Then $1 = (c+1)(c+3)$, that is $c^2 + 4c + 2 = 0$, giving $c = -2 \pm \sqrt{2}$. The two solutions are

$$y = x - 2 + \sqrt{2} \qquad \text{and} \qquad y = x - 2 - \sqrt{2}.$$

Stationary points of y_4. The gradient is zero exactly when $u = -1$ or $u = -3$, so any stationary point lies on

$$y = x - 1 \qquad \text{or} \qquad y = x - 3.$$

Between these lines we have $-3 < y - x < -1$, so $u + 1 < 0 < u + 3$ and the gradient $(u+1)(u+3)$ is *negative*. (Outside them, it is positive.)

Where the curve lives. At $x = 0$, $u = -2$, which lies between $-2 - \sqrt{2} \approx -3.41$ and $-2 + \sqrt{2} \approx -0.59$. So at $x = 0$ the curve is between the two straight-line solutions, and since solution curves never cross,

$$x - 2 - \sqrt{2} < y_4(x) < x - 2 + \sqrt{2} \quad \text{for all } x.$$

The band between the lines of stationary points ($-3 < u < -1$) sits inside this band.

> **Ask yourself** The curve is trapped between two parallel lines, but where does it go inside the band?

Track $u = y_4 - x$. It satisfies

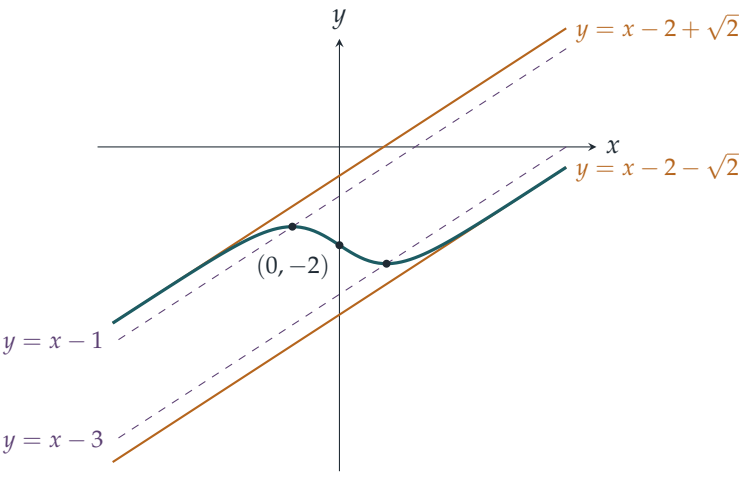
$$\frac{\mathrm{d}u}{\mathrm{d}x} = (u+1)(u+3) - 1 = u^2 + 4u + 2 = \left(u + 2 - \sqrt{2}\right)\left(u + 2 + \sqrt{2}\right),$$

which is negative while $-2 - \sqrt{2} < u < -2 + \sqrt{2}$, which is always. So u is strictly decreasing and bounded, and therefore has limits as $x \to \pm\infty$. At such a limit $\frac{\mathrm{d}u}{\mathrm{d}x}$ must tend to 0 (otherwise u would change without bound), so the limits are the roots: $u \to -2 + \sqrt{2}$ as $x \to -\infty$ and $u \to -2 - \sqrt{2}$ as $x \to +\infty$.

So, reading from left to right:

- far to the left the curve hugs $y = x - 2 + \sqrt{2}$ from below, with gradient close to 1;
- as u falls through -1 it meets $y = x - 1$ in a single point, a *local maximum* (the gradient changes from positive to negative);
- it falls through $(0, -2)$ with gradient $(-1)(1) = -1$;
- as u falls through -3 it meets $y = x - 3$ in a single point, a *local minimum*;
- then it rises again, hugging $y = x - 2 - \sqrt{2}$ from above with gradient tending to 1.

Each of $u = -1$ and $u = -3$ is passed exactly once because u is strictly decreasing, so there are exactly two stationary points. One more nice feature: replacing (x, y) by $(-x, -4 - y)$ leaves the equation and the starting point $(0, -2)$ unchanged, so the curve has half-turn symmetry about $(0, -2)$, and the maximum and minimum are mirror images of each other through that point.



Checks

- In part (i), substitute back: $y = (k+2)e^x - x - 2$ gives $y' = (k+2)e^x - 1$ and $y + x + 1 = (k+2)e^x - 1$. Setting $k = -2$ recovers the straight line.
- For $k = 0$ the minimum $(-\ln 2, \ln 2 - 1) \approx (-0.693, -0.307)$ does lie on $y = -x - 1$.
- In part (ii), a step-by-step numerical solution from $(0, -2)$ (confirmed by computer) stays between the two lines, reaches a maximum near $(-0.623, -1.623)$ on $y = x - 1$ and a minimum near $(0.623, -2.377)$ on $y = x - 3$, and by $x = \pm 6$ is within about 10^{-7} of the asymptotes. Those two points are images of each other in $(0, -2)$, as the symmetry predicts.

Common slips

- Forgetting to use the non-crossing property: "$k < -2$ means no stationary points" needs the curve to stay below $y = -x - 2$ for *all* x, not just at $x = 0$.
- Sketching y_4 crossing one of the straight-line solutions, or with the maximum and minimum the wrong way round. The sign of $(u+1)(u+3)$ between the dashed lines settles which is which.
- In finding m and c in part (ii), dividing by $m - 1$ without first ruling out (or in) $m = 1$.

Beyond STEP

The note at the start of the question is the uniqueness half of the Picard–Lindelöf theorem: for a smooth right-hand side, exactly one solution passes through each point, so solution curves cannot cross. Part (ii) is an *autonomous* equation in disguise, $u' = u^2 + 4u + 2$, and the way we found where the curve goes, from the equilibria and

the sign of u' in between and with no formula, is the start of the qualitative theory of differential equations. Boyce and DiPrima, *Elementary Differential Equations*, covers both uniqueness and this phase-line picture.

WORKED SOLUTION

STEP II 2019, Question 7

Pure • Vectors and Scalar Products

STEP II 2019, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

For unit vectors, lengths and angles both come from scalar products: $|\mathbf{u}-\mathbf{v}|^2 = 2-2\,\mathbf{u}\cdot\mathbf{v}$. So the trick is to split the zero-sum condition, with some vectors on one side and the rest on the other, and take the length squared of each side. That turns the vector equation into equations between scalar products, and those tell us the shape.

Solution

Throughout we use the fact that for unit vectors $\mathbf{u}$, $\mathbf{v}$,

$$|\mathbf{u}\pm\mathbf{v}|^2 = \mathbf{u}\cdot\mathbf{u} \pm 2\,\mathbf{u}\cdot\mathbf{v} + \mathbf{v}\cdot\mathbf{v} = 2 \pm 2\,\mathbf{u}\cdot\mathbf{v}.$$

Part (i)

Since $\mathbf{a}+\mathbf{b} = -\mathbf{c}$, taking the length squared of both sides gives

$$2 + 2\,\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}+\mathbf{b}|^2 = |\mathbf{c}|^2 = 1,$$

so $\mathbf{a}\cdot\mathbf{b} = -\frac{1}{2}$, as required.

The condition is symmetric in $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, so in the same way (writing $\mathbf{b}+\mathbf{c} = -\mathbf{a}$ and $\mathbf{c}+\mathbf{a} = -\mathbf{b}$) we get $\mathbf{b}\cdot\mathbf{c} = \mathbf{c}\cdot\mathbf{a} = -\frac{1}{2}$. Hence

$$AB^2 = |\mathbf{a}-\mathbf{b}|^2 = 2 - 2\left(-\tfrac{1}{2}\right) = 3,$$

and likewise $BC^2 = CA^2 = 3$. All three sides have length $\sqrt{3}$, so **triangle ABC is equilateral**. (Each pair of position vectors makes an angle of $120°$, and O is both the circumcentre and the centroid of the triangle, since $\frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}) = \mathbf{0}$.)

Part (ii)

Write the condition as $\mathbf{a}_1 + \mathbf{a}_2 = -(\mathbf{a}_3 + \mathbf{a}_4)$ and take the length squared of both sides:

$$2 + 2\,\mathbf{a}_1\cdot\mathbf{a}_2 = 2 + 2\,\mathbf{a}_3\cdot\mathbf{a}_4 \quad\Longrightarrow\quad \mathbf{a}_1\cdot\mathbf{a}_2 = \mathbf{a}_3\cdot\mathbf{a}_4.$$

Grouping the vectors in the other two possible ways gives, in the same manner,

$$\mathbf{a}_1 \cdot \mathbf{a}_3 = \mathbf{a}_2 \cdot \mathbf{a}_4, \qquad \mathbf{a}_1 \cdot \mathbf{a}_4 = \mathbf{a}_2 \cdot \mathbf{a}_3.$$

Since $A_iA_j^2 = 2 - 2\,\mathbf{a}_i \cdot \mathbf{a}_j$, this says that *opposite edges are equal*: $A_1A_2 = A_3A_4$, $A_1A_3 = A_2A_4$ and $A_1A_4 = A_2A_3$.

(a) The four points in a plane

? **Ask yourself** Does the plane of the four points have to pass through the origin?

Yes. The centroid of the four points is $\frac{1}{4}(\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4) = \mathbf{0}$, and the centroid of points in a plane lies in that plane. So the plane contains O, and since every A_i is at distance 1 from O, the four points lie on the circle of radius 1 centred at O in that plane. From now on we work in this plane.

Let $\mathbf{w} = \mathbf{a}_1 + \mathbf{a}_2$, so that $\mathbf{a}_3 + \mathbf{a}_4 = -\mathbf{w}$.

If $\mathbf{w} = \mathbf{0}$, then $\mathbf{a}_2 = -\mathbf{a}_1$ and $\mathbf{a}_4 = -\mathbf{a}_3$.

If $\mathbf{w} \neq \mathbf{0}$, dot $\mathbf{a}_3$ with $\mathbf{a}_3 + \mathbf{a}_4 = -\mathbf{w}$ and use (ii):

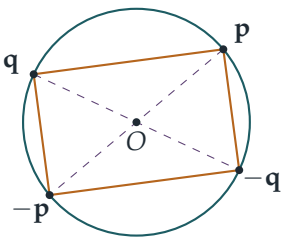
$$\mathbf{a}_3 \cdot \mathbf{w} = -(1 + \mathbf{a}_3 \cdot \mathbf{a}_4) = -(1 + \mathbf{a}_1 \cdot \mathbf{a}_2) = -\tfrac{1}{2}|\mathbf{w}|^2.$$

So $\mathbf{a}_3$ is a unit vector with a fixed component along $\mathbf{w}$: its tip lies on a fixed line perpendicular to $\mathbf{w}$, and a line meets a circle in at most two points. Now $-\mathbf{a}_1$ and $-\mathbf{a}_2$ are two such vectors, as

$$(-\mathbf{a}_1) \cdot \mathbf{w} = -(1 + \mathbf{a}_1 \cdot \mathbf{a}_2) = -\tfrac{1}{2}|\mathbf{w}|^2,$$

and similarly for $-\mathbf{a}_2$; they are different because $A_1 \neq A_2$. Hence $\mathbf{a}_3 = -\mathbf{a}_1$ or $\mathbf{a}_3 = -\mathbf{a}_2$, and then $\mathbf{a}_4 = -\mathbf{w} - \mathbf{a}_3$ is the other one.

In every case the four position vectors are $\mathbf{p}, -\mathbf{p}, \mathbf{q}, -\mathbf{q}$ for some unit vectors $\mathbf{q} \neq \pm\mathbf{p}$. So the four points are the ends of two diameters of the circle. The quadrilateral they form has diagonals that bisect each other (at O), so it is a parallelogram, and its diagonals are equal (both of length 2), so it is a **rectangle**. It is a square exactly when the two diameters are perpendicular.



(Strictly, the vertices should be taken in order around the circle. If the labels happen to go $A_1 = \mathbf{p}$, $A_2 = \mathbf{q}$, $A_3 = -\mathbf{q}$, $A_4 = -\mathbf{p}$, joining them in that order traces two sides and both diagonals of this rectangle.)

(b) A regular tetrahedron Now all six edges A_iA_j are equal, so all six scalar products $\mathbf{a}_i \cdot \mathbf{a}_j$ ($i \neq j$) take the same value, k say. Dotting $\mathbf{a}_1$ with the zero sum,

$$0 = \mathbf{a}_1 \cdot (\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4) = 1 + 3k,$$

so $k = -\frac{1}{3}$. Then each edge has

$$A_1A_2^2 = 2 - 2k = 2 + \tfrac{2}{3} = \tfrac{8}{3}, \qquad \text{so the side length is} \qquad \sqrt{\tfrac{8}{3}} = \frac{2\sqrt{6}}{3}.$$

Checks

- The tetrahedron with vertices $\frac{1}{\sqrt{3}}(1,1,1)$, $\frac{1}{\sqrt{3}}(1,-1,-1)$, $\frac{1}{\sqrt{3}}(-1,1,-1)$, $\frac{1}{\sqrt{3}}(-1,-1,1)$ is regular, has unit position vectors summing to $\mathbf{0}$, and has edge $\frac{1}{\sqrt{3}}|(0,2,2)| = \frac{2\sqrt{2}}{\sqrt{3}} = \frac{2\sqrt{6}}{3} \approx 1.633$.
- Compare with (i): three unit vectors at 120° give side $\sqrt{3} \approx 1.732$, slightly longer, which makes sense since the tetrahedron's vertices are "more crowded" on the sphere.
- A computer check on thousands of random sets of four unit vectors with zero sum confirmed all three equalities of scalar products in (ii), and that in the planar case the points always come in opposite pairs $\pm\mathbf{p}$, $\pm\mathbf{q}$.

Common slips

- In (i), stopping at $\mathbf{a} \cdot \mathbf{b} = -\frac{1}{2}$: you need all three scalar products (or all three side lengths) before concluding that the triangle is equilateral.
- In (a), concluding "parallelogram" from equal opposite sides without checking how the vertices are arranged, or forgetting to show that the plane passes through O.

Beyond STEP

Sets of unit vectors summing to zero with all pairwise scalar products equal are the vertices of a *regular simplex*: in n dimensions there are $n+1$ of them, with $\mathbf{a}_i \cdot \mathbf{a}_j = -\frac{1}{n}$, which gives $-\frac{1}{2}$ and $-\frac{1}{3}$ above. Such configurations appear in chemistry (the 109.5° bond angle of methane is $\cos^{-1}(-\frac{1}{3})$) and in coding theory. The matrix of all scalar products $\mathbf{a}_i \cdot \mathbf{a}_j$, called the *Gram matrix*, determines a configuration of vectors up to rotation, which is the idea behind this whole question. Strang's *Introduction to Linear Algebra* is a good place to read more.

WORKED SOLUTION

STEP II 2019, Question 8

Pure • Multiplicative Matrix Functions

STEP II 2019, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 7.3, *What Makes a Determinant*: https://beyondstep.keithmcnulty.org/ch07/determinant-axioms.html

The key idea

Everything follows from the single rule $\mathrm{f}(\mathbf{MN}) = \mathrm{f}(\mathbf{M})\mathrm{f}(\mathbf{N})$, applied to well-chosen products. The trick each time is to find two different ways of writing the same matrix, or a matrix that is a product of itself with something else, and then compare the values of f. The determinant is one such function, so it's a useful guide to what the answers ought to be, but we must not assume that f *is* the determinant.

Solution

Part (i)

Since $\mathbf{MI} = \mathbf{M}$,

$$\mathrm{f}(\mathbf{M}) = \mathrm{f}(\mathbf{MI}) = \mathrm{f}(\mathbf{M})\,\mathrm{f}(\mathbf{I}).$$

We are told $\mathrm{f}(\mathbf{M}) \neq 0$, so we may divide by it: $\mathrm{f}(\mathbf{I}) = 1$.

(Without that assumption we could not conclude this: the function that is identically zero is multiplicative too.)

Part (ii)

Multiplying out, $\mathbf{J}^2 = \mathbf{I}$, so

$$\mathrm{f}(\mathbf{J})^2 = \mathrm{f}(\mathbf{J}^2) = \mathrm{f}(\mathbf{I}) = 1.$$

Hence $\mathrm{f}(\mathbf{J}) = \pm 1$, and since $\mathrm{f}(\mathbf{J}) \neq 1$ we get $\mathrm{f}(\mathbf{J}) = -1$.

? **Ask yourself** What does multiplying by $\mathbf{J}$ do to a matrix?

Multiplying on the left by $\mathbf{J}$ swaps the rows; multiplying on the right swaps the columns:

$$\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}a&b\\c&d\end{pmatrix}=\begin{pmatrix}c&d\\a&b\end{pmatrix},\qquad \begin{pmatrix}c&d\\a&b\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix}=\begin{pmatrix}d&c\\b&a\end{pmatrix}.$$

Applying f to each and using $\mathrm{f}(\mathbf{J})=-1$,

$$\mathrm{f}\begin{pmatrix}c&d\\a&b\end{pmatrix}=-\mathrm{f}\begin{pmatrix}a&b\\c&d\end{pmatrix},\qquad \mathrm{f}\begin{pmatrix}d&c\\b&a\end{pmatrix}=-\mathrm{f}\begin{pmatrix}c&d\\a&b\end{pmatrix}.$$

The first gives $\mathrm{f}\begin{pmatrix}a&b\\c&d\end{pmatrix}=-\mathrm{f}\begin{pmatrix}c&d\\a&b\end{pmatrix}$, and the second, multiplied by -1, then shows this also equals $\mathrm{f}\begin{pmatrix}d&c\\b&a\end{pmatrix}$, as required.

Part (iii)

Let the first row of $\mathbf{A}$ be $(a\ \ b)$ and the second row λ times it. Let $\mathbf{B}$ be the matrix with two equal rows, and take $k=\lambda$ in $\mathbf{K}$. Multiplying on the left by $\mathbf{K}$ scales the second row by k, so

$$\mathbf{A}=\begin{pmatrix}a&b\\\lambda a&\lambda b\end{pmatrix}=\begin{pmatrix}1&0\\0&\lambda\end{pmatrix}\begin{pmatrix}a&b\\a&b\end{pmatrix}=\mathbf{KB}.$$

Now swapping the rows of $\mathbf{B}$ leaves it unchanged, so by part (ii) (with $c=a$, $d=b$)

$$\mathrm{f}(\mathbf{B})=-\mathrm{f}(\mathbf{B}),$$

which forces $\mathrm{f}(\mathbf{B})=0$. Therefore $\mathrm{f}(\mathbf{A})=\mathrm{f}(\mathbf{K})\,\mathrm{f}(\mathbf{B})=0$. This works for every real λ, including $\lambda=0$.

Part (iv)

First compute the three matrices. For $k\neq 0$, $\mathbf{K}^{-1}=\begin{pmatrix}1&0\\0&1/k\end{pmatrix}$, and

$$\mathbf{P}^2=\begin{pmatrix}1&2\\0&1\end{pmatrix},\qquad \mathbf{P}^{-1}=\begin{pmatrix}1&-1\\0&1\end{pmatrix},\qquad \mathbf{K}^{-1}\mathbf{P}\mathbf{K}=\begin{pmatrix}1&0\\0&1/k\end{pmatrix}\begin{pmatrix}1&k\\0&k\end{pmatrix}=\begin{pmatrix}1&k\\0&1\end{pmatrix}.$$

Next, conjugating by $\mathbf{K}$ does not change f. Indeed $\mathrm{f}(\mathbf{K}^{-1})\mathrm{f}(\mathbf{K})=\mathrm{f}(\mathbf{K}^{-1}\mathbf{K})=\mathrm{f}(\mathbf{I})=1$, so

$$\mathrm{f}(\mathbf{K}^{-1}\mathbf{P}\mathbf{K})=\mathrm{f}(\mathbf{K}^{-1})\,\mathrm{f}(\mathbf{P})\,\mathrm{f}(\mathbf{K})=\mathrm{f}(\mathbf{P}).$$

? **Ask yourself** Which values of k turn $\mathbf{K}^{-1}\mathbf{P}\mathbf{K}$ into $\mathbf{P}^2$ and $\mathbf{P}^{-1}$?

Taking $k=2$ gives $\mathbf{K}^{-1}\mathbf{P}\mathbf{K}=\mathbf{P}^2$, so

$$\mathrm{f}(\mathbf{P})=\mathrm{f}(\mathbf{P}^2)=\mathrm{f}(\mathbf{P})^2,$$

and so $\mathrm{f}(\mathbf{P})=0$ or 1.

Taking $k = -1$ gives $\mathbf{K}^{-1}\mathbf{P}\mathbf{K} = \mathbf{P}^{-1}$, so $\mathrm{f}(\mathbf{P}^{-1}) = \mathrm{f}(\mathbf{P})$. But $\mathrm{f}(\mathbf{P})\,\mathrm{f}(\mathbf{P}^{-1}) = \mathrm{f}(\mathbf{I}) = 1$, so

$$\mathrm{f}(\mathbf{P})^2 = 1,$$

and in particular $\mathrm{f}(\mathbf{P}) \neq 0$. Combining the two results,

$$\mathrm{f}(\mathbf{P}) = 1.$$

Checks

The determinant satisfies every hypothesis: it is multiplicative, $\det \mathbf{I} = 1$ and $\det \mathbf{J} = -1$. It agrees with all our conclusions: swapping rows or columns changes its sign, a matrix with proportional rows has determinant 0, and $\det \mathbf{P} = 1$. A less obvious example is $\mathrm{f}(\mathbf{M}) = \operatorname{sgn}(\det \mathbf{M})\,|\det \mathbf{M}|^{0.7}$. A computer check on thousands of random matrices confirmed the multiplicative rule, the row and column swap identities and $\mathrm{f}(\mathbf{A}) = 0$ for proportional rows, as well as $\mathrm{f}(\mathbf{P}) = 1$, for both functions.

Common slips

- Assuming $\mathrm{f} = \det$. The question asks you to deduce the properties from multiplicativity alone; the determinant is only a guide.
- Using $\mathrm{f}(\mathbf{K}^{-1}) = 1/\mathrm{f}(\mathbf{K})$ without justification. It needs $\mathrm{f}(\mathbf{I}) = 1$ from part (i) and $k \neq 0$, so that $\mathbf{K}$ is invertible.
- Stopping at $\mathrm{f}(\mathbf{P})^2 = \mathrm{f}(\mathbf{P})$ and forgetting to rule out $\mathrm{f}(\mathbf{P}) = 0$.

Beyond STEP

A map such as f that respects multiplication is a *homomorphism* from the group of invertible 2×2 matrices to the non-zero reals. Part (iv) is the start of a classic argument: the shear matrices like $\mathbf{P}$ are products of commutators, so every such homomorphism sends them to 1, and f ends up depending only on the determinant. You can also show that the determinant is the only function that is linear in each row, alternating, and equal to 1 at $\mathbf{I}$. Strang's *Introduction to Linear Algebra* builds the determinant from properties like these, and Allenby's *Rings, Fields and Groups* covers homomorphisms.

WORKED SOLUTION

STEP II 2019, Question 9

Mechanics • Projectile Distance from Origin

STEP II 2019, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 11.10, *A Ball That Comes Back Towards You*: https://beyondstep.keithmcnulty.org/ch11/coming-back.html

The key idea

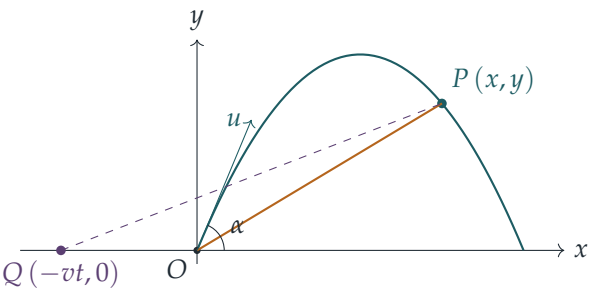
A distance is awkward to differentiate, but its square is not: OP is increasing exactly when $\frac{\mathrm{d}}{\mathrm{d}t}(OP^2) > 0$. Differentiating $x^2 + y^2$ gives t times a quadratic in t, so the whole question becomes "when is this quadratic positive?", which its discriminant answers. Part (ii) is the same calculation with Q's motion added to the horizontal separation.

Solution

Model P and Q as particles, ignore air resistance, and take O as the origin with x horizontal (in the direction of P's motion) and y vertically up. Write $s = \sin\alpha$ and $c = \cos\alpha$, both positive since $0 < \alpha < \frac{1}{2}\pi$. At time t,

$$x = uct, \qquad y = ust - \tfrac{1}{2}gt^2,$$

and P lands when $y = 0$ again, at $T = \dfrac{2us}{g}$.



Part (i)

Let $r = OP$, so $r^2 = x^2 + y^2$ and, using $\dot{x} = uc$ and $\dot{y} = us - gt$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\right) = 2(x\dot{x} + y\dot{y}) &= 2\left[u^2c^2t + \left(ust - \tfrac{1}{2}gt^2\right)(us - gt)\right] \\ &= 2t\left[u^2c^2 + u^2s^2 - \tfrac{3}{2}usgt + \tfrac{1}{2}g^2t^2\right] \\ &= t\left(g^2t^2 - 3usg\,t + 2u^2\right),\end{aligned}$$

using $c^2 + s^2 = 1$. Since $r > 0$ during the flight, $\frac{\mathrm{d}}{\mathrm{d}t}(r^2) = 2r\dot{r}$ has the same sign as $\dot{r}$, and since $t > 0$ the sign is that of

$$f(t) = g^2t^2 - 3usg\,t + 2u^2.$$

This quadratic has positive leading coefficient and discriminant

$$9u^2s^2g^2 - 8u^2g^2 = u^2g^2\left(9s^2 - 8\right).$$

If $s < \frac{2\sqrt{2}}{3}$, then $9s^2 < 8$ (as $s > 0$), so the discriminant is negative and $f(t) > 0$ for every t. Hence $\dot{r} > 0$ throughout the flight: OP is increasing.

If $s > \frac{2\sqrt{2}}{3}$, the discriminant is positive and f has two distinct roots

$$t_{1,2} = \frac{u\left(3s \mp \sqrt{9s^2 - 8}\right)}{2g},$$

with $f(t) < 0$ for $t_1 < t < t_2$. Both roots are positive (their sum $3us/g$ and product $2u^2/g^2$ are positive). We must check that this interval actually occurs while P is in the air.

> **Ask yourself** The quadratic goes negative somewhere, but how do we know that happens before P lands?

Evaluate f at landing:

$$f(T) = g^2 \cdot \frac{4u^2s^2}{g^2} - 3usg \cdot \frac{2us}{g} + 2u^2 = 2u^2(1 - s^2) = 2u^2c^2 > 0.$$

So T is not between the roots. But the vertex of f is at $t = \frac{3us}{2g} < \frac{2us}{g} = T$, so T lies to the right of the vertex; being outside $[t_1, t_2]$, it must satisfy $T > t_2$. Hence $0 < t_1 < t_2 < T$, and for $t_1 < t < t_2$ we have $\dot{r} < 0$: OP is decreasing at some time before P lands, as required.

Part (ii)

Q slides on the smooth ground with constant speed v away from P's side, so at time t it is at $(-vt, 0)$. The horizontal separation of P and Q is $x + vt = wt$, where

$$w = uc + v > 0,$$

and the vertical separation is y. So $QP^2 = w^2t^2 + y^2$, and exactly as before

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(QP^2\right) &= 2\left[w^2t + \left(ust - \tfrac{1}{2}gt^2\right)(us - gt)\right] \\ &= t\left(g^2t^2 - 3usg\,t + 2w^2 + 2u^2s^2\right).\end{aligned}$$

The sign during the flight is that of $h(t) = g^2t^2 - 3usg\,t + 2(w^2 + u^2s^2)$, whose discriminant is

$$9u^2s^2g^2 - 8g^2\left(w^2 + u^2s^2\right) = g^2\left(u^2s^2 - 8w^2\right).$$

This is negative exactly when $8w^2 > u^2s^2$, that is (both w and us being positive) when $2\sqrt{2}\,w > us$. Substituting $w = uc + v$,

$$2\sqrt{2}\,(uc + v) > us \iff 2\sqrt{2}\,v > \left(\sin\alpha - 2\sqrt{2}\cos\alpha\right)u.$$

So under the given condition $h(t) > 0$ for all t, hence $\frac{\mathrm{d}}{\mathrm{d}t}(QP^2) > 0$ for $t > 0$ and QP is increasing throughout the flight of P, as required.

The same argument as in part (i) (now $h(T) = 2w^2 > 0$) shows that when the inequality is reversed strictly, QP does decrease for a while before P lands, so the condition is in fact also necessary, apart from the boundary case of equality.

Checks

- Setting $v = 0$ in part (ii) gives $2\sqrt{2}\cos\alpha > \sin\alpha$, i.e. $\tan\alpha < 2\sqrt{2}$, i.e. $\sin\alpha < \frac{2\sqrt{2}}{3}$: part (i) again, as it should be since Q then stays at O.
- The critical angle is $\arcsin\frac{2\sqrt{2}}{3} \approx 70.5^\circ$, independent of u and g, as dimensional analysis demands (the only dimensionless quantity available is α).
- With $u = 10\,\mathrm{m\,s^{-1}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ and $\sin\alpha = 0.97$, the roots are $t_1 \approx 1.14\,\mathrm{s}$ and $t_2 \approx 1.83\,\mathrm{s}$, both before landing at $T \approx 1.98\,\mathrm{s}$. Stepping through the flight by computer confirms that OP falls between these times, and that across many random u, α, v the distance QP is increasing exactly when the part (ii) inequality holds.

Common slips

- Stopping at "the discriminant is positive" in part (i): you must show the dip in OP happens before P lands, not after.
- Putting Q at $(vt, 0)$: Q moves away from P's side, so its horizontal separation from P is $uct + vt$.
- Forgetting the factor t, or dividing by it without noting $t > 0$.

Beyond STEP

Studying the sign of $\frac{\mathrm{d}}{\mathrm{d}t}|\mathbf{r}|^2 = 2\,\mathbf{r}\cdot\dot{\mathbf{r}}$ is the standard way to ask whether a body is approaching or receding, and it reappears in orbital mechanics and in the relative motion of two bodies, where part (ii) is really a question about the position of P relative to Q. The neat fact that a projectile thrown more steeply than about 70.5° comes back towards its thrower for part of its flight is a favourite of mechanics texts. For more, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 2019, Question 10

Mechanics • Rigid Body Equilibrium

STEP II 2019, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

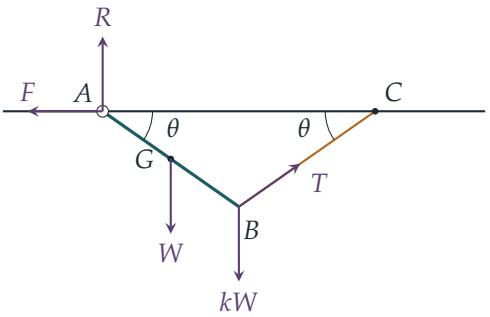
The key idea

Since $AB = BC = 2a$ and A, C both lie on the rail, triangle ABC is isosceles, so the string also makes angle θ with the rail. With that geometry in hand this is a standard rigid-body equilibrium problem: taking moments about A gives the tension without involving the unknown forces at the ring, and resolving then gives the normal reaction and friction. The final part compares two thresholds on k: which one is reached first as k grows?

Solution

Model and forces. The rod is uniform and rigid, so its weight W acts at its midpoint G, with $AG = a$. The ring is light, so the rail's forces on the ring are passed straight on to the rod at A: a normal reaction R vertically upwards and a friction force F along the rail. The string is light and inextensible, with tension T acting at B along BC. The extra load kW acts vertically downwards at B.

In triangle ABC the sides AB and BC are both $2a$, so the base angles at A and C are equal: the string makes angle θ with the rail. Horizontally, B is $2a\cos\theta$ from A, and C is $4a\cos\theta$ from A.



We take F to act along the rail away from C; we shall see that it must, because the string pulls the rod towards C.

Part (i)

Take moments about A, which removes R and F. The weight and the load have moment arms $a\cos\theta$ and $2a\cos\theta$. For the tension, split it at B into $T\cos\theta$ horizontally (towards C) and $T\sin\theta$ vertically upwards. The vertical part has arm $2a\cos\theta$ and the horizontal part has arm $2a\sin\theta$ (the depth of B below the rail), and both turn the rod the same way, anticlockwise in the diagram. So

$$T\sin\theta\cdot 2a\cos\theta + T\cos\theta\cdot 2a\sin\theta = W\cdot a\cos\theta + kW\cdot 2a\cos\theta,$$
$$4aT\sin\theta\cos\theta = aW(1+2k)\cos\theta.$$

Since $\cos\theta \neq 0$ for $0 < \theta < 90^\circ$, we may divide by $a\cos\theta$:

$$T = \frac{(2k+1)W}{4\sin\theta}.$$

This holds whatever the ring does, as long as it does not slip. The string breaks if $T > W$, that is, if $(2k+1)W > 4W\sin\theta$ (multiplying by $4\sin\theta > 0$), which is

$$2k+1 > 4\sin\theta,$$

as required.

Part (ii)

Resolve for the whole rod. Vertically,

$$R + T\sin\theta = W + kW \quad\Longrightarrow\quad R = (1+k)W - \frac{(2k+1)W}{4} = \frac{(2k+3)W}{4}.$$

Horizontally, the only forces are F and the horizontal part of the tension, so

$$F = T\cos\theta = \frac{(2k+1)W\cos\theta}{4\sin\theta}.$$

This is positive, confirming that friction acts away from C. Note also that $R > 0$, so the ring really is pressed onto the rail from above.

Ask yourself When exactly does a ring on a rough rail slip?

Friction can supply at most μR. If equilibrium needs $F > \mu R$, friction cannot hold the ring and it slips. So the ring slips if

$$\frac{(2k+1)W\cos\theta}{4\sin\theta} > \mu\,\frac{(2k+3)W}{4}.$$

Multiplying by $4\sin\theta/(W\cos\theta) > 0$ gives

$$2k+1 > (2k+3)\mu\tan\theta,$$

as required.

Part (iii)

Write $t = \tan\theta$ and $s = \sin\theta$. Each condition says that k has passed a threshold.

The string. By part (i), it breaks once $2k + 1 > 4s$, that is, once

$$k > k_{\text{break}} = \frac{4s - 1}{2}.$$

The ring. Rearranging the condition in part (ii),

$$2k + 1 > (2k + 3)\mu t \iff 2k(1 - \mu t) > 3\mu t - 1.$$

Here we use $\mu t < 1$: dividing by $2(1 - \mu t) > 0$ keeps the inequality the same way round, so the ring slips once

$$k > k_{\text{slip}} = \frac{3\mu t - 1}{2(1 - \mu t)}.$$

(Without $\mu t < 1$ the ring would never slip. If $\mu t \geq 1$ then $(2k + 3)\mu t \geq 2k + 3 > 2k + 1$ for every k. That is why the condition is given.)

As k increases from zero, whichever threshold is lower is crossed first. So the ring slips before the string breaks exactly when $k_{\text{slip}} < k_{\text{break}}$. Multiplying first by $1 - \mu t > 0$,

$$\begin{aligned}
\frac{3\mu t - 1}{1 - \mu t} < 4s - 1 &\iff 3\mu t - 1 < (4s - 1)(1 - \mu t) \\
&\iff 3\mu t - 1 < 4s - 1 - 4s\mu t + \mu t \\
&\iff 2\mu t + 4s\mu t < 4s \\
&\iff \mu t(1 + 2s) < 2s \\
&\iff \mu < \frac{2\sin\theta}{\tan\theta\,(1 + 2\sin\theta)} = \frac{2\cos\theta}{1 + 2\sin\theta},
\end{aligned}$$

dividing by $t(1 + 2s) > 0$ in the last step. This is the required condition.

The rod is in equilibrium to begin with, so at $k = 0$ neither threshold has yet been passed. Both thresholds are crossed only once, because each condition is "k greater than something". So comparing the two thresholds really does decide which happens first.

Checks

- Taking moments about B instead gives the same answer. The forces at A have moment $R \cdot 2a\cos\theta - F \cdot 2a\sin\theta$ about B, and the weight has moment $Wa\cos\theta$. Substituting the values of R and F gives $\frac{1}{2}(2k + 3)Wa\cos\theta - \frac{1}{2}(2k + 1)Wa\cos\theta = Wa\cos\theta$, which balances the weight exactly.
- Take $\theta = 30^\circ$. Then $k_{\text{break}} = \frac{1}{2}$, and the critical friction coefficient is $2\cos 30^\circ / (1 + 2\sin 30^\circ) = \frac{\sqrt{3}}{2} \approx 0.866$, so $\mu t = 0.866 \times 0.577 = \frac{1}{2}$. With this μ, $k_{\text{slip}} = \frac{3/2 - 1}{2 \cdot \frac{1}{2}} = \frac{1}{2}$ as well, so the two thresholds coincide, as they should at the boundary.

- We confirmed by computer that all the forces and moments balance for many random θ and k. We also increased k in small steps for random θ and μ: the ring slipped first exactly when $\mu < 2\cos\theta/(1+2\sin\theta)$.

Common slips

- Forgetting the horizontal part of the tension in the moment about A. It has arm $2a\sin\theta$ and doubles the moment of the tension, which is where the 4 in $4\sin\theta$ comes from.
- Taking $R = (1+k)W$. The string carries part of the weight, so the normal reaction is smaller than the total load.
- Dividing by $1-\mu\tan\theta$ without saying that it is positive. That is exactly what the given condition $\mu\tan\theta < 1$ is for.

Beyond STEP

This is a statically determinate problem: there are three unknowns (T, R, F) and three equilibrium equations for a rigid body in a plane. Part (iii) shows the typical way such a structure fails, by comparing the load at which each constraint gives way. Engineers ask the same question about which member of a structure fails first. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and torque carefully and goes on to rotational dynamics.

WORKED SOLUTION

STEP II 2019, Question 11

Probability and Statistics • Counting Triangles

STEP II 2019, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Three lengths form a triangle of non-zero area exactly when the two shorter ones add up to more than the longest. So everything reduces to counting pairs below a fixed largest value, and the cleanest way to count is to fix the middle number n_2 and ask how many values of n_1 are allowed. Part (i) does the counting; parts (ii) and (iii) just divide by the right number of equally likely choices.

Solution

Part (i)

Write $m = n_3$. Fix $n_2 = k$, with $k \leq m-1$. Then n_1 must satisfy

$$m - k < n_1 < k,$$

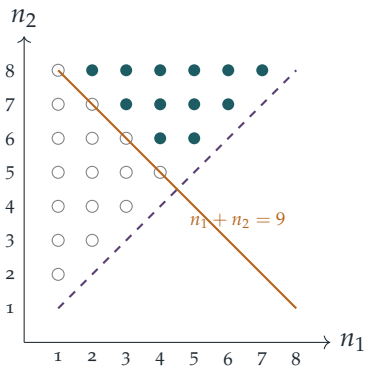
so n_1 runs over $m-k+1, \ldots, k-1$, giving $2k-m-1$ choices whenever this is positive. (The condition $n_1 > 0$ is automatic since $m - k \geq 1$.)

For $m = 9$ the count is $2k-10$, positive for $k = 6, 7, 8$:

$$2 + 4 + 6 = 12.$$

For $m = 10$ the count is $2k-11$, positive for $k = 6, 7, 8, 9$:

$$1 + 3 + 5 + 7 = 16.$$



The figure shows the case $n_3 = 9$: the filled dots lie above the line $n_1 + n_2 = 9$ and above the diagonal $n_1 = n_2$, and there are $2 + 4 + 6 = 12$ of them, one column of the triangle for each n_2.

Odd case, $n_3 = 2n + 1$**.** The count for $n_2 = k$ is $2k - 2n - 2 = 2(k - n - 1)$, positive for $k = n + 2, \ldots, 2n$. Putting $j = k - n - 1$, which runs from 1 to $n - 1$,

$$\sum_{j=1}^{n-1} 2j = (n-1)n = n(n-1).$$

Even case, $n_3 = 2n$**.** The count for $n_2 = k$ is $2k - 2n - 1$, positive for $k = n + 1, \ldots, 2n - 1$. With $j = k - n$ running from 1 to $n - 1$,

$$\sum_{j=1}^{n-1} (2j - 1) = (n-1)^2,$$

the sum of the first $n - 1$ odd numbers. Both formulas agree with $12 = 4 \cdot 3$ and $16 = 4^2$.

Part (ii)

The rod of length N is the longest of the three, so the rods form a triangle of non-zero area if and only if the two chosen lengths $a < b$ satisfy $a + b > N$. (The other two triangle inequalities, such as $a + N > b$, hold automatically because $N > b$.) So the favourable choices are exactly the pairs counted in part (i) with $n_3 = N$.

The two rods are chosen from the $N - 1$ remaining, so there are $\binom{N-1}{2}$ equally likely choices.

For $N = 2n + 1$: there are $\binom{2n}{2} = n(2n - 1)$ choices, so

$$\mathrm{P}(\text{triangle}) = \frac{n(n-1)}{n(2n-1)} = \frac{n-1}{2n-1},$$

as required.

For $N = 2n$: there are $\binom{2n-1}{2} = (2n-1)(n-1)$ choices. For this to be non-zero we need $n \geq 2$ (with $N = 2$ there is only one other rod, so two cannot be chosen). Then

$$\mathrm{P}(\text{triangle}) = \frac{(n-1)^2}{(2n-1)(n-1)} = \frac{n-1}{2n-1}.$$

Rather surprisingly, the answer is the same formula as in the odd case, even though N is one smaller.

Part (iii)

Any three distinct rods have a longest one, of length m say, with $3 \leq m \leq 2M+1$. The choice forms a triangle exactly when the other two lengths form one of the pairs counted in part (i) with $n_3 = m$. Each favourable choice is counted once, under its longest rod, so the number of favourable choices is

$$T = \sum_{m=2}^{2M+1} c(m),$$

where $c(m)$ is the count from part (i). (Including $m = 2$ costs nothing, since $c(2) = 0$.)

Ask yourself How can we add up $c(m)$ when it has a different formula for odd and even m?

Pair them up. For each $n = 1, \ldots, M$ take the even value $m = 2n$ and the odd value $m = 2n+1$ together; together these cover $m = 2, 3, \ldots, 2M+1$ exactly once. Their combined contribution is

$$(n-1)^2 + n(n-1) = (n-1)(2n-1) = 2n^2 - 3n + 1.$$

Using the given formula for $\sum k^2$ and $\sum k = \frac{1}{2}M(M+1)$,

$$\begin{aligned} T &= \sum_{n=1}^{M} \left(2n^2 - 3n + 1\right) \\ &= \tfrac{1}{3}M(M+1)(2M+1) - \tfrac{3}{2}M(M+1) + M \\ &= \tfrac{1}{6}M\left[2(2M^2 + 3M + 1) - 9(M+1) + 6\right] \\ &= \tfrac{1}{6}M\left(4M^2 - 3M - 1\right) \\ &= \tfrac{1}{6}M(4M+1)(M-1). \end{aligned}$$

The total number of equally likely choices of three rods is

$$\binom{2M+1}{3} = \frac{(2M+1)(2M)(2M-1)}{6}.$$

Dividing, and cancelling $M/6$,

$$\mathrm{P}(\text{triangle}) = \frac{M(4M+1)(M-1)}{2M(2M+1)(2M-1)} = \frac{(4M+1)(M-1)}{2(2M+1)(2M-1)},$$

as required.

Checks

- $M = 1$: the rods are $1, 2, 3$ and $1 + 2 = 3$ gives a degenerate triangle, so the probability should be 0, and the formula gives 0.
- $M = 2$: rods 1 to 5. The triangles are $\{2, 3, 4\}$, $\{2, 4, 5\}$ and $\{3, 4, 5\}$, so the probability is $3/10$; the formula gives $\frac{9 \cdot 1}{2 \cdot 5 \cdot 3} = \frac{3}{10}$.
- As $M \to \infty$ the probability tends to $\frac{4}{8} = \frac{1}{2}$. This matches the continuous version: for three independent uniform lengths on $[0, 1]$, the chance that the two smaller sum to more than the largest is $\frac{1}{2}$.
- All the counts in part (i) and the probabilities in parts (ii) and (iii) were confirmed by a computer check that listed every pair or triple directly, for n up to 39 and M up to 19 (for example, $M = 3$ gives $13/35$).

Common slips

- Allowing $n_1 + n_2 = n_3$: three rods with lengths $2, 3, 5$ lie flat and have zero area, so the inequality must be strict.
- In part (ii), using $\binom{N}{2}$ instead of $\binom{N-1}{2}$: the rod of length N has already been taken.
- In part (iii), summing only the odd values of m (or only the even ones): the longest rod can have any length from 3 to $2M + 1$.

Beyond STEP

Splitting a count according to the largest element, and then summing, is a standard technique in combinatorics, and the same idea gives the continuous answer $\frac{1}{2}$ by integrating over the value of the maximum. Counting lattice points in a region, as in the figure, and comparing with its area is the bridge between discrete and continuous probability. Blitzstein and Hwang, *Introduction to Probability*, has many problems of this kind, including the classic "broken stick" triangle problem.

WORKED SOLUTION

STEP II 2019, Question 12

Probability and Statistics • Continuous Distributions and Quantiles

STEP II 2019, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The cumulative distribution function here is simply x^n on $[0,1]$, so every quantile can be written down exactly: the value with a fraction p of the distribution below it is $p^{1/n}$. Comparing the mean with a quantile then becomes a comparison of powers, and raising both sides to the nth power turns it into a statement about $\left(1+\frac{1}{n}\right)^n$, which is exactly what the binomial expansion is built to estimate.

Solution

Part (i)

Integrating against the density,

$$\mu = \mathrm{E}(X) = \int_0^1 x \cdot nx^{n-1}\,\mathrm{d}x = \int_0^1 nx^n\,\mathrm{d}x = \frac{n}{n+1},$$
$$\mathrm{E}(X^2) = \int_0^1 nx^{n+1}\,\mathrm{d}x = \frac{n}{n+2}.$$

Hence

$$\sigma^2 = \mathrm{E}(X^2) - \mu^2 = \frac{n}{n+2} - \frac{n^2}{(n+1)^2} = \frac{n\left[(n+1)^2 - n(n+2)\right]}{(n+1)^2(n+2)}$$
$$= \frac{n\left[n^2+2n+1-n^2-2n\right]}{(n+1)^2(n+2)} = \frac{n}{(n+1)^2(n+2)},$$

as required.

Part (ii)

For $0 \le x \le 1$ the distribution function is

$$\mathrm{F}(x) = \mathrm{P}(X \le x) = \int_0^x nt^{n-1}\,\mathrm{d}t = x^n.$$

With $n = 2$, $\mathrm{F}(x) = x^2$. The lower quartile q_1 satisfies $q_1^2 = \frac{1}{4}$ and the upper quartile q_3 satisfies $q_3^2 = \frac{3}{4}$; both lie in $[0, 1]$, so we take positive roots:

$$q_1 = \tfrac{1}{2}, \qquad q_3 = \tfrac{\sqrt{3}}{2}, \qquad q_3 - q_1 = \frac{\sqrt{3}-1}{2}.$$

From part (i), $\sigma^2 = \dfrac{2}{9 \cdot 4} = \dfrac{1}{18}$, so $\sigma = \dfrac{1}{3\sqrt{2}} = \dfrac{\sqrt{2}}{6}$ and $2\sigma = \dfrac{\sqrt{2}}{3}$.

We want to show $\dfrac{\sqrt{3}-1}{2} < \dfrac{\sqrt{2}}{3}$. Multiplying by 6, this is equivalent to

$$3\sqrt{3} - 3 < 2\sqrt{2} \iff 3\sqrt{3} < 3 + 2\sqrt{2}.$$

Both sides of the last inequality are positive, so it is equivalent to the inequality between their squares:

$$27 < 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2} \iff 10 < 12\sqrt{2},$$

which is true because $\sqrt{2} > 1$ gives $12\sqrt{2} > 12 > 10$. Every step was an equivalence, so working back up the chain, the interquartile range is less than 2σ.

Ask yourself Why is it safe to square both sides here, but not in general?

Squaring preserves an inequality $a < b$ only when a and b are non-negative (for example $-3 < 2$ but $9 > 4$). That is why we first rearranged to $3\sqrt{3} < 3 + 2\sqrt{2}$, where both sides are clearly positive, rather than squaring $3\sqrt{3} - 3 < 2\sqrt{2}$ with a subtraction still inside.

Part (iii)

By the binomial theorem,

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2}\, x^2 + \cdots,$$

and the $(k+1)$th term is

$$\binom{n}{k} x^k = \frac{n(n-1)\cdots(n-k+1)}{k!}\, x^k.$$

Median and lower quartile. Since $\mathrm{F}(x) = x^n$, the median m satisfies $m^n = \frac{1}{2}$ and the lower quartile q satisfies $q^n = \frac{1}{4}$, so

$$m = \left(\tfrac{1}{2}\right)^{1/n}, \qquad q = \left(\tfrac{1}{4}\right)^{1/n}.$$

All of μ, m, q are positive, and $t \mapsto t^n$ is strictly increasing for $t > 0$, so we may compare them by comparing their nth powers. Also

$$\mu^n = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1 + \frac{1}{n}\right)^n}.$$

Write $A = \left(1+\frac{1}{n}\right)^n$. Then

$$\mu < m \iff \mu^n < \tfrac{1}{2} \iff A > 2, \qquad \mu > q \iff \mu^n > \tfrac{1}{4} \iff A < 4.$$

μ is less than the median. Put $x = \frac{1}{n}$ in the expansion. Every term is non-negative, so A is at least the sum of its first three terms:

$$A \geq 1 + n\cdot\frac{1}{n} + \frac{n(n-1)}{2}\cdot\frac{1}{n^2} = 2 + \frac{n-1}{2n} > 2,$$

where the last inequality is strict because $n > 1$. (For $n \geq 2$ the third term really is present in the expansion.) Hence $\mu < m$.

μ is greater than the lower quartile. With $x = \frac{1}{n}$, the $(k+1)$th term is

$$\binom{n}{k}\frac{1}{n^k} = \frac{1}{k!}\cdot\frac{n}{n}\cdot\frac{n-1}{n}\cdots\frac{n-k+1}{n} \leq \frac{1}{k!},$$

since each of the k fractions $\frac{n-j}{n}$ $(0 \leq j \leq k-1)$ lies between 0 and 1. (For $k = 0$ the term is $1 = \frac{1}{0!}$.) Summing over $k = 0, 1, \ldots, n$,

$$A \leq 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} < 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots < 4,$$

the middle inequality holding because the infinite series contains further positive terms, and the last being the result we were allowed to assume. Hence $A < 4$, so $\mu > q$.

Putting the two together, the lower quartile is less than μ, which is less than the median, for every integer $n > 1$.

Ask yourself Why does comparing nth powers settle the comparison of μ with m?

For positive numbers a and b, $a < b$ if and only if $a^n < b^n$, because t^n is strictly increasing on $t > 0$. Our three numbers are all positive, so nothing is lost by raising to the nth power, and doing so removes the awkward nth roots in m and q.

Checks

- For $n = 2$: $\mu = \frac{2}{3}$, and $\sigma^2 = \frac{2}{9\cdot 4} = \frac{1}{18}$, which matches $\mathrm{E}(X^2) - \mu^2 = \frac{1}{2} - \frac{4}{9}$.
- Numerically, for $n = 2$ the interquartile range is 0.366 and $2\sigma = 0.471$.
- For $n = 2$ the three values are $q = 0.5$, $\mu \approx 0.667$, $m \approx 0.707$; for $n = 10$ they are $q \approx 0.871$, $\mu \approx 0.909$, $m \approx 0.933$. The ordering $q < \mu < m$ was confirmed by computer for every n from 2 to 2000.
- As $n \to \infty$, $A = \left(1+\frac{1}{n}\right)^n \to \mathrm{e} \approx 2.718$, which sits comfortably between 2 and 4.

Common slips

- Squaring $3\sqrt{3} - 3 < 2\sqrt{2}$ directly, without first checking that both sides are positive.
- Getting the direction wrong when taking reciprocals: $\mu^n < \frac{1}{2}$ corresponds to $A > 2$, not $A < 2$.
- Claiming the $(k+1)$th term equals $\frac{1}{k!}$ rather than being at most $\frac{1}{k!}$.

Beyond STEP

The quantity $\left(1 + \frac{1}{n}\right)^n$ increases with n to the limit e, and the bound you proved, term by term against $\sum 1/k!$, is the first step in showing that $\mathrm{e} = \sum_{k \geq 0} 1/k!$, a standard result in a first analysis course. The distribution itself is a $\mathrm{Beta}(n, 1)$ distribution; it is also the distribution of the maximum of n independent uniform random variables on $[0, 1]$, since $\mathrm{P}(\max \leq x) = x^n$. For more on both threads, see Blitzstein and Hwang, *Introduction to Probability*, and Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2019, Question 1

Pure • Coupled Differential Equations

STEP III 2019, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

For $k = 1$ the right-hand sides are exactly what you get by multiplying $x + \mathrm{i}y$ by a complex constant, so the pair of equations collapses into a single equation $\dot{z} = cz$ for $z = x + \mathrm{i}y$. That turns the motion into a rotation combined with exponential decay: a spiral. For $k = 0$ the coupling goes one way only: x doesn't depend on y, so we solve for x first and then feed it into the equation for y.

Solution

Part (i)

With $k = 1$ we have $\dot{x} = -x - y$ and $\dot{y} = x - y$. Put $z = x + \mathrm{i}y$. Then

$$\begin{aligned}\dot{z} = \dot{x} + \mathrm{i}\dot{y} &= (-x - y) + \mathrm{i}(x - y) \\ &= (-1 + \mathrm{i})(x + \mathrm{i}y) = (-1 + \mathrm{i})z,\end{aligned}$$

as we can confirm by expanding $(-1 + \mathrm{i})(x + \mathrm{i}y) = -x - \mathrm{i}y + \mathrm{i}x - y$. With $z(0) = 1$, the solution is

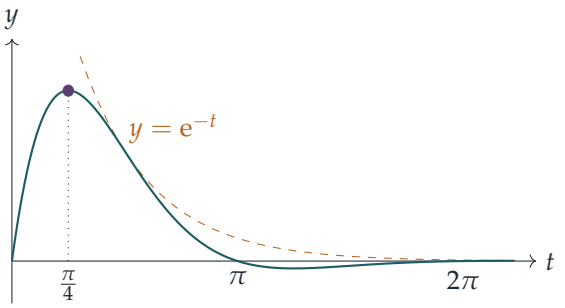
$$z = \mathrm{e}^{(-1+\mathrm{i})t} = \mathrm{e}^{-t}(\cos t + \mathrm{i}\sin t),$$

so

$$x = \mathrm{e}^{-t}\cos t, \qquad y = \mathrm{e}^{-t}\sin t.$$

(If you prefer to avoid complex numbers: differentiate the first equation and eliminate y to get $\ddot{x} + 2\dot{x} + 2x = 0$, with auxiliary roots $-1 \pm \mathrm{i}$, and use $x(0) = 1$, $\dot{x}(0) = -1$.)

Graph of y against t. This is a sine wave inside the envelope $\pm\mathrm{e}^{-t}$, so it's a damped oscillation. It is zero at $t = n\pi$, and its turning points are where $\dot{y} = x - y = \mathrm{e}^{-t}(\cos t - \sin t) = 0$, i.e. $t = \frac{\pi}{4} + n\pi$. Each successive peak is smaller by a factor $\mathrm{e}^{-\pi} \approx 0.04$, so after the first hump the oscillations are barely visible on a true scale.



The path. In polar form, $z = e^{-t}e^{it}$ has modulus $r = e^{-t}$ and argument $\theta = t$. So the path is the spiral

$$r = e^{-\theta}, \qquad \theta \geq 0,$$

starting at $(1, 0)$ and winding anticlockwise into the origin, getting there only as $t \to \infty$. At the start $\dot{x} = -1$ and $\dot{y} = 1$, so it leaves $(1, 0)$ moving up and to the left. Each turn shrinks the distance from the origin by a factor $e^{-2\pi} \approx 0.002$, so on a true scale only the first turn is visible.

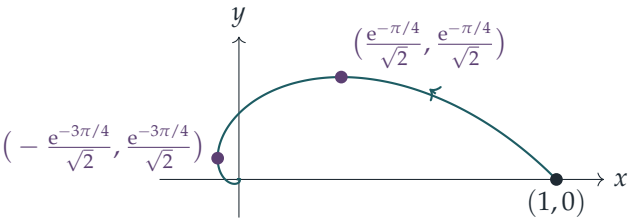
Greatest y. The turning points of y are at $t = \frac{\pi}{4} + n\pi$. At $t = \frac{\pi}{4}$ we get $y = e^{-\pi/4}/\sqrt{2} > 0$. For every $t \geq \frac{\pi}{2}$ we have $y \leq e^{-t} \leq e^{-\pi/2}$, and $e^{-\pi/2} < e^{-\pi/4}/\sqrt{2}$ because this is equivalent to $\sqrt{2} < e^{\pi/4} \approx 2.19$. On $[0, \frac{\pi}{2}]$ the only turning point is $t = \frac{\pi}{4}$. So the greatest value of y is at

$$\left(\frac{e^{-\pi/4}}{\sqrt{2}}, \frac{e^{-\pi/4}}{\sqrt{2}}\right) \approx (0.322,\ 0.322).$$

Least x. Now $\dot{x} = -x - y = -e^{-t}(\cos t + \sin t)$, which is zero when $\tan t = -1$, i.e. $t = \frac{3\pi}{4} + n\pi$. Where $x < 0$ we need $\cos t < 0$, which picks out $t = \frac{3\pi}{4} + 2n\pi$, giving $x = -e^{-t}/\sqrt{2}$; since e^{-t} is decreasing, this is most negative for the smallest such t, namely $t = \frac{3\pi}{4}$. (The least value must occur at one of these turning points, since $x(0) = 1 > 0$ and $x \to 0$ as $t \to \infty$.) So the least value of x is at

$$\left(-\frac{e^{-3\pi/4}}{\sqrt{2}}, \frac{e^{-3\pi/4}}{\sqrt{2}}\right) \approx (-0.067,\ 0.067).$$

Both points lie on the lines $y = \pm x$, as you'd expect: on the spiral, y is greatest where the path is heading in the x-direction, and x is least where it is heading in the y-direction.



Part (ii)

With $k = 0$ the first equation is $\dot{x} = -x$ on its own, so with $x(0) = 1$,

$$x = \mathrm{e}^{-t}.$$

Then $\dot{y} + y = \mathrm{e}^{-t}$. Multiplying by the integrating factor e^{t} gives $\frac{\mathrm{d}}{\mathrm{d}t}(y\mathrm{e}^{t}) = 1$, so $y\mathrm{e}^{t} = t + c$, and $y(0) = 0$ gives $c = 0$:

$$y = t\mathrm{e}^{-t}.$$

Ask yourself How do we turn this into an equation for the path?

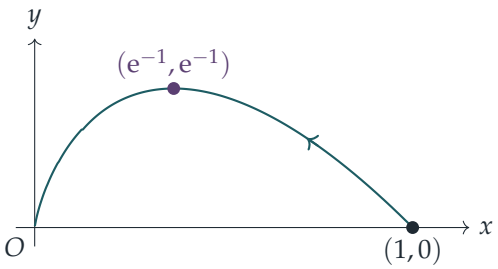
Eliminate t: since $x = \mathrm{e}^{-t}$, we have $t = -\ln x$, and so

$$y = -x \ln x, \qquad 0 < x \leq 1.$$

As t runs from 0 to ∞, x decreases from 1 towards 0, so the particle starts at $(1, 0)$ and moves leftwards along this curve, approaching the origin as $t \to \infty$.

Features for the sketch:

- $\frac{\mathrm{d}y}{\mathrm{d}x} = -\ln x - 1$, which is zero at $x = \mathrm{e}^{-1}$. So the highest point is $(\mathrm{e}^{-1}, \mathrm{e}^{-1})$, reached at $t = 1$.
- At $(1, 0)$ the gradient is -1, matching $\dot{x} = -1$, $\dot{y} = 1$ at $t = 0$.
- As $x \to 0^{+}$, $x \ln x \to 0$, so $y \to 0$; but the gradient $-\ln x - 1 \to +\infty$, so the curve comes into the origin tangent to the y-axis.
- $y > 0$ for $0 < x < 1$, so the whole path lies above the x-axis.



Notice the contrast with part (i): with $k = 0$ there's no longer any rotation, so the particle never crosses an axis. It simply curves round into the origin.

Checks

- Substitute back: for $x = \mathrm{e}^{-t}\cos t$, $\dot{x} = -\mathrm{e}^{-t}\cos t - \mathrm{e}^{-t}\sin t = -x - y$, as required; $\dot{y}$ works the same way. Both solutions also satisfy $x(0) = 1$, $y(0) = 0$.

- Stepping the equations forward numerically in small time steps reproduces both solutions, and confirms the greatest y in part (i) is about 0.3224 at $t \approx 0.785$, the least x is about -0.0670 at $t \approx 2.356$, and in part (ii) y peaks at $1/\mathrm{e} \approx 0.3679$ at $t = 1$.
- Sanity check on (ii): $y = t\mathrm{e}^{-t}$ is the familiar "repeated root" form, which makes sense since the matrix of the system has the repeated eigenvalue -1.

Common slips

- Taking the least value of x at $t = -\frac{\pi}{4}$ or $\frac{7\pi}{4}$: $\dot{x} = 0$ there too, but $x > 0$ there (it's a local maximum). Check the sign of $\cos t$.
- Forgetting that the spiral only reaches the origin as $t \to \infty$, or drawing it winding clockwise: $\theta = t$ increases, so it winds anticlockwise.
- In (ii), missing the vertical tangent at the origin: $y \to 0$ but $\mathrm{d}y/\mathrm{d}x \to \infty$.

Beyond STEP

Any linear system $\dot{\mathbf{x}} = A\mathbf{x}$ is governed by the eigenvalues of A. Here $A = \begin{pmatrix} -1 & -k \\ 1 & -1 \end{pmatrix}$ has eigenvalues $-1 \pm \sqrt{-k}$: for $k = 1$ they are the complex pair $-1 \pm \mathrm{i}$, giving a *stable spiral*; for $k = 0$ there is the repeated eigenvalue -1, giving a *degenerate node*; and for $k < 0$ you'd get two real eigenvalues. Classifying fixed points this way (phase-plane analysis) is a core first-year topic. Boyce and DiPrima, *Elementary Differential Equations*, covers it with plenty of pictures.

WORKED SOLUTION

STEP III 2019, Question 2

Pure • Functional Equations

STEP III 2019, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 3.8, *Functions from Their Addition Laws*: https://beyondstep.keithmcnulty.org/ch03/addition-laws.html

The key idea

A functional equation tells us everything about a function once we know how it behaves near one point. Putting $x = y = 0$ pins down the value at 0; then writing the difference quotient at a general x and using the functional equation turns it into the difference quotient *at* 0, where we know the limit is k. That gives a differential equation, which we solve with the initial value we found.

Solution

Part (i)

The value at 0**.** Put $x = y = 0$ in the functional equation: $\mathrm{f}(0) = \mathrm{f}(0)^2$, so $\mathrm{f}(0) = 0$ or $\mathrm{f}(0) = 1$.

Suppose $\mathrm{f}(0) = 0$. Then putting $y = 0$ gives $\mathrm{f}(x) = \mathrm{f}(x)\mathrm{f}(0) = 0$ for every x, so f is the zero function and $\mathrm{f}'(0) = 0$. This contradicts $\mathrm{f}'(0) = k \neq 0$. Hence $\mathrm{f}(0) = 1$.

The derivative. From $(*)$ and the functional equation with $y = h$,

$$\mathrm{f}'(x) = \lim_{h\to 0} \frac{\mathrm{f}(x)\mathrm{f}(h) - \mathrm{f}(x)}{h} = \mathrm{f}(x) \lim_{h\to 0} \frac{\mathrm{f}(h) - 1}{h}.$$

Since $\mathrm{f}(0) = 1$, the last limit is $\lim_{h\to 0} \dfrac{\mathrm{f}(0+h) - \mathrm{f}(0)}{h}$, which is $\mathrm{f}'(0) = k$ by $(*)$ at $x = 0$. So

$$\mathrm{f}'(x) = k\,\mathrm{f}(x),$$

as required. (We could take $\mathrm{f}(x)$ outside the limit because it does not depend on h.)

Solving. Rather than divide by f (which needs $f \neq 0$), consider $u(x) = e^{-kx} f(x)$. Then

$$u'(x) = e^{-kx}\big(f'(x) - k\,f(x)\big) = 0,$$

so u is constant, equal to $u(0) = f(0) = 1$. Therefore

$$f(x) = e^{kx}.$$

This does satisfy everything: $e^{k(x+y)} = e^{kx}e^{ky}$ and its derivative at 0 is k.

Part (ii)

The value at 0. Put $x = y = 0$:

$$g(0) = \frac{2g(0)}{1+g(0)^2} \quad \Longrightarrow \quad g(0)\big(g(0)^2 - 1\big) = 0,$$

so $g(0) \in \{0, 1, -1\}$. Since $|g(x)| < 1$ for all x, we must have $g(0) = 0$. Notice also that $|g| < 1$ guarantees $1 + g(x)g(y) > 0$, so the functional equation never divides by zero.

The difference quotient. Using the functional equation with $y = h$ and putting everything over one denominator,

$$g(x+h) - g(x) = \frac{g(x) + g(h) - g(x) - g(x)^2 g(h)}{1 + g(x)g(h)} = \frac{g(h)\big(1 - g(x)^2\big)}{1 + g(x)g(h)}.$$

Dividing by h,

$$\frac{g(x+h) - g(x)}{h} = \big(1 - g(x)^2\big) \cdot \frac{g(h)}{h} \cdot \frac{1}{1 + g(x)g(h)}.$$

Ask yourself What happens to each factor on the right as $h \to 0$?

The first factor does not involve h. Since $g(0) = 0$, the middle factor is $\dfrac{g(0+h) - g(0)}{h}$, which tends to $g'(0) = k$. For the last factor we need $g(h) \to g(0) = 0$: this holds because g is differentiable, hence continuous, at 0. So the last factor tends to 1, and

$$g'(x) = k\big(1 - g(x)^2\big).$$

Solving. Because $|g(x)| < 1$, the factor $1 - g(x)^2$ is never zero, so we may separate the variables:

$$\int \frac{dg}{1 - g^2} = \int k\,dx.$$

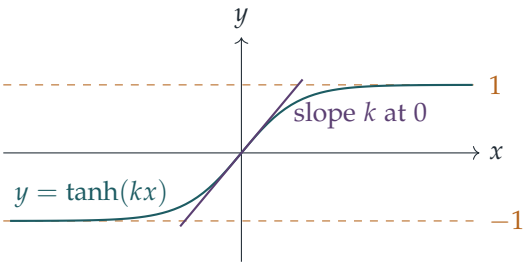
By partial fractions, $\dfrac{1}{1-g^2} = \dfrac{1}{2}\Big(\dfrac{1}{1+g} + \dfrac{1}{1-g}\Big)$, and both $1 \pm g$ are positive, so

$$\frac{1}{2}\ln\frac{1+g(x)}{1-g(x)} = kx + c.$$

At $x = 0$ the left-hand side is $\frac{1}{2}\ln 1 = 0$, so $c = 0$. Hence $\dfrac{1+\mathrm{g}}{1-\mathrm{g}} = \mathrm{e}^{2kx}$, and solving for g,

$$\mathrm{g}(x) = \frac{\mathrm{e}^{2kx}-1}{\mathrm{e}^{2kx}+1} = \tanh(kx).$$

This indeed works: $|\tanh(kx)| < 1$, the addition formula $\tanh(a+b) = \dfrac{\tanh a + \tanh b}{1+\tanh a \tanh b}$ is exactly the functional equation, and the derivative of $\tanh(kx)$ at 0 is k.



Checks

- In (i), $\mathrm{f}(x) = \mathrm{e}^{kx}$ with $k = 1$, $x = 0.3$, $y = 0.5$: $\mathrm{e}^{0.3}\mathrm{e}^{0.5} = 1.3499 \times 1.6487 = 2.2255 = \mathrm{e}^{0.8}$.
- In (ii), with $k = 1$, $x = 0.3$, $y = 0.5$: $\tanh 0.3 = 0.29131$, $\tanh 0.5 = 0.46212$, and $(0.29131 + 0.46212)/(1 + 0.29131 \times 0.46212) = 0.66404 = \tanh 0.8$.
- Both answers were also confirmed by computer for many random k, x, y, including the derivative relations $\mathrm{f}' = k\mathrm{f}$ and $\mathrm{g}' = k(1-\mathrm{g}^2)$.
- The two parts are linked: $\tanh(kx) = \dfrac{\mathrm{e}^{2kx}-1}{\mathrm{e}^{2kx}+1}$ is built from the exponential of part (i), with $2k$ in place of k.

Common slips

- Discarding $\mathrm{f}(0) = 0$ or $\mathrm{g}(0) = \pm 1$ without a reason. The conditions $k \neq 0$ and $|\mathrm{g}| < 1$ are there precisely to rule these out.
- Letting $\mathrm{g}(h) \to 0$ inside the denominator without saying why: it needs continuity of g at 0, which comes from differentiability.
- Dividing by f or $1 - \mathrm{g}^2$ without checking it is non-zero.

Beyond STEP

These are examples of *Cauchy-type functional equations*. A first analysis course shows that the exponential equation $\mathrm{f}(x+y) = \mathrm{f}(x)\mathrm{f}(y)$ forces $\mathrm{f}(x) = \mathrm{e}^{kx}$ (or $\mathrm{f} \equiv 0$) under

assumptions much weaker than differentiability, such as continuity at a single point, while without any regularity there are wild solutions. The tanh equation is the addition law for velocities in special relativity, where $\tanh^{-1}$ of the velocity (the rapidity) simply adds. For the analysis behind this, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2019, Question 3

Pure • Invariant Lines of Matrices

STEP III 2019, Question 3

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

A point (x, y) is invariant exactly when $(\mathbf{A} - \mathbf{I})$ sends it to zero, which gives two linear equations. Everything in parts (i) and (ii) comes from asking when those two equations describe the same line, and the condition for that is $(a-1)(d-1) = bc$, which says $\det(\mathbf{A} - \mathbf{I}) = 0$. In part (iii) we substitute a general point of the line into the transformation and compare coefficients.

Solution

Part (i)

The point (x, y) is invariant when $ax + by = x$ and $cx + dy = y$, that is,

$$(a-1)x = -by, \tag{1}$$
$$(d-1)y = -cx. \tag{2}$$

Multiplying (1) and (2) together gives $(a-1)(d-1)xy = bc\,xy$, so

$$\big((a-1)(d-1) - bc\big)xy = 0$$

for every point (x, y) of L_1, as required.

For the second result, we want a point of L_1 with $xy \neq 0$, so that we can divide by xy.

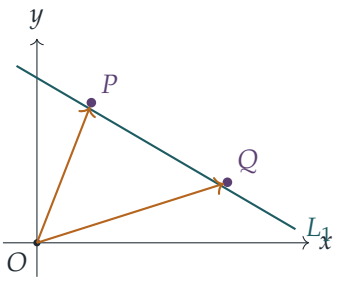
? **Ask yourself** What if every point of L_1 has $x = 0$ or $y = 0$?

The set $xy = 0$ is just the two coordinate axes. A line lying inside the union of the axes has to be one of them, so unless L_1 is an axis it contains a point with $xy \neq 0$. Dividing by xy then gives $(a-1)(d-1) = bc$.

That leaves the axes. If L_1 is the y-axis, every $(0, y)$ is invariant, and (1) and (2) give $by = 0$ and $(d-1)y = 0$ for all y. So $b = 0$ and $d = 1$, and both sides of

$(a-1)(d-1) = bc$ are 0. If L_1 is the x-axis then in the same way $c = 0$ and $a = 1$, and again both sides are 0. So $(a-1)(d-1) = bc$ in every case.

Now suppose L_1 misses the origin. Take two distinct points P and Q on L_1, with position vectors $\mathbf{p}$ and $\mathbf{q}$.



Since O, P and Q do not lie on one line, $\mathbf{p}$ and $\mathbf{q}$ are not parallel, so every vector can be written as $\mathbf{v} = \lambda\mathbf{p} + \mu\mathbf{q}$. By linearity,

$$\mathbf{Av} = \lambda\mathbf{Ap} + \mu\mathbf{Aq} = \lambda\mathbf{p} + \mu\mathbf{q} = \mathbf{v},$$

so every point of the plane is invariant and $\mathbf{A} = \mathbf{I}$. (Then $a = d = 1$ and $b = c = 0$, which agrees with $(a-1)(d-1) = bc$.)

Part (ii)

Suppose $(a-1)(d-1) = bc$. We need a line on which both (1) and (2) hold.

Case $b \neq 0$**.** Take the line $(a-1)x + by = 0$, which is a genuine line through the origin because $b \neq 0$. Every point on it satisfies (1). On this line $y = -\frac{(a-1)x}{b}$, so

$$cx + (d-1)y = cx - \frac{(a-1)(d-1)}{b}\,x = cx - \frac{bc}{b}\,x = 0,$$

and (2) holds too. So $(a-1)x + by = 0$ is a line of invariant points.

Case $b = 0$**.** Now the condition becomes $(a-1)(d-1) = 0$, so $a = 1$ or $d = 1$.

- If $a \neq 1$, then $d = 1$. Equation (1) reads $(a-1)x = 0$, so $x = 0$, and (2) reads $cx = 0$, which holds when $x = 0$. The line of invariant points is the y-axis, $x = 0$.
- If $a = 1$, then (1) holds for every point, and we only need (2): $cx + (d-1)y = 0$.
  - If c and $d-1$ are not both zero, this is a line through the origin, and it is a line of invariant points.
  - If $c = 0$ and $d = 1$, then $\mathbf{A} = \mathbf{I}$, every point is invariant and every line is a line of invariant points.

So whenever $(a-1)(d-1) = bc$ there is a line of invariant points.

Part (iii)

Write L_2 as $y = mx + k$ with $k \neq 0$, since L_2 does not pass through the origin. A general point of L_2 is $(x,\ mx + k)$, and its image is

$$(X, Y) = \big((a + bm)x + bk,\ (c + dm)x + dk\big).$$

The line is invariant exactly when $Y = mX + k$ for every real x:

$$(c + dm)x + dk = m(a + bm)x + mbk + k.$$

Both sides are linear in x and agree for all x, so the coefficients must match:

$$\text{constant terms:} \quad dk = mbk + k, \tag{3}$$

$$\text{coefficients of } x: \quad c + dm = am + bm^2. \tag{4}$$

? **Ask yourself** Where do we use the fact that L_2 misses the origin?

We use it in (3): because $k \neq 0$ we may divide by k, which gives $d - 1 = mb$. Substituting $d = 1 + mb$ into (4),

$$c = am + bm^2 - dm = am + bm^2 - m - bm^2 = m(a - 1).$$

Therefore

$$bc = bm(a - 1) = (d - 1)(a - 1),$$

which is the required result. (If $k = 0$, equation (3) gives nothing, which is why a line through the origin can be invariant without the condition. A rotation through π is an example: it maps every line through O to itself, but $(a - 1)(d - 1) = 4 \neq 0 = bc$.)

Checks

- Try $\mathbf{A} = \begin{pmatrix} 3 & 2 \\ -1 & 0 \end{pmatrix}$: here $(a - 1)(d - 1) = -2 = bc$. Part (ii) predicts the invariant line $2x + 2y = 0$, that is $y = -x$, and indeed $\mathbf{A}(1, -1)^{\mathrm{T}} = (1, -1)^{\mathrm{T}}$.
- For part (iii), take $m = 2$, $k = 1$, $a = 3$, $b = 1$. Then $d = 1 + mb = 3$ and $c = m(a - 1) = 4$. The point $(1, 3)$ on $y = 2x + 1$ maps to $(6, 13)$, and $13 = 2 \times 6 + 1$. Also $(a - 1)(d - 1) = 4 = bc$.
- Thousands of random matrices were confirmed by computer: every invariant line $y = mx + k$ with $k \neq 0$ found had $d - 1 = mb$, $c = m(a - 1)$ and $(a - 1)(d - 1) = bc$, and every line of fixed points missing the origin came from $\mathbf{A} = \mathbf{I}$.

Common slips

- Dividing by xy in part (i) without dealing with the case where L_1 is a coordinate axis.

- Missing a sub-case in part (ii) when $b = 0$, especially $\mathbf{A} = \mathbf{I}$, where every line works.
- In part (iii), confusing an *invariant line* (points stay on the line) with a *line of invariant points* (points stay put), and so setting $X = x$.

Beyond STEP

The condition $(a-1)(d-1) = bc$ is $\det(\mathbf{A} - \mathbf{I}) = 0$, which says that 1 is an eigenvalue of $\mathbf{A}$. A line of invariant points through the origin is then the eigenspace for that eigenvalue. Part (iii) shows that an invariant line missing the origin also forces the eigenvalue 1. The relations $c = m(a-1)$ and $d - 1 = mb$ say that the second row of $\mathbf{A} - \mathbf{I}$ is m times the first, so $\mathbf{A} - \mathbf{I}$ has rank at most 1 and moves every point parallel to L_2. Eigenvalues, eigenvectors and the geometry of linear maps are covered in Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP III 2019, Question 4

Pure • Roots and Coefficients

STEP III 2019, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The alternating signs are exactly the signs in Vieta's formulas: a monic polynomial with roots $r_1, \ldots, r_n$ is $x^n - e_1 x^{n-1} + e_2 x^{n-2} - \cdots + (-1)^n e_n$, where e_k is the sum of all products of k of the roots. So "reflexive" just says that $e_k(a_1, \ldots, a_n) = a_k$ for every k. Small cases are then a few simultaneous equations, and the sum of squares of the roots, $e_1^2 - 2e_2$, gives the identity in (ii). Once $a_n = 0$ is forced, a factor of x comes out and we can peel the polynomial down.

Solution

Throughout, write e_k for the kth elementary symmetric function of $a_1, \ldots, a_n$. By Vieta, P has roots $a_1, \ldots, a_n$ exactly when

$$e_k(a_1, \ldots, a_n) = a_k \qquad \text{for } k = 1, 2, \ldots, n.$$

Part (i)

Degree 1. $\mathrm{P}(x) = x - a_1$ has root a_1, so every such polynomial is reflexive: $\mathrm{P}(x) = x - a$ for any real a.

Degree 2. We need $a_1 + a_2 = a_1$ and $a_1 a_2 = a_2$. The first gives $a_2 = 0$, and then the second holds automatically. So $\mathrm{P}(x) = x^2 - ax = x(x - a)$ for any real a; its roots a and 0 are indeed a_1 and a_2.

Degree 3. We need

$$\begin{aligned} a_1 + a_2 + a_3 &= a_1, \\ a_1 a_2 + a_1 a_3 + a_2 a_3 &= a_2, \\ a_1 a_2 a_3 &= a_3. \end{aligned}$$

The first gives $a_3 = -a_2$. Then the second becomes $a_1(a_2 + a_3) + a_2 a_3 = a_2$, that is $-a_2^2 = a_2$, so $a_2 = 0$ or $a_2 = -1$.

- If $a_2 = 0$ then $a_3 = 0$ and the third equation reads $0 = 0$. So a_1 is arbitrary: $\mathrm{P}(x) = x^3 - ax^2 = x^2(x-a)$.
- If $a_2 = -1$ then $a_3 = 1$ and the third equation gives $-a_1 = 1$, so $a_1 = -1$. This gives

$$\mathrm{P}(x) = x^3 + x^2 - x - 1 = (x+1)^2(x-1),$$

whose roots $-1, -1, 1$ are indeed a_1, a_2, a_3.

So the reflexive polynomials of degree at most 3 are

$$x - a, \qquad x^2 - ax, \qquad x^3 - ax^2 \quad (a \in \mathbb{R}), \qquad \text{and} \qquad x^3 + x^2 - x - 1.$$

Part (ii)

The sum of the squares of the roots is

$$a_1^2 + a_2^2 + \cdots + a_n^2 = e_1^2 - 2e_2,$$

since squaring e_1 gives each a_i^2 once and each product $a_i a_j$ $(i < j)$ twice. For a reflexive polynomial $e_1 = a_1$ and $e_2 = a_2$, so

$$a_1^2 + a_2^2 + \cdots + a_n^2 = a_1^2 - 2a_2, \qquad \text{that is} \qquad 2a_2 = -a_2^2 - a_3^2 - \cdots - a_n^2,$$

as required. (This works for every $n \geq 2$.)

Now suppose the a_k are integers, $a_n \neq 0$ and, for contradiction, $n > 3$. Moving a_2^2 across and completing the square,

$$(a_2 + 1)^2 = 1 - \left(a_3^2 + a_4^2 + \cdots + a_n^2\right).$$

? **Ask yourself** How do integers squeeze this equation?

The left side is at least 0, so $a_3^2 + \cdots + a_n^2 \leq 1$. But a_n is a non-zero integer, so $a_n^2 \geq 1$. Hence the sum is exactly 1: $a_n^2 = 1$ and every other term is zero, so $a_3 = a_4 = \cdots = a_{n-1} = 0$, and also $a_2 = -1$. Since $n \geq 4$, the list $a_3, \ldots, a_{n-1}$ is not empty, so in particular $a_3 = 0$. Now the last Vieta equation says the product of the roots is a_n:

$$a_n = e_n = a_1 a_2 a_3 \cdots a_n = 0,$$

because $a_3 = 0$. This contradicts $a_n \neq 0$. Therefore $n \leq 3$.

Part (iii)

Removing a factor of x. Suppose P is reflexive of degree $n \geq 2$ with $a_n = 0$. Then $\mathrm{P}(x) = x\,\mathrm{Q}(x)$ with

$$\mathrm{Q}(x) = x^{n-1} - a_1 x^{n-2} + \cdots + (-1)^{n-1} a_{n-1}.$$

The roots of P are $a_1, \ldots, a_{n-1}, a_n$ with $a_n = 0$, and the factor x accounts for one root 0. So the roots of Q are the rest, $a_1, \ldots, a_{n-1}$: Q is reflexive, with the same (integer)

coefficients. Conversely, if Q is reflexive of degree m with coefficients $a_1, \ldots, a_m$, then $x\,\mathrm{Q}(x)$ has coefficients $a_1, \ldots, a_m, 0$ and roots $a_1, \ldots, a_m, 0$, so it is reflexive too.

Peeling down. Start with any reflexive P with integer coefficients. While its degree is at least 2 and its last coefficient is 0, remove a factor of x. This stops at $\mathrm{P}(x) = x^m \mathrm{R}(x)$, with $m \geq 0$ and R an integer reflexive polynomial of degree k that either has $k = 1$ or has non-zero last coefficient a_k.

- If $a_k \neq 0$, then $k \leq 3$ by part (ii) (for $k \leq 3$ there is nothing to prove). From the list in part (i): with $k = 1$ we get $x - a$ with $a \neq 0$; with $k = 2$ the last coefficient is always 0, so nothing; with $k = 3$ the only one with $a_3 \neq 0$ is $x^3 + x^2 - x - 1$.
- If $k = 1$ and $a_1 = 0$, then $\mathrm{R}(x) = x$.

Folding $\mathrm{R}(x) = x$ into the case $x - a$ with $a = 0$, and using the converse above to see that every multiple by x^m is reflexive, the reflexive polynomials with integer coefficients are exactly

$$x^m(x - a) \quad (m \geq 0,\ a \in \mathbb{Z}),$$
$$x^m(x^3 + x^2 - x - 1) = x^m(x + 1)^2(x - 1) \quad (m \geq 0).$$

Checks

- $x^2(x+1)^2(x-1) = x^5 + x^4 - x^3 - x^2$ has $a_1, \ldots, a_5 = -1, -1, 1, 0, 0$ and roots $-1, -1, 1, 0, 0$: it is reflexive. The identity in (ii) reads $2(-1) = -(1 + 1 + 0 + 0)$, which holds.
- A computer search over every integer choice of $a_1, \ldots, a_n$ in $\{-3, \ldots, 3\}$ with $n \leq 7$ found no reflexive polynomials other than those listed; a search over small rationals for $n = 2, 3$ confirmed the list in part (i).

Common slips

- Forgetting the families with a free parameter: $x^2 - ax$ and $x^3 - ax^2$ are reflexive for every real a, not just for special values.
- In (ii), finding $a_n = \pm 1$, $a_2 = -1$ and stopping. The contradiction comes from the product of the roots, which vanishes because some $a_3, \ldots, a_{n-1}$ is zero when $n \geq 4$.
- In (iii), saying "$a_n = 0$ so divide by x" without checking that the quotient is again reflexive: one zero root goes with the factor x, and the remaining roots match the remaining coefficients.

Beyond STEP

The equations $e_k(a_1, \ldots, a_n) = a_k$ are a small example of a polynomial system whose solutions form an algebraic variety, here a union of lines and isolated points. The fact

that "sum of squares $= e_1^2 - 2e_2$" is the first of Newton's identities, which link power sums of the roots to the coefficients and underlie the theory of symmetric functions. The step "integers that are squares and sum to at most 1" is a typical move in Diophantine arguments. Davenport's *The Higher Arithmetic* and Allenby's *Rings, Fields and Groups* are good places to explore these ideas further.

WORKED SOLUTION

STEP III 2019, Question 5

Pure • Integration by Substitution

STEP III 2019, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The function in (i) maps $[0,\infty)$ smoothly and increasingly onto $[0,1)$. So scaling it by a constant turns a y-integral over part of $[0,c)$ into an x-integral over part of $[0,\infty)$. It also makes c^2-y^2 a perfect square in disguise, which removes the awkward square root. Once the root has gone we're left with a rational integrand, and we can do that with the arctangent integral given in the note. The reciprocal substitution $y=1/u$ links the later integrals back to the first one.

Solution

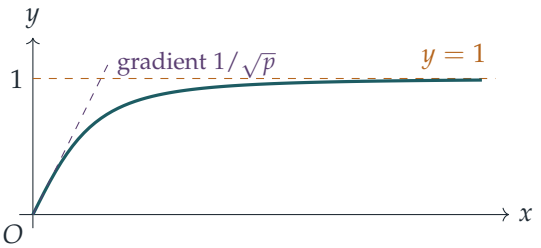
Part (i)

With $f(x)=x(x^2+p)^{-1/2}$ and $p>0$, we have $f(0)=0$ and

$$f'(x)=(x^2+p)^{-1/2}-x^2(x^2+p)^{-3/2}=\frac{p}{(x^2+p)^{3/2}}>0,$$

$$f''(x)=-\frac{3px}{(x^2+p)^{5/2}}<0 \quad \text{for } x>0.$$

So the curve leaves the origin with gradient $1/\sqrt{p}$ and rises steadily, bending downwards (it's concave). Since $f(x)=1/\sqrt{1+p/x^2}$ for $x>0$, we get $f(x)\to 1$ from below as $x\to\infty$, so $y=1$ is an asymptote that the curve never reaches.



Part (ii)

As printed, the substitution has b in the numerator. That doesn't work: it makes $b^2 - y^2$ neat but leaves the square root of $c^2(x^2+p) - b^2x^2$, which isn't a perfect square. The substitution that does work (and the one the examiners intended) has c in the numerator:

$$y = \frac{cx}{\sqrt{x^2+p}}, \qquad p > 0.$$

By part (i), this is an increasing map from $x \in [0, \infty)$ onto $y \in [0, c)$, which is the range where $\sqrt{c^2 - y^2}$ is defined. Now

$$\begin{aligned} c^2 - y^2 &= \frac{c^2(x^2+p) - c^2x^2}{x^2+p} = \frac{c^2 p}{x^2+p}, \\ b^2 - y^2 &= \frac{b^2 p + (b^2 - c^2)x^2}{x^2+p}, \\ \frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{cp}{(x^2+p)^{3/2}} \quad \text{(from part (i)).} \end{aligned}$$

Since $c > 0$ and $p > 0$, $\sqrt{c^2 - y^2} = c\sqrt{p}/\sqrt{x^2+p}$. Substituting,

$$I = \int \frac{x^2+p}{b^2 p + (b^2-c^2)x^2} \cdot \frac{\sqrt{x^2+p}}{c\sqrt{p}} \cdot \frac{cp}{(x^2+p)^{3/2}} \,\mathrm{d}x = \int \frac{\sqrt{p}}{b^2 p + (b^2-c^2)x^2} \,\mathrm{d}x.$$

Choosing $p = 1$ gives

$$I = \int \frac{1}{b^2 + (b^2-c^2)x^2} \,\mathrm{d}x,$$

which is the required result. The substitution is $y = cx/\sqrt{x^2+1}$, and its inverse is $x = y/\sqrt{c^2 - y^2}$.

The first integral. Take $b = \sqrt{3}$ and $c = \sqrt{2}$. Then $x = y/\sqrt{2-y^2}$, so $y = 1$ gives $x = 1$, and $x \to \infty$ as $y \to \sqrt{2}$. The original integrand blows up at $y = \sqrt{2}$ like $(2-y^2)^{-1/2}$, which is integrable. After the substitution the integral is a convergent improper integral on $[1, \infty)$:

$$\begin{aligned} \int_1^{\sqrt{2}} \frac{\mathrm{d}y}{(3-y^2)\sqrt{2-y^2}} &= \int_1^{\infty} \frac{\mathrm{d}x}{3+x^2} = \frac{1}{\sqrt{3}} \Big[\tan^{-1} \frac{x}{\sqrt{3}} \Big]_1^{\infty} \\ &= \frac{1}{\sqrt{3}} \Big(\frac{\pi}{2} - \frac{\pi}{6} \Big) = \frac{\pi}{3\sqrt{3}}. \end{aligned}$$

Ask yourself The new integral has $3y^2 - 1$ and $2y^2 - 1$ where the old one had $3 - y^2$ and $2 - y^2$. What substitution swaps the roles of y^2 and the constant?

The second integral. Put $y = 1/u$ with $u > 0$, so that $\mathrm{d}y = -\mathrm{d}u/u^2$. Then

$$3y^2 - 1 = \frac{3 - u^2}{u^2}, \qquad \sqrt{2y^2 - 1} = \frac{\sqrt{2 - u^2}}{u},$$

and the limits $y = \frac{1}{\sqrt{2}}$ and $y = 1$ become $u = \sqrt{2}$ and $u = 1$. So

$$\int_{1/\sqrt{2}}^{1} \frac{y\,\mathrm{d}y}{(3y^2 - 1)\sqrt{2y^2 - 1}} = \int_{\sqrt{2}}^{1} \frac{1}{u} \cdot \frac{u^2}{3 - u^2} \cdot \frac{u}{\sqrt{2 - u^2}} \cdot \left(-\frac{\mathrm{d}u}{u^2}\right) = \int_{1}^{\sqrt{2}} \frac{\mathrm{d}u}{(3 - u^2)\sqrt{2 - u^2}}.$$

This is exactly the first integral, so its value is also $\dfrac{\pi}{3\sqrt{3}}$.

Part (iii)

The same reciprocal substitution $y = 1/u$ turns the integrand into $u/\big((3 - u^2)\sqrt{2 - u^2}\big)$. The extra factor of u now invites $w = \sqrt{2 - u^2}$, since then $3 - u^2 = 1 + w^2$. Combining the two steps gives a single substitution:

$$w = \frac{\sqrt{2y^2 - 1}}{y} = \sqrt{2 - \frac{1}{y^2}}, \qquad \text{so} \qquad w^2 = 2 - \frac{1}{y^2}, \quad 2w\,\mathrm{d}w = \frac{2}{y^3}\,\mathrm{d}y.$$

For $y \in [\frac{1}{\sqrt{2}}, 1]$, w increases from 0 to 1. Also $\mathrm{d}y = wy^3\,\mathrm{d}w$, $\sqrt{2y^2 - 1} = wy$, and $3y^2 - 1 = y^2\big(3 - \frac{1}{y^2}\big) = y^2(1 + w^2)$. Hence

$$\int_{1/\sqrt{2}}^{1} \frac{\mathrm{d}y}{(3y^2 - 1)\sqrt{2y^2 - 1}} = \int_0^1 \frac{wy^3\,\mathrm{d}w}{y^2(1 + w^2) \cdot wy} = \int_0^1 \frac{\mathrm{d}w}{1 + w^2} = \frac{\pi}{4}.$$

(The singularity at $y = 1/\sqrt{2}$ is again only of the integrable type $(2y^2 - 1)^{-1/2}$, and the substitution removes it.)

This is the part (ii) method in disguise. Writing $x = 1/w$ gives $y = x/\sqrt{2(x^2 - 1)}$, which is the general substitution with $c^2 = \frac{1}{2}$ and $p = -1$. That's why part (i) allowed p to be negative.

Checks

- $\pi/(3\sqrt{3}) \approx 0.6046$ and $\pi/4 \approx 0.7854$. Both values were confirmed by numerical integration of the original integrals, after a trigonometric substitution to smooth out the endpoint singularities.
- Rough size check for the first integral: on $[1, \sqrt{2}]$ we have $1 \le 3 - y^2 \le 2$, so the integral lies between $\frac{1}{2}$ and 1 times $\int_1^{\sqrt{2}} (2 - y^2)^{-1/2}\,\mathrm{d}y = \frac{\pi}{4}$. That gives a range of 0.39 to 0.79, and 0.6046 sits comfortably inside it.
- Check the part (ii) identity at one point: with $b = \sqrt{3}$, $c = \sqrt{2}$ and $x = 1$ we get $y = 1$. The integrand is $\frac{1}{2}$, $\mathrm{d}y/\mathrm{d}x = \sqrt{2}/2^{3/2} = \frac{1}{2}$, and their product is $\frac{1}{4} = 1/(3 + 1)$, as expected.

Common slips

- Persisting with the printed b in the substitution and getting stuck: the square root you need to remove is $\sqrt{c^2 - y^2}$, so c must be the scale factor.
- Converting the limits wrongly. Here $y = \sqrt{2} = c$ corresponds to $x = \infty$, not to a finite value of x.
- Dropping the minus sign from $\mathrm{d}y = -\mathrm{d}u/u^2$, or forgetting to swap the limits back, which gives a negative answer for a positive integrand.

Beyond STEP

The substitutions here are rationalising substitutions. They turn an integrand that involves the square root of a quadratic into a rational function, and partial fractions or $\tan^{-1}$ then finish the job. In first-year analysis you meet the theory behind the improper integrals used here: when an integral with an infinite limit or an unbounded integrand converges, and why substitution is still valid for such integrals. Bartle and Sherbert's *Introduction to Real Analysis* covers both carefully.

WORKED SOLUTION

STEP III 2019, Question 6

Pure • Circles in the Argand Diagram

STEP III 2019, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The opening expression is $|z-a|^2 - r^2$ in disguise, so the locus is the set of points at distance r from a. For the image under $\omega = 1/z$, substitute $z = 1/\omega$, clear the fractions, and then rearrange the result back into the *same* shape as the original equation. Once it's in that shape, we can read the centre and radius straight off by matching coefficients. Comparing centres and radii then settles when C' and C are the same circle.

Solution

Since $(z-a)^* = z^* - a^*$,

$$|z-a|^2 = (z-a)(z^*-a^*) = zz^* - a^*z - az^* + aa^*,$$

so the given equation says exactly that $|z-a|^2 = r^2$, that is, $|z-a| = r$ (as $r > 0$). Hence C is the circle with **centre** a **and radius** r.

More generally, an equation $ww^* - bw^* - b^*w + bb^* - s^2 = 0$ with $s > 0$ is the circle with centre b and radius s. We'll use this as a template.

Part (i)

First, $z = 0$ is not on C: that would need $|a| = r$, i.e. $a^*a = r^2$, which is excluded. So $\omega = 1/z$ is defined for every point of C, and $\omega \neq 0$. Also $z = 1/\omega$, so Q lies on C' exactly when $\omega \neq 0$ and

$$\left|\frac{1}{\omega} - a\right| = r \iff |1 - a\omega| = r|\omega| \iff (1-a\omega)(1-a^*\omega^*) = r^2\omega\omega^*.$$

Expanding and collecting terms,

$$(a^*a - r^2)\,\omega\omega^* - a\omega - a^*\omega^* + 1 = 0.$$

Write $k = a^*a - r^2 = |a|^2 - r^2$, which is a non-zero real number by hypothesis. Dividing by k,

$$\omega\omega^* - \frac{a^*}{k}\,\omega^* - \frac{a}{k}\,\omega + \frac{1}{k} = 0.$$

This matches the template with $b = a^*/k$ (so $b^* = a/k$, since k is real), provided $bb^* - s^2 = 1/k$, i.e.

$$s^2 = \frac{|a|^2}{k^2} - \frac{1}{k} = \frac{|a|^2 - k}{k^2} = \frac{r^2}{k^2}.$$

This is positive, so C' is a genuine circle, with

$$\text{centre } \frac{a^*}{|a|^2 - r^2}, \qquad \text{radius } \frac{r}{\big||a|^2 - r^2\big|}.$$

(Note that $\omega = 0$ doesn't satisfy the equation, since it would give $1 = 0$. So every point of this circle really is $1/z$ for some z on C.)

Ask yourself Two circles are the same exactly when…?

They are the same exactly when they have the same centre and the same radius. So $C' = C$ requires

$$\frac{r}{|k|} = r \qquad \text{and} \qquad \frac{a^*}{k} = a.$$

The first gives $|k| = 1$, i.e. $k^2 = 1$:

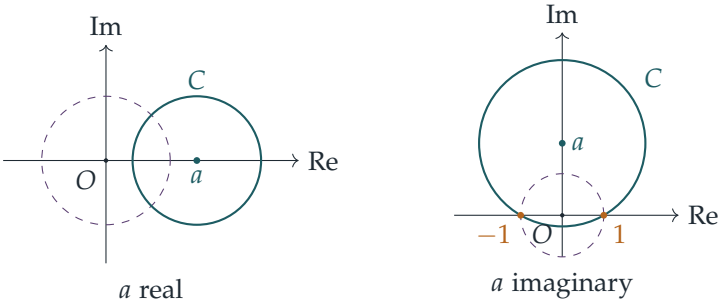
$$\big(|a|^2 - r^2\big)^2 = 1.$$

So $k = 1$ or $k = -1$, and then the second condition gives:

- if $k = 1$: $a^* = a$, so a is **real**;
- if $k = -1$: $a^* = -a$, so a is **imaginary** (purely imaginary).

(If $a = 0$ it is both, and then $r = 1$.) Conversely, either of these, together with the radius condition, does make the centres and radii agree.

Sketches. If a is real, then $|a|^2 = r^2 + 1 > r^2$, so the centre is on the real axis at distance $\sqrt{r^2+1}$ from O, and the origin lies *outside* C. The circle meets the real axis at $a - r$ and $a + r$, whose product is $a^2 - r^2 = 1$, as it must be, since $1/z$ swaps them. If a is imaginary, then $|a|^2 = r^2 - 1$, so this case needs $r \geq 1$. The centre is on the imaginary axis and the origin lies *inside* C. Writing $a = \mathrm{i}c$, the real-axis crossings satisfy $x^2 + c^2 = r^2$, so $x = \pm 1$. In both cases C crosses the unit circle (dashed) at right angles, because $|a|^2 = r^2 \pm 1$ (the left picture has $a = \sqrt{2}$, $r = 1$; the right has $a = \mathrm{i}\sqrt{3}$, $r = 2$).



If a is real it may of course be negative, giving the mirror image of the left picture in the imaginary axis; similarly, an imaginary a may lie below the real axis.

Part (ii)

Now $z = 1/\omega^*$, so Q is on the locus exactly when

$$\left|\frac{1}{\omega^*} - a\right| = r \iff |1 - a\omega^*| = r|\omega^*| \iff |1 - a^*\omega| = r|\omega|,$$

taking the conjugate inside the first modulus (a number and its conjugate have the same modulus). This is the equation of part (i) with a replaced by a^*. So, by part (i), the locus is the circle with

$$\text{centre } \frac{a}{k}, \qquad \text{radius } \frac{r}{|k|}, \qquad k = |a|^2 - r^2.$$

For this to be C we need $|k| = 1$ and $a/k = a$. If $a \neq 0$, the second condition forces $k = 1$, and that's all: the conditions are exactly

$$|a|^2 = r^2 + 1,$$

which says nothing about the argument of a. (If $a = 0$ we need $r = 1$, as before.)

? **Ask yourself** Can you find a specific a that is neither real nor imaginary?

Take $a = 1 + \mathrm{i}$ and $r = 1$. Then $|a|^2 = 2 = r^2 + 1$, so the locus of Q is C, but a is neither real nor purely imaginary. So the answer is **no**: it is not the case that a must be real or imaginary.

Checks

- For $a = \sqrt{5}$, $r = 2$ (so $|a|^2 - r^2 = 1$), the points $z = \sqrt{5} \pm 2$ on C go to $1/(\sqrt{5} \pm 2) = \sqrt{5} \mp 2$, which are again on C.
- In part (ii) with $a = 1 + \mathrm{i}$, $r = 1$: the point $z = 1$ is on C, and $1/z^* = 1$ is too. The point $z = 2 + \mathrm{i}$ gives $1/z^* = (2 + \mathrm{i})/5 = 0.4 + 0.2\mathrm{i}$, and $|(0.4 + 0.2\mathrm{i}) - (1 + \mathrm{i})|^2 = 0.36 + 0.64 = 1$. But $1/z = 0.4 - 0.2\mathrm{i}$ is not on C, so this circle is not fixed by the map in part (i).

- The image circles in both parts were also confirmed by computer, by mapping many points of random circles and measuring their distances from the predicted centres.

Common slips

- Writing the centre of C' as $a/(|a|^2 - r^2)$ instead of $a^*/(|a|^2 - r^2)$. Reading off the centre needs the coefficient of ω^*, not of ω.
- Forgetting the modulus in the radius $r/\left||a|^2 - r^2\right|$. Without it, the case $|a| < r$ gives a negative radius.
- Answering "yes" in part (ii) by analogy with part (i), without noticing that the conjugate removes the condition $a^* = \pm a$.

Beyond STEP

The map $z \mapsto 1/z^*$ is *inversion in the unit circle*, and $z \mapsto 1/z$ is that inversion followed by reflection in the real axis. Both belong to the family of Möbius transformations, which always send circles and lines to circles and lines. Part (ii) is the classical fact that a circle is fixed by inversion exactly when it cuts the circle of inversion at right angles, i.e. $|a|^2 = r^2 + 1$ (or is the unit circle itself). Needham's *Visual Complex Analysis* develops these ideas beautifully, with pictures.

WORKED SOLUTION

STEP III 2019, Question 7

Pure • Curve Sketching

STEP III 2019, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The equation contains only x^2 and y^2, so it is a quadratic in x^2 (or in y^2). That gives two tools at once: a discriminant, which tells us which values of y the curve can reach, and an explicit formula for x^2 in terms of y. For the shape near the origin we keep only the lowest-degree terms; far away we keep only the highest-degree terms. In the special case $a = b$ the equation factorises completely.

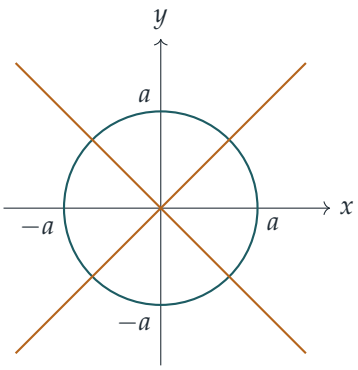
Solution

Part (i)

With $a = b$, move everything to one side and group the quartic and quadratic terms:

$$\begin{aligned} 0 &= (y^4 - x^4) - a^2(y^2 - x^2) \\ &= (y^2 - x^2)(y^2 + x^2) - a^2(y^2 - x^2) \\ &= (y^2 - x^2)(x^2 + y^2 - a^2). \end{aligned}$$

So a point lies on the curve exactly when $y = \pm x$ or $x^2 + y^2 = a^2$: the Devil's Curve is the pair of lines $y = x$ and $y = -x$ together with the circle of radius a centred at the origin.



Part (ii)(a)

Now $a = 2$, $b = \sqrt{5}$, and the curve is $y^4 - 5y^2 = x^4 - 4x^2$. Fix $y \geq 0$ and write $X = x^2$. A point (x, y) with $x \geq 0$ lies on the curve exactly when $X \geq 0$ satisfies

$$X^2 - 4X - (y^4 - 5y^2) = 0.$$

For this to have a real root, the discriminant must be non-negative:

$$\begin{aligned} 16 + 4(y^4 - 5y^2) &= 4(y^4 - 5y^2 + 4) = 4(y^2 - 1)(y^2 - 4) \\ &= 4(y+1)(y+2)(y-1)(y-2) \geq 0. \end{aligned}$$

Since $y \geq 0$, the factors $y + 1$ and $y + 2$ are positive, so we need $(y - 1)(y - 2) \geq 0$, that is,

$$0 \leq y \leq 1 \quad \text{or} \quad y \geq 2.$$

Conversely, every such y does occur. The roots are

$$X = 2 \pm \sqrt{D}, \qquad D = (y^2 - 1)(y^2 - 4),$$

and the larger one is at least 2, so it is a valid x^2. We will use this formula for x^2 below.

Part (ii)(b)

Close to the origin, the quartic terms y^4 and x^4 are tiny compared with y^2 and x^2. So the curve is approximately

$$-5y^2 \approx -4x^2, \qquad \text{that is} \qquad y \approx \tfrac{2}{\sqrt{5}}\, x \quad (x, y \geq 0).$$

The curve leaves the origin along the line of gradient $2/\sqrt{5} \approx 0.89$, a little below the line $y = x$. (The formula from (a) agrees: on the root $X = 2 - \sqrt{D}$, a binomial expansion gives $x^2 = \frac{5}{4}y^2 + O(y^4)$.)

Very far from the origin, the highest-degree terms dominate: $y^4 \approx x^4$, so $y \approx x$. We can do better. Write the equation as

$$(y^2 - x^2)(y^2 + x^2) = 5y^2 - 4x^2 \quad \Longrightarrow \quad y^2 - x^2 = \frac{5y^2 - 4x^2}{y^2 + x^2}.$$

Since $y/x \to 1$, the right-hand side tends to $\frac{5-4}{2} = \frac{1}{2}$. So $y^2 \approx x^2 + \frac{1}{2}$, which gives

$$y \approx x + \frac{1}{4x}.$$

The curve has the line $y = x$ as an asymptote and approaches it from *above*. In fact it never meets that line away from the origin: putting $y = x$ in the equation gives $x^2 = 0$.

Part (ii)(c)

Differentiating $y^4 - 5y^2 = x^4 - 4x^2$ implicitly with respect to x,

$$(4y^3 - 10y)\frac{\mathrm{d}y}{\mathrm{d}x} = 4x^3 - 8x \qquad \Longrightarrow \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2x(x^2-2)}{y(2y^2-5)}.$$

Tangent parallel to the x-axis: we need the numerator to be zero and the denominator not. So $x = 0$ or $x = \sqrt{2}$.

- $x = 0$: then $y^4 = 5y^2$, so $y = 0$ or $y = \sqrt{5}$. At $(0, \sqrt{5})$ the denominator is $\sqrt{5}(10 - 5) \neq 0$, so this point counts.
- $x = \sqrt{2}$: then $y^4 - 5y^2 = 4 - 8$, so $y^4 - 5y^2 + 4 = 0$, which gives $y = 1$ or $y = 2$. The denominators there are -3 and 6, both non-zero.

? **Ask yourself** Why don't we count the origin, where the numerator is also zero?

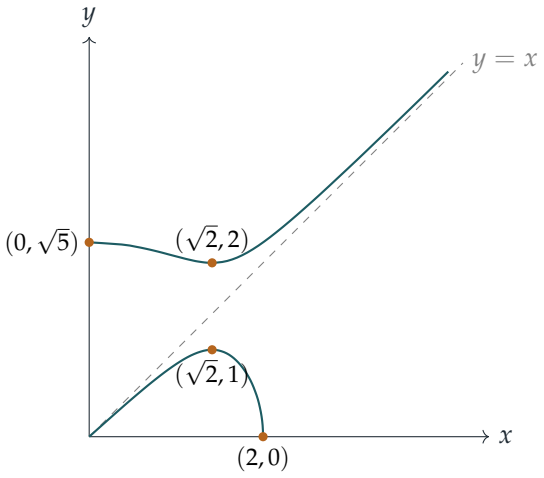
At $(0, 0)$ the denominator vanishes too, so the formula gives $0/0$ and tells us nothing. Part (b) has already settled the question: the curve passes through the origin with gradient $2/\sqrt{5}$, not 0.

So the tangent is horizontal at $(0, \sqrt{5})$, $(\sqrt{2}, 1)$ and $(\sqrt{2}, 2)$.

Tangent parallel to the y-axis: we need the denominator to be zero and the numerator not, so $y = 0$ or $y^2 = \frac{5}{2}$. But $\sqrt{5/2} \approx 1.58$ lies strictly between 1 and 2, which part (a) rules out. With $y = 0$ we get $x^4 = 4x^2$, so $x = 0$ (the origin again, excluded as above) or $x = 2$. At $(2, 0)$ the numerator is $2 \cdot 2 \cdot 2 = 8 \neq 0$. So the only such point is $(2, 0)$.

The sketch for $x \geq 0, y \geq 0$ Using $x^2 = 2 \pm \sqrt{D}$, there are two pieces.

- **A loop for $0 \leq y \leq 1$.** The root $2 - \sqrt{D}$ runs from 0 (at $y = 0$) up to 2 (at $y = 1$), and $2 + \sqrt{D}$ runs from 4 down to 2. So the curve leaves the origin with gradient $2/\sqrt{5}$, rises to a maximum at $(\sqrt{2}, 1)$ and comes down to meet the x-axis vertically at $(2, 0)$.
- **A wing for $y \geq 2$.** When $2 \leq y \leq \sqrt{5}$, the root $2 - \sqrt{D}$ runs from 2 down to 0; beyond $y = \sqrt{5}$ it is negative, so it is not a valid x^2. The root $2 + \sqrt{D}$ increases from 2 without bound. So this branch starts at $(0, \sqrt{5})$ with a horizontal tangent, dips to a minimum at $(\sqrt{2}, 2)$, and then rises, approaching $y = x$ from above.

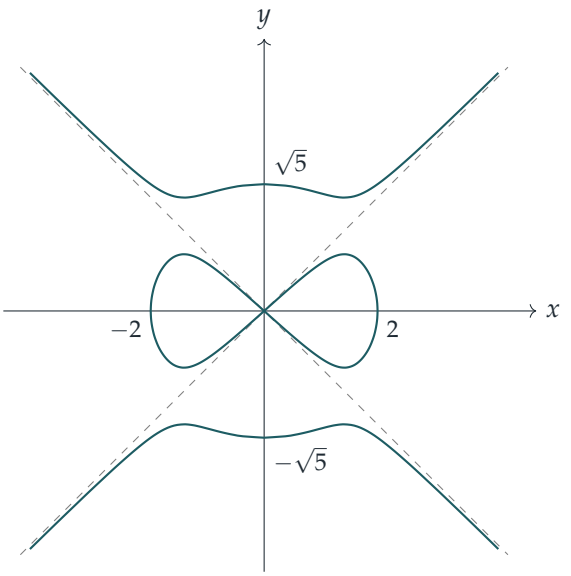


Part (iii)

The equation involves only x^2 and y^2, so the curve is symmetric in the y-axis (replace x by $-x$) and in the x-axis (replace y by $-y$). Reflecting the quarter-curve in both axes:

- The loop becomes a figure-of-eight between $x = -2$ and $x = 2$, with $|y| \leq 1$. Its two lobes cross at the origin along the lines $y = \pm\frac{2}{\sqrt{5}}x$, with turning points at $(\pm\sqrt{2}, \pm 1)$.
- The wing becomes two branches, one with $y \geq 2$ and one with $y \leq -2$. Each has a local maximum or minimum at $(0, \pm\sqrt{5})$ and turning points at $(\pm\sqrt{2}, \pm 2)$, and both approach the asymptotes $y = \pm x$ from outside.

No part of the curve lies in the strip $1 < |y| < 2$.



Checks

- Each marked point satisfies the equation. For example, at $(\sqrt{2}, 2)$ both sides equal -4, and at $(0, \sqrt{5})$ both sides equal 0.
- The point $(1, 1.5)$ is in the forbidden strip. There the left-hand side is $2.25 \times (-2.75) \approx -6.19$, while the right-hand side $x^2(x^2-4)$ can never be less than -4 (its minimum, at $x^2 = 2$). This is the discriminant argument in miniature.
- On the wing at $y = 100$, the formula gives $x^2 = 2 + \sqrt{(10^4-1)(10^4-4)}$. Then $4x(y-x) \approx 1.0002$, matching $y - x \approx \frac{1}{4x}$. Near the origin, the ratio y/x on the loop tends to $0.8944 \approx 2/\sqrt{5}$.

Common slips

- Listing $(0, 0)$ as a point with a horizontal (or vertical) tangent. There $\mathrm{d}y/\mathrm{d}x$ is $0/0$, and the true gradient comes from (b).
- Drawing the loop leaving the origin along $y = x$, or the wing crossing $y = x$. The loop starts at gradient $2/\sqrt{5} < 1$, and the wing stays strictly above $y = x$.
- Forgetting the condition $x^2 \geq 0$, which cuts the inner root off at $y = \sqrt{5}$.

Beyond STEP

The Devil's Curve is an *algebraic curve*: the zero set of a polynomial $F(x, y)$. The origin is a *singular point*, where both partial derivatives F_x and F_y vanish. The lowest-degree terms of F give the tangent lines there, which is exactly what we did in (b), and the

highest-degree terms give the directions of the asymptotes. At university these ideas reappear in the implicit function theorem and in algebraic geometry. Siklos, *Advanced Problems in Mathematics*, has more STEP-style curve-sketching problems with careful commentary.

WORKED SOLUTION

STEP III 2019, Question 8

Pure • Vectors and 3D Geometry

STEP III 2019, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

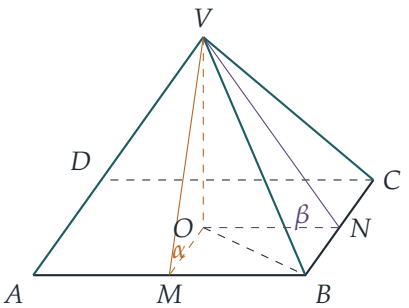
Put coordinates on the pyramid. Each sloping face contains a base edge, which is horizontal, so its normal is perpendicular to that edge. That forces the normal into a vertical plane, where the angle between the face and the base can be read off directly. Then the angle between two faces comes from a dot product of normals, and the rest is careful trigonometric algebra plus one use of AM–GM.

Solution

Let the centre of the base be the origin O, with $\mathbf{i}$ along AB, $\mathbf{j}$ along BC and $\mathbf{k}$ up. Take

$$A=(-a,-b,0),\quad B=(a,-b,0),\quad C=(a,b,0),\quad D=(-a,b,0),\quad V=(0,0,h),$$

with $a,b,h>0$.



Let $M=(0,-b,0)$ be the midpoint of AB and $N=(a,0,0)$ the midpoint of BC. Since $VA=VB$, the line VM lies in face AVB and is perpendicular to AB; so is OM in the base. So the angle between the face and the base is angle VMO, and

$$\tan\alpha=\frac{h}{b},\qquad\text{and similarly}\qquad\tan\beta=\frac{h}{a}.$$

Part (i)

Face AVB contains the directions $\overrightarrow{AB} \parallel \mathbf{i}$ and $\overrightarrow{MV} = b\,\mathbf{j} + h\,\mathbf{k}$. A normal is

$$\mathbf{i} \times (b\,\mathbf{j} + h\,\mathbf{k}) = -h\,\mathbf{j} + b\,\mathbf{k}.$$

Dividing by its length $\sqrt{h^2 + b^2}$ and using $\tan\alpha = h/b$ gives the unit normal

$$\mathbf{n}_1 = -\sin\alpha\,\mathbf{j} + \cos\alpha\,\mathbf{k}.$$

Its $\mathbf{j}$-component is negative, so it points out of the pyramid (face AVB is on the side $y < 0$).

In exactly the same way, face BVC contains $\mathbf{j}$ and $\overrightarrow{NV} = -a\,\mathbf{i} + h\,\mathbf{k}$, and its outward unit normal is

$$\mathbf{n}_2 = \sin\beta\,\mathbf{i} + \cos\beta\,\mathbf{k}.$$

Hence

$$\mathbf{n}_1 \cdot \mathbf{n}_2 = \cos\alpha\cos\beta.$$

Ask yourself Is the angle between the normals the angle between the faces, or its supplement?

Look at a cross-section perpendicular to the common edge VB. The two faces appear as two lines meeting at the interior angle $\pi - \theta$, and each outward normal is perpendicular to its line and points away from the solid. Turning one face onto the other through the interior angle turns the outward normal through the same angle, but in the cross-section the two normals both point *outwards*, so they make the angle $\pi - (\pi - \theta) = \theta$. (Picture a cube: faces at $\frac{\pi}{2}$, normals at $\frac{\pi}{2}$; flatten the pyramid and the faces approach angle π while the normals approach angle 0.) Therefore

$$\cos\theta = \mathbf{n}_1 \cdot \mathbf{n}_2 = \cos\alpha\cos\beta,$$

as required. Since $0 < \cos\alpha\cos\beta < 1$, θ is acute, consistent with the faces meeting at an obtuse angle.

Part (ii)

The foot of the perpendicular from V is O, so the angle between BV and the base is angle VBO, with

$$\cot\phi = \frac{OB}{h} = \frac{\sqrt{a^2 + b^2}}{h}.$$

Therefore

$$\cot^2\phi = \frac{a^2}{h^2} + \frac{b^2}{h^2} = \cot^2\beta + \cot^2\alpha,$$

using $\cot\beta = a/h$ and $\cot\alpha = b/h$. This is the first result.

For the second, write $p = \cos^2\alpha$, $q = \cos^2\beta$, so that $\cos^2\theta = pq$ by part (i). Then

$$\cot^2\phi = \frac{p}{1-p} + \frac{q}{1-q} = \frac{p + q - 2pq}{(1-p)(1-q)}.$$

Since $\cos^2\phi = \dfrac{\cot^2\phi}{1+\cot^2\phi}$, we need

$$1+\cot^2\phi = \frac{(1-p)(1-q)+p+q-2pq}{(1-p)(1-q)} = \frac{1-pq}{(1-p)(1-q)},$$

and dividing,

$$\cos^2\phi = \frac{p+q-2pq}{1-pq} = \frac{\cos^2\alpha+\cos^2\beta-2\cos^2\theta}{1-\cos^2\theta}.$$

(The denominator is positive since $\cos\theta < 1$.)

Now $\cos\alpha$ and $\cos\beta$ are positive, so by the AM–GM inequality

$$\cos^2\alpha+\cos^2\beta \;\geq\; 2\cos\alpha\cos\beta = 2\cos\theta,$$

and since $1-\cos^2\theta > 0$,

$$\cos^2\phi \;\geq\; \frac{2\cos\theta-2\cos^2\theta}{1-\cos^2\theta}.$$

? **Ask yourself** How does this lower bound tell us anything about ϕ and θ?

Simplify it and compare it with $\cos^2\theta$. Cancelling the positive factor $1-\cos\theta$,

$$\frac{2\cos\theta(1-\cos\theta)}{(1-\cos\theta)(1+\cos\theta)} = \frac{2\cos\theta}{1+\cos\theta}.$$

Writing $c = \cos\theta \in (0,1)$,

$$\frac{2c}{1+c} - c^2 = \frac{c\,(2-c-c^2)}{1+c} = \frac{c\,(1-c)(2+c)}{1+c} > 0.$$

So $\cos^2\phi \geq \dfrac{2c}{1+c} > \cos^2\theta$. Both ϕ and θ lie in $(0, \frac{\pi}{2})$, where cosine is positive and decreasing, so $\cos\phi > \cos\theta$ gives $\phi < \theta$.

Checks

- Square base with h equal to half a side ($a = b = h$): $\alpha = \beta = \frac{\pi}{4}$, so $\cos\theta = \frac{1}{2}$ and $\theta = 60^\circ$. Directly, $\cot\phi = \sqrt{2}$, so $\cot^2\phi = 2 = 1+1$ and $\phi \approx 35.26^\circ < 60^\circ$. Also $\cos^2\phi = \frac{2}{3}$, matching $(\frac{1}{2}+\frac{1}{2}-\frac{1}{2})/(1-\frac{1}{4}) = \frac{2}{3}$.
- A computer check over 10^5 random pyramids, measuring the angle between the faces directly from a cross-section through VB, agreed with $\cos\theta = \cos\alpha\cos\beta$ and with both formulas for ϕ, and always gave $\phi < \theta$.

Common slips

- Taking an inward normal for one face and an outward normal for the other: the dot product then has the wrong sign and gives $\cos(\pi-\theta)$.

- Measuring α along the edge VA rather than along the slant height VM; the angle between planes must be measured perpendicular to their line of intersection.
- Stopping at $\cos^2\phi \geq \dfrac{2\cos\theta}{1+\cos\theta}$ without showing that this exceeds $\cos^2\theta$.

Beyond STEP

The angle between two planes equals the angle between their normals (up to taking the supplement), which is the start of how normals are used to describe surfaces in multivariable calculus and differential geometry. The identity $\cos\theta = \cos\alpha\cos\beta$ is also a disguised form of the spherical Pythagoras theorem: the three normals $\mathbf{k}$, $\mathbf{n}_1$, $\mathbf{n}_2$ form a right-angled spherical triangle. Strang's *Introduction to Linear Algebra* is a good place to deepen your feel for dot and cross products.

WORKED SOLUTION

STEP III 2019, Question 9

Mechanics • Particle on Moving Hemisphere

STEP III 2019, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

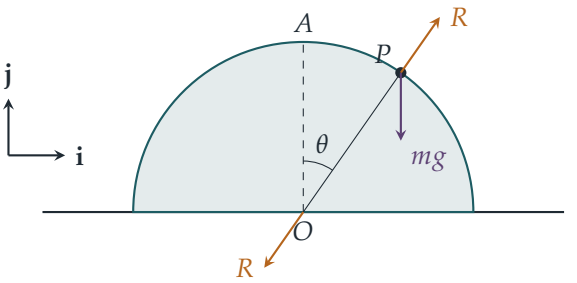
www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Both surfaces are smooth, so no external force on the system (particle plus hemisphere) has a horizontal component: horizontal momentum is conserved, and that ties the hemisphere's speed to the particle's angular speed. Since nothing does work except gravity, energy is conserved too, and those two laws give part (ii). For leaving the surface, the clean test is that the normal reaction becomes zero; the suggested vector is chosen so that the unknown $\ddot{\theta}$ disappears when we take a component.

Solution

Model: P is a particle, the hemisphere is rigid, and all contacts are smooth. While they touch, the only force between P and the hemisphere is a normal reaction R along OP, and the table pushes the hemisphere vertically.



The reaction on the hemisphere (drawn at O for clarity) is equal and opposite to the one on P.

Part (i)

At time t the centre O has moved to $-s\,\mathbf{i}$, and P is at distance a from it in direction $\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$. So

$$\mathbf{r} = (a\sin\theta - s)\,\mathbf{i} + a\cos\theta\,\mathbf{j},$$

and differentiating with respect to t (chain rule on $\sin\theta$ and $\cos\theta$) gives

$$\dot{\mathbf{r}} = (a\dot{\theta}\cos\theta - \dot{s})\,\mathbf{i} - a\dot{\theta}\sin\theta\,\mathbf{j},$$

as required.

The external forces on the system are the two weights and the table's reaction, all vertical, so the total horizontal momentum stays at its initial value, which is zero (we take the small initial displacement to give negligible speed). The hemisphere's velocity is $-\dot{s}\,\mathbf{i}$, so

$$m(a\dot{\theta}\cos\theta - \dot{s}) - M\dot{s} = 0 \quad\Longrightarrow\quad \dot{s} = \frac{m}{m+M}\,a\dot{\theta}\cos\theta = (1-k)\,a\dot{\theta}\cos\theta,$$

because $1 - k = \dfrac{m}{m+M}$. Substituting into the $\mathbf{i}$-component of $\dot{\mathbf{r}}$,

$$a\dot{\theta}\cos\theta - (1-k)a\dot{\theta}\cos\theta = k\,a\dot{\theta}\cos\theta, \qquad \text{so} \qquad \dot{\mathbf{r}} = a\dot{\theta}\,(k\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}).$$

Part (ii)

The reactions do no net work: the table's reaction is perpendicular to the hemisphere's motion, and the pair of forces $\pm R$ between P and the hemisphere do no total work because P stays on the surface (their relative velocity is along the surface, perpendicular to R). So kinetic plus potential energy is conserved. P has dropped by $a(1-\cos\theta)$, and initially everything is (essentially) at rest, so

$$\tfrac{1}{2}ma^2\dot{\theta}^2\left(k^2\cos^2\theta + \sin^2\theta\right) + \tfrac{1}{2}M\dot{s}^2 = mga(1-\cos\theta).$$

Using part (i), $M\dot{s}^2 = M(1-k)^2a^2\dot{\theta}^2\cos^2\theta$. The total coefficient of $a^2\dot{\theta}^2\cos^2\theta$ is then

$$mk^2 + M(1-k)^2 = \frac{mM^2 + Mm^2}{(m+M)^2} = \frac{mM}{m+M} = mk.$$

Hence $\tfrac{1}{2}ma^2\dot{\theta}^2\left(k\cos^2\theta + \sin^2\theta\right) = mga(1-\cos\theta)$, and dividing by $\tfrac{1}{2}ma$,

$$a\dot{\theta}^2\left(k\cos^2\theta + \sin^2\theta\right) = 2g(1-\cos\theta).$$

Part (iii)

Write $\mathbf{u} = \sin\theta\,\mathbf{i} + k\cos\theta\,\mathbf{j}$. Differentiating the last result of part (i),

$$\ddot{\mathbf{r}} = a\ddot{\theta}\,(k\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}) - a\dot{\theta}^2\,(k\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}).$$

The first bracket is perpendicular to $\mathbf{u}$, since $k\cos\theta\sin\theta - \sin\theta\,k\cos\theta = 0$. So

$$\ddot{\mathbf{r}}\cdot\mathbf{u} = -a\dot{\theta}^2\left(k\sin^2\theta + k\cos^2\theta\right) = -k\,a\dot{\theta}^2.$$

Ask yourself Why is this particular vector useful?

It kills the $\ddot{\theta}$ term, which we know nothing about, leaving only $\dot{\theta}^2$, which part (ii) gives us. Now compare with Newton's second law for P:

$$m\ddot{\mathbf{r}} = -mg\,\mathbf{j} + R(\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}).$$

Taking the component along $\mathbf{u}$ and equating with the result above,

$$-mk\,a\dot{\theta}^2 = -mgk\cos\theta + R\left(\sin^2\theta + k\cos^2\theta\right),$$

so

$$R\left(\sin^2\theta + k\cos^2\theta\right) = mk\left(g\cos\theta - a\dot{\theta}^2\right).$$

The particle leaves the hemisphere when R falls to zero. Since $k > 0$, at $\theta = \alpha$ this gives

$$a\dot{\theta}^2 = g\cos\alpha.$$

The cubic. Put $c = \cos\alpha$ and use part (ii) at $\theta = \alpha$, with $\sin^2\alpha = 1 - c^2$:

$$gc\left(kc^2 + 1 - c^2\right) = 2g(1 - c) \quad\Longrightarrow\quad (1-k)\,c^3 - 3c + 2 = 0.$$

Let $f(x) = (1-k)x^3 - 3x + 2$, where $0 < 1 - k < 1$ because m and M are both positive.

- $f'(x) = 3\left((1-k)x^2 - 1\right) < 0$ for $-1 \le x \le 1$, so f is strictly decreasing there and has at most one root in $[-1, 1]$.
- $f\left(\frac{2}{3}\right) = \frac{8}{27}(1-k) - 2 + 2 = \frac{8}{27}(1-k) > 0$ and $f(1) = -k < 0$.

So the only root in $[-1, 1]$ lies strictly between $\frac{2}{3}$ and 1. As $\cos\alpha$ must lie in $[-1, 1]$, it is this root, and

$$\cos\alpha > \tfrac{2}{3}.$$

(This is consistent with the formula for R: substituting part (ii) shows $g\cos\theta - a\dot{\theta}^2$ has the sign of $-f(\cos\theta)$, which is positive at $\theta = 0$ and first vanishes at this root, so P really does stay in contact until then.)

Checks

- If $M \to \infty$ then $k \to 1$, the hemisphere stays put, and the cubic becomes $-3c + 2 = 0$: the familiar $\cos\alpha = \frac{2}{3}$ for a particle sliding off a fixed sphere. The moving hemisphere makes the particle leave earlier, at a larger $\cos\alpha$.
- For equal masses, $1 - k = \frac{1}{2}$ and the cubic $c^3 - 6c + 4 = 0$ factorises as $(c-2)(c^2 + 2c - 2) = 0$, giving $\cos\alpha = \sqrt{3} - 1 \approx 0.732$, which exceeds $\frac{2}{3}$.
- A step-by-step numerical simulation of the full motion by computer, for several mass ratios, confirmed the relations in parts (i) and (ii) along the way and found the reaction vanishing at $\cos\theta \approx 0.732$ when $m = M$, matching the cubic.

Common slips

- Using the fixed-sphere condition $a\dot{\theta}^2 = g\cos\theta$ without justification: here the hemisphere accelerates, so the particle's acceleration is not just circular motion about a fixed point. It happens to hold at the instant of leaving only because $R = 0$ means the hemisphere has no horizontal acceleration at that moment.
- Forgetting the hemisphere's kinetic energy $\frac{1}{2}M\dot{s}^2$ in the energy equation.
- Showing only that a root lies between $\frac{2}{3}$ and 1, without ruling out another root in $[-1, 1]$.

Beyond STEP

This is a two-degree-of-freedom system, and Lagrangian mechanics handles it systematically: write the kinetic and potential energies in terms of θ and s, and the conservation of horizontal momentum appears automatically because s does not appear in the Lagrangian (an "ignorable" coordinate, linked to a symmetry by Noether's theorem). Constraint forces such as R are found afterwards, just as we did here. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to build the Newtonian intuition first.

WORKED SOLUTION

STEP III 2019, Question 10

Mechanics • Oblique Collisions

STEP III 2019, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

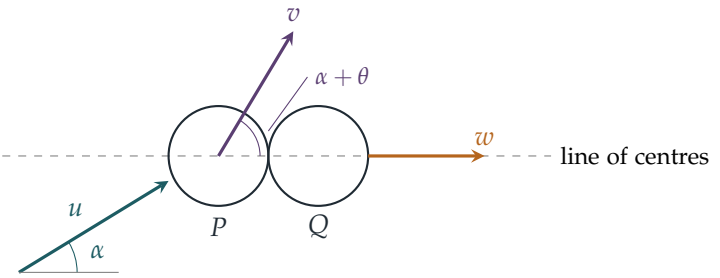
www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Split every velocity into a component along the line of centres and a component perpendicular to it. Because the spheres are smooth, the impulse acts only along the line of centres, so P's perpendicular component is unchanged and Q moves off along that line. Conservation of momentum and Newton's experimental law then apply along the line of centres just as in a one-dimensional collision. The rest is trigonometry.

Solution

Model P and Q as smooth spheres of equal mass m on a smooth horizontal table. Take the line of centres at impact as the x-direction, pointing from P to Q, and let y be perpendicular to it in the plane of motion, on the side of P's initial velocity.



Part (i)

Before the collision P has components $u\cos\alpha$ along the line of centres and $u\sin\alpha$ perpendicular to it.

The contact force between smooth spheres acts along the line of centres. So there is no impulse on P perpendicular to that line, and P's perpendicular component is unchanged. Also, Q starts at rest and receives an impulse only along the line of centres,

so it moves off along that line with speed w.

> **Ask yourself** Why does P's new direction make the angle $\alpha + \theta$ with the line of centres, and not $\alpha - \theta$?

The impulse on P points backwards along the line of centres. It reduces P's x-component, from $u\cos\alpha$ to the value v_x found below, and leaves the y-component alone. The velocity therefore swings away from the line of centres, and the angle it makes with that line grows from α to $\alpha + \theta$. We'll confirm below that $v_x > 0$ when $e < 1$, so $0 < \alpha + \theta < \frac{\pi}{2}$.

After the collision, P's components are $v\cos(\alpha+\theta)$ along the line of centres and $v\sin(\alpha+\theta)$ perpendicular to it. Equating the perpendicular components gives

$$u\sin\alpha = v\sin(\alpha+\theta),$$

as required.

Along the line of centres, conservation of momentum (the masses are equal, so we cancel m) gives

$$u\cos\alpha = v\cos(\alpha+\theta) + w. \tag{1}$$

Substituting $u = v\sin(\alpha+\theta)/\sin\alpha$ from the first result,

$$\begin{aligned} w &= \frac{v\sin(\alpha+\theta)\cos\alpha}{\sin\alpha} - v\cos(\alpha+\theta) \\ &= \frac{v\big[\sin(\alpha+\theta)\cos\alpha - \cos(\alpha+\theta)\sin\alpha\big]}{\sin\alpha} = \frac{v\sin\theta}{\sin\alpha}, \end{aligned}$$

using the expansion of $\sin\big((\alpha+\theta)-\alpha\big)$. So

$$w = \frac{v\sin\theta}{\sin\alpha}.$$

This has a neat interpretation. Vector momentum conservation says $\mathbf{u} = \mathbf{v} + \mathbf{w}$, and resolving perpendicular to $\mathbf{u}$ gives $0 = v\sin\theta - w\sin\alpha$.

Part (ii)

Newton's experimental law along the line of centres says that the speed of separation is e times the speed of approach:

$$w - v\cos(\alpha+\theta) = e\,u\cos\alpha. \tag{2}$$

Substitute $w = v\sin\theta/\sin\alpha$ and $u = v\sin(\alpha+\theta)/\sin\alpha$ from part (i):

$$\frac{v\sin\theta}{\sin\alpha} - v\cos(\alpha+\theta) = \frac{e\,v\sin(\alpha+\theta)\cos\alpha}{\sin\alpha}.$$

Multiplying through by $\sin\alpha/v$ (allowed since $v > 0$ and $\sin\alpha \neq 0$) and rearranging gives

$$\sin\theta = \cos(\theta+\alpha)\sin\alpha + e\sin(\theta+\alpha)\cos\alpha,$$

as required.

To find $\tan\theta$, expand both compound angles:

$$\sin\theta = (\cos\theta\cos\alpha - \sin\theta\sin\alpha)\sin\alpha + e(\sin\theta\cos\alpha + \cos\theta\sin\alpha)\cos\alpha.$$

Collecting the $\sin\theta$ terms on the left:

$$\sin\theta\left(1 + \sin^2\alpha - e\cos^2\alpha\right) = (1+e)\cos\theta\sin\alpha\cos\alpha.$$

Divide by $\cos\theta\cos^2\alpha$ and write $t = \tan\alpha$, using $1/\cos^2\alpha = 1 + t^2$:

$$\tan\theta\left(1 + t^2 + t^2 - e\right) = (1+e)\,t, \qquad \text{so} \qquad \tan\theta = \frac{(1+e)\tan\alpha}{1 - e + 2\tan^2\alpha}.$$

The denominator is positive because $e < 1$, so $\tan\theta > 0$ and θ really is an acute deflection.

The maximum of $\tan\theta$

As α ranges over acute angles, $t = \tan\alpha$ takes every value in $(0, \infty)$. For $t > 0$ we can divide top and bottom by t:

$$\tan\theta = \frac{1+e}{\dfrac{1-e}{t} + 2t}.$$

Ask yourself How do we maximise this without calculus?

Make the denominator as small as possible. Both $(1-e)/t$ and $2t$ are positive, so by the AM–GM inequality

$$\frac{1-e}{t} + 2t \geq 2\sqrt{\frac{1-e}{t}\cdot 2t} = 2\sqrt{2(1-e)},$$

with equality exactly when $(1-e)/t = 2t$, that is, when $\tan\alpha = \sqrt{(1-e)/2}$. This is a genuine acute angle, so the minimum is attained. Therefore the maximum value of $\tan\theta$ is

$$\tan\theta_{\max} = \frac{1+e}{2\sqrt{2(1-e)}}.$$

(Differentiating with respect to t gives the same stationary point, $2t^2 = 1 - e$.)

Checks

- Elastic-like limit $e \to 1$: the spheres exchange their components along the line of centres, so P keeps only $u\sin\alpha$ and moves at right angles to the line of centres. Then $\alpha + \theta = \frac{\pi}{2}$, and $\tan\theta = \cot\alpha$. The formula gives $\tan\theta = 2t/(2t^2) = 1/t$, which agrees. The maximum blows up, as it should, because $\theta \to \frac{\pi}{2}$ when $\alpha \to 0$.

- With $e = \frac{1}{2}$ the maximum is $\frac{3/2}{2\sqrt{1}} = \frac{3}{4}$, attained at $\tan\alpha = \frac{1}{2}$ (about $26.6°$). Scanning α finely confirms 0.75. A computer simulation of many random collisions also reproduced all four results of the question.
- Small α (nearly head-on): $\tan\theta \approx \frac{1+e}{1-e}\alpha \to 0$, so a head-on collision causes no deflection, as expected.

Common slips

- Applying Newton's experimental law to whole speeds instead of to the components along the line of centres. Restitution acts only in the direction of the impulse.
- Writing P's new angle as $\alpha - \theta$. The impulse removes some of the component along the line of centres, so P turns away from that line.
- Forgetting that $e < 1$ is what makes $1 - e + 2\tan^2\alpha$ positive, and that $\tan\alpha$ ranges over all of $(0, \infty)$, so the AM–GM equality case is actually achieved.

Beyond STEP

Oblique collisions of smooth spheres are the simplest example of scattering. In this problem the same two conservation laws are applied along and across a line, and that idea scales up to particle physics and the kinetic theory of gases. For elastic collisions of equal masses the result is the famous right angle between the outgoing velocities. In the centre-of-mass frame every collision just rotates (and, here, shrinks) the relative velocity, which is a slicker way to see all of these results. Kleppner and Kolenkow's *An Introduction to Mechanics* develops collisions from this viewpoint.

WORKED SOLUTION

STEP III 2019, Question 11

Probability and Statistics • Poisson Thinning

STEP III 2019, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Everything rests on conditioning on the number of customers who arrive, or who take sand, and then summing a Poisson series. Part (i) is the classic "thinning" of a Poisson distribution: when the binomial and Poisson probabilities are multiplied together, the sum over the unseen total collapses to an exponential. Once we know that the number of sand-takers is Poisson, parts (ii) and (iii) only need the expectation of $(1-k)^Y$, and that is another exponential series.

Solution

Let N be the number of customers in a day, so $N \sim \text{Po}(\lambda)$, and let Y be the number who take sand. We assume customers decide independently of each other and of N, so that given $N = n$ we have $Y \sim \text{B}(n, p)$.

Part (i)

For $y = 0, 1, 2, \ldots$, conditioning on N (only $n \geq y$ contributes):

$$\begin{aligned}
\text{P}(Y = y) &= \sum_{n=y}^{\infty} \text{e}^{-\lambda} \frac{\lambda^n}{n!} \binom{n}{y} p^y (1-p)^{n-y} \\
&= \text{e}^{-\lambda} \frac{(p\lambda)^y}{y!} \sum_{n=y}^{\infty} \frac{(\lambda(1-p))^{n-y}}{(n-y)!} \qquad \text{since } \frac{\lambda^n}{n!} \cdot \frac{n!}{y!\,(n-y)!} = \frac{\lambda^y \lambda^{n-y}}{y!\,(n-y)!} \\
&= \text{e}^{-\lambda} \frac{(p\lambda)^y}{y!} \, \text{e}^{\lambda(1-p)} = \text{e}^{-p\lambda} \frac{(p\lambda)^y}{y!},
\end{aligned}$$

where we put $m = n - y$ and used $\sum_{m \geq 0} x^m / m! = \text{e}^x$. This is the $\text{Po}(p\lambda)$ probability, so $Y \sim \text{Po}(p\lambda)$, as required.

Part (ii)

Each customer who takes sand leaves a fraction $1-k$ of what was there. So after Y sand-takers the mass left is $S(1-k)^Y$, and the mass taken is

$$T = S\big(1-(1-k)^Y\big).$$

By part (i),

$$\mathrm{E}\big[(1-k)^Y\big] = \sum_{y=0}^{\infty}(1-k)^y\,\mathrm{e}^{-p\lambda}\,\frac{(p\lambda)^y}{y!} = \mathrm{e}^{-p\lambda}\,\mathrm{e}^{p\lambda(1-k)} = \mathrm{e}^{-kp\lambda}.$$

Hence, by linearity of expectation,

$$\mathrm{E}(T) = S\big(1-\mathrm{E}[(1-k)^Y]\big) = \big(1-\mathrm{e}^{-kp\lambda}\big)S,$$

which is the required result.

Part (iii)

> **Ask yourself** How do we turn "a proportion k of the sand" into a probability about one particular grain?

Since there are a great many grains and none is singled out, whenever someone takes a proportion k of the remaining sand, a grain that is still in the bag is taken with probability k, independently of what happened earlier. (Equivalently, the chance that the golden grain is in a given portion equals that portion's share of the mass.)

Let G be the event that the assistant takes the golden grain. Given $Y=y$, the grain survives all y customers with probability $(1-k)^y$, and then the assistant takes it with probability k. So

$$\mathrm{P}(G \mid Y=y) = k(1-k)^y,$$

and, using the expectation found in part (ii),

$$\mathrm{P}(G) = k\,\mathrm{E}\big[(1-k)^Y\big] = k\,\mathrm{e}^{-kp\lambda}.$$

This agrees with part (ii): the probability that the golden grain survives the day is $\mathrm{e}^{-kp\lambda}$, which is exactly the expected fraction of sand remaining.

The case $k=0$. Then $\mathrm{P}(G)=0$. This makes sense: with $k=0$ nobody, the assistant included, takes any sand, so the grain stays in the bag.

The limit $k\to 1$. Then $\mathrm{P}(G)\to \mathrm{e}^{-p\lambda}$, which is $\mathrm{P}(Y=0)$. When k is close to 1, the first customer who takes sand takes nearly all of it, golden grain almost certainly included. The assistant essentially gets the grain only if no customer took sand, and then takes (almost) the whole bag.

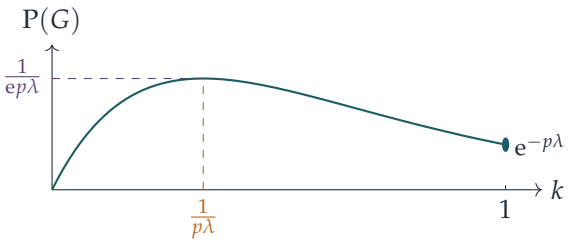
Maximising. Write $\mu = p\lambda$ and $f(k) = k\,\mathrm{e}^{-\mu k}$ for $0\le k<1$. Then

$$f'(k) = \mathrm{e}^{-\mu k}(1-\mu k),$$

which is positive for $k < 1/\mu$ and negative for $k > 1/\mu$. Since $\mu > 1$, the point $k = 1/\mu$ lies in $(0,1)$, so f increases up to it and decreases after it, and it gives the maximum over $[0,1)$. Hence the probability is maximised when

$$k = \frac{1}{p\lambda},$$

and the maximum probability is $\dfrac{1}{\mathrm{e}\,p\lambda}$. (If $p\lambda \le 1$ then f increases throughout $[0,1)$ and there is no maximum, only the supremum $\mathrm{e}^{-p\lambda}$ as $k \to 1$. That is why the question restricts to $p\lambda > 1$.)



The graph shows $\mathrm{P}(G) = k\,\mathrm{e}^{-kp\lambda}$ in the case $p\lambda = 3$.

Checks

- Part (i) passes the obvious tests: $p = 1$ gives back $\mathrm{Po}(\lambda)$, and $p = 0$ gives $Y = 0$ for certain.
- In part (ii), $k = 0$ gives zero sand taken, and as $\lambda \to \infty$ the expected mass taken tends to S, as it should.
- With $\lambda = 3$, $p = 0.6$, $k = 0.4$ the formula gives $\mathrm{P}(G) = 0.4\,\mathrm{e}^{-0.72} \approx 0.1947$. A computer simulation of 200 000 days gave 0.1953.
- With $p\lambda = 3$, the largest value of $k\,\mathrm{e}^{-3k}$ on a fine grid occurs at $k \approx 0.3333$ and equals $0.1226 \approx 1/(3\mathrm{e})$.

Common slips

- Writing the sand taken as kSY, as though each customer took a proportion k of the *original* sand. Each customer takes a share of what is *left*, so the amounts form a geometric sequence.
- Computing $\mathrm{E}[(1-k)^Y]$ as $(1-k)^{\mathrm{E}(Y)} = (1-k)^{p\lambda}$. The expectation of a non-linear function is not the function of the expectation.
- Forgetting to check that $1/(p\lambda)$ actually lies in the allowed range $0 \le k < 1$. This is exactly where the condition $p\lambda > 1$ is used.

Beyond STEP

Part (i) is the thinning property of the Poisson distribution. The expectation $\mathrm{E}[s^Y] = \mathrm{e}^{\mu(s-1)}$ that appears in parts (ii) and (iii) is the *probability generating function* of $\mathrm{Po}(\mu)$, evaluated at $s = 1 - k$. Generating functions make thinning almost a one-liner: if $\mathrm{E}[s^N] = \mathrm{e}^{\lambda(s-1)}$, then $\mathrm{E}[s^Y] = \mathrm{E}[(1-p+ps)^N] = \mathrm{e}^{\lambda p(s-1)}$. In the theory of Poisson processes this becomes the statement that independently marking the points of a Poisson process gives Poisson processes again. Grimmett and Welsh, *Probability: An Introduction*, treats generating functions and these ideas clearly.

WORKED SOLUTION

STEP III 2019, Question 12

Probability and Statistics • Random Subsets

STEP III 2019, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 9.12, *Random Subsets*: https://beyondstep.keithmcnulty.org/ch09/random-subsets.html

The key idea

Choosing a subset of S uniformly at random is the same as going through the integers $1, 2, \ldots, n$ one at a time and tossing a fair coin for each to decide whether it goes in. Once we see this, every event in the question becomes a statement about each integer separately, the integers behave independently, and the answer is a single small probability raised to the power n.

Solution

Counting the subsets

A subset of S is fixed completely by deciding, for each of the n integers, whether it is in or out. There are 2 choices for each integer, made independently, and different patterns of choices give different subsets. So the number of subsets, counting $\varnothing$ (everything out) and S (everything in), is

$$2 \times 2 \times \cdots \times 2 = 2^n.$$

? **Ask yourself** Why may we treat the integers as independent coin tosses?

Let A be uniform on T. Each of the 2^n in/out patterns is one subset, so each pattern has probability $2^{-n} = (\frac{1}{2})^n$. That is exactly the probability of a given pattern when we toss n independent fair coins. So the events "$1 \in A$", "$2 \in A$", ..., "$n \in A$" are independent, each with probability $\frac{1}{2}$. Since $A_1, \ldots, A_m$ are chosen independently of each other too, the mn events "$i \in A_k$" are all independent, each with probability $\frac{1}{2}$.

Part (i)

Exactly half of the 2^n subsets contain 1 (pair each subset containing 1 with the same subset with 1 removed). So

$$\mathrm{P}(1 \in A_1) = \tfrac{1}{2}.$$

Part (ii)

For an integer i, let E_i be the event that i is *not* in both A_1 and A_2. Then $A_1 \cap A_2 = \varnothing$ exactly when every E_i happens. By independence,

$$\mathrm{P}(i \in A_1 \text{ and } i \in A_2) = \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4}, \qquad \text{so} \qquad \mathrm{P}(E_i) = \tfrac{3}{4}.$$

The event E_i depends only on the coins for integer i, so $E_1, \ldots, E_n$ are independent, and

$$\mathrm{P}(A_1 \cap A_2 = \varnothing) = \mathrm{P}(E_1) \cdots \mathrm{P}(E_n) = \left(\tfrac{3}{4}\right)^n,$$

as required.

The same argument works for any number of sets. An integer i lies in $A_1 \cap A_2 \cap A_3$ with probability $(\frac{1}{2})^3 = \frac{1}{8}$, so it avoids the intersection with probability $\frac{7}{8}$, and

$$\mathrm{P}(A_1 \cap A_2 \cap A_3 = \varnothing) = \left(\tfrac{7}{8}\right)^n.$$

In general i lies in all m sets with probability 2^{-m}, so

$$\mathrm{P}(A_1 \cap A_2 \cap \cdots \cap A_m = \varnothing) = \left(1 - \frac{1}{2^m}\right)^n.$$

Part (iii)

Now $A_1 \subseteq A_2$ means that no integer is in A_1 but missing from A_2. For each integer i there are four equally likely possibilities for (in A_1?, in A_2?), and only one of them, "in A_1, not in A_2", is forbidden. So each integer behaves well with probability $\frac{3}{4}$, independently, and

$$\mathrm{P}(A_1 \subseteq A_2) = \left(\tfrac{3}{4}\right)^n.$$

(This matches part (ii): $A_1 \subseteq A_2$ exactly when A_1 and the complement of A_2 are disjoint, and the complement of a uniform random subset is again uniform.)

For $A_1 \subseteq A_2 \subseteq A_3$, look at the membership pattern of one integer i in (A_1, A_2, A_3), writing 1 for "in" and 0 for "out". The chain condition says that once i is in some A_k it is in all the later ones too, so the pattern must be one of

$$000, \quad 001, \quad 011, \quad 111.$$

That is 4 of the 8 equally likely patterns, so each integer is fine with probability $\frac{4}{8} = \frac{1}{2}$, and

$$\mathrm{P}(A_1 \subseteq A_2 \subseteq A_3) = \left(\tfrac{1}{2}\right)^n.$$

For a chain of m sets, an allowed pattern is a block of 0s followed by a block of 1s: it is decided by the position where the 1s start, which is one of $1, 2, \ldots, m$, or "never". That gives $m+1$ allowed patterns out of 2^m, so

$$\mathrm{P}(A_1 \subseteq A_2 \subseteq \cdots \subseteq A_m) = \left(\frac{m+1}{2^m}\right)^n.$$

Checks

- With $m = 2$ the general formulas give $(1 - \frac{1}{4})^n$ and $(\frac{3}{4})^n$, and with $m = 3$ they give $(\frac{7}{8})^n$ and $(\frac{4}{8})^n$, matching the special cases. With $m = 1$ they give $(\frac{1}{2})^n$ (only $A_1 = \varnothing$ works) and 1 (one set is always a chain), as they should.
- With $n = 1$ everything can be listed by hand. For example, $A_1 = A_2 = \{1\}$ is the only one of the four choices with a non-empty intersection, giving $\frac{3}{4}$.
- A computer check listing every choice of sets confirmed all the formulas for small n and m. For example, with $n = 3$, $\mathrm{P}(A_1 \cap A_2 = \varnothing) = \frac{27}{64}$, and $\mathrm{P}(A_1 \subseteq A_2 \subseteq A_3) = \frac{1}{8}$.

Common slips

- Answering $\left(\frac{1}{2}\right)^n$ for $\mathrm{P}(A_1 \subseteq A_2)$ by arguing that each element of A_1 must lie in A_2, without allowing for the size of A_1 itself being random.
- Counting the chain patterns as m rather than $m+1$, by forgetting the pattern in which the integer is in none of the sets.

Beyond STEP

Representing a random subset as a string of independent coin tosses (its *indicator vector*) is a standard tool in combinatorics, and the chain count here is the start of the theory of *posets* and the Boolean lattice. The same ideas give the expected number of chains, or the probability that random sets cover S, with very little extra work. Grimmett and Welsh, *Probability: An Introduction*, develops independence and indicator variables in this spirit.

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